

# Thesis Plan

Decentralized Multi-Agent Reinforcement Learning for Power Grid Topology Control

Corentin Plumet

August 27, 2026

# Contents

|          |                                                                           |           |
|----------|---------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Thesis Plan</b>                                                        | <b>1</b>  |
| 1.1      | Working Title . . . . .                                                   | 1         |
| 1.2      | Central Thesis Question . . . . .                                         | 1         |
| 1.3      | Research Questions . . . . .                                              | 1         |
| 1.4      | Expected Contributions . . . . .                                          | 1         |
| <b>2</b> | <b>Introduction</b>                                                       | <b>3</b>  |
| 2.1      | Topology control . . . . .                                                | 3         |
| 2.2      | Why it has to be automated . . . . .                                      | 3         |
| 2.3      | Why decentralized agents . . . . .                                        | 4         |
| 2.4      | Reinforcement learning for topology control . . . . .                     | 4         |
| 2.5      | Starting point: MARL2Grid and MAPPO . . . . .                             | 5         |
| 2.6      | Outline . . . . .                                                         | 5         |
| <b>3</b> | <b>Establishing a Baseline</b>                                            | <b>6</b>  |
| 3.1      | Evaluation Protocol . . . . .                                             | 6         |
| 3.2      | IEEE 14-Bus Experiments . . . . .                                         | 6         |
| 3.2.1    | MLP Baseline . . . . .                                                    | 6         |
| 3.2.2    | Initial Do-Nothing Bias . . . . .                                         | 8         |
| <b>4</b> | <b>Sparse Intervention Control</b>                                        | <b>10</b> |
| 4.1      | Evaluation-Time Rho Heuristics . . . . .                                  | 10        |
| 4.2      | Intervention-Gated Actor . . . . .                                        | 11        |
| 4.3      | Fixed Sparse-Action Penalty . . . . .                                     | 12        |
| 4.4      | Adaptive Intervention Budget . . . . .                                    | 12        |
| 4.5      | Behavioral Cloning from Heuristics . . . . .                              | 13        |
| 4.6      | Results . . . . .                                                         | 13        |
| 4.7      | Preliminary WCCI Stress Test . . . . .                                    | 14        |
| <b>5</b> | <b>Methods</b>                                                            | <b>16</b> |
| 5.1      | Physical Scaling and Voltage-Angle Representation . . . . .               | 16        |
| 5.1.1    | Physical scaling . . . . .                                                | 16        |
| 5.1.2    | Angle representation . . . . .                                            | 16        |
| 5.1.3    | Use across the experiments . . . . .                                      | 17        |
| 5.2      | Graph Representations . . . . .                                           | 17        |
| 5.2.1    | Busbar Representation . . . . .                                           | 17        |
| 5.2.2    | Disaggregated Representation . . . . .                                    | 18        |
| 5.2.3    | Disaggregated Line-Node Representation . . . . .                          | 20        |
| 5.2.4    | Local actor information boundary and the global state graph . . . . .     | 22        |
| 5.2.5    | Graph actor and centralized critic . . . . .                              | 24        |
| 5.3      | Substation Representation: Direct Edges and Summary Nodes . . . . .       | 25        |
| 5.3.1    | Direct same-substation edges (e) . . . . .                                | 26        |
| 5.3.2    | Substation summary nodes (n) . . . . .                                    | 27        |
| 5.4      | Pooling and Graph Readout . . . . .                                       | 27        |
| 5.4.1    | Mean and maximum pooling . . . . .                                        | 28        |
| 5.4.2    | Virtual-node graph readout . . . . .                                      | 29        |
| 5.5      | Candidate-Action Scoring from the Disaggregated Line-Node Graph . . . . . | 30        |

|          |                                                                  |           |
|----------|------------------------------------------------------------------|-----------|
| 5.5.1    | From action indices to candidate scoring . . . . .               | 30        |
| 5.5.2    | What each action touches . . . . .                               | 31        |
| 5.5.3    | Pooling the touched nodes . . . . .                              | 32        |
| 5.5.4    | Scoring and the idle action . . . . .                            | 33        |
| 5.5.5    | Global-maximum and action-delta extension . . . . .              | 33        |
| 5.5.6    | Cost and current limits . . . . .                                | 34        |
| 5.6      | Action-Space Reduction for Large Grids . . . . .                 | 34        |
| <b>6</b> | <b>Graph-Design Screening</b>                                    | <b>36</b> |
| 6.1      | Preliminary GNN Encoder Viability . . . . .                      | 36        |
| 6.2      | Shared Protocol . . . . .                                        | 39        |
| 6.2.1    | Endpoint statistics . . . . .                                    | 39        |
| 6.2.2    | Compute . . . . .                                                | 39        |
| 6.3      | Screen A: Encoder Depth and Width . . . . .                      | 39        |
| 6.4      | Screen B: Graph Readout Scope and Reducer . . . . .              | 41        |
| 6.5      | Screen C: Structural Augmentations of the Busbar Graph . . . . . | 42        |
| 6.6      | Screen D: Generator and Load Message Direction . . . . .         | 44        |
| 6.7      | Screen E: Candidate-Action Scoring . . . . .                     | 45        |
| 6.7.1    | Why the descriptor hurt: a normalization defect . . . . .        | 46        |
| 6.7.2    | Replication with Improved Graph Features . . . . .               | 47        |
| <b>7</b> | <b>What the Shared Encoder Reads</b>                             | <b>49</b> |
| 7.1      | Measurement protocol . . . . .                                   | 49        |
| 7.2      | What the encoder reads . . . . .                                 | 50        |
| 7.3      | Results . . . . .                                                | 50        |
| 7.4      | Two failure modes . . . . .                                      | 51        |
| 7.5      | Correcting the inputs . . . . .                                  | 52        |
| 7.5.1    | Running standardization makes the mismatch worse . . . . .       | 52        |
| 7.5.2    | Physical scaling for the power channels . . . . .                | 53        |
| 7.5.3    | The angle displacement is a gauge artifact . . . . .             | 53        |
| 7.5.4    | The corrected distribution . . . . .                             | 54        |
| <b>8</b> | <b>WCCI36: Target Environment and Controls</b>                   | <b>56</b> |
| 8.1      | Target environment . . . . .                                     | 56        |
| 8.1.1    | Removing maintenance . . . . .                                   | 56        |
| 8.1.2    | Reduced action spaces . . . . .                                  | 57        |
| 8.2      | Evaluation protocol . . . . .                                    | 57        |
| 8.3      | Inputs used for transfer . . . . .                               | 57        |
| 8.4      | Controls trained on WCCI . . . . .                               | 57        |
| <b>9</b> | <b>Zero-Shot Transfer and Target Adaptation</b>                  | <b>59</b> |
| 9.1      | Comparison rules . . . . .                                       | 59        |
| 9.2      | What controls zero-shot transfer? . . . . .                      | 59        |
| 9.3      | Policy changes and the local heuristic . . . . .                 | 61        |
| 9.4      | Does target training help? . . . . .                             | 62        |
| 9.5      | Limits of the study . . . . .                                    | 62        |
| <b>A</b> | <b>Supplementary Results</b>                                     | <b>63</b> |
| A.1      | Sparse Intervention Control . . . . .                            | 63        |
| A.2      | Screen A: Encoder Depth and Width . . . . .                      | 65        |
| A.3      | Screen B: Graph Readout Scope and Reducer . . . . .              | 66        |
| A.4      | Screen C: Structural Augmentations of the Busbar Graph . . . . . | 67        |
| A.5      | Screen D: Generator and Load Message Direction . . . . .         | 69        |
| A.6      | Screen E: Candidate-Action Scoring . . . . .                     | 70        |
| A.7      | Encoder Input Distributions . . . . .                            | 71        |
| A.8      | Deterministic Feature Scales . . . . .                           | 72        |
| A.9      | WCCI Result Atlas . . . . .                                      | 73        |
| A.9.1    | Evaluation coverage . . . . .                                    | 73        |
| A.9.2    | Zero-shot matrices . . . . .                                     | 74        |

|          |                                               |           |
|----------|-----------------------------------------------|-----------|
| A.9.3    | Target-trained controls . . . . .             | 76        |
| A.9.4    | Per-chronic diagnostic . . . . .              | 79        |
| <b>B</b> | <b>GINE Graph-Actor Architecture</b>          | <b>80</b> |
| B.1      | Shared input and encoder pipeline . . . . .   | 80        |
| B.2      | GINE message passing . . . . .                | 80        |
| B.3      | Pooling and graph output . . . . .            | 81        |
| B.4      | Agent-specific action head . . . . .          | 81        |
| B.5      | Exact node inputs by representation . . . . . | 82        |
| B.6      | Exact edge inputs by representation . . . . . | 82        |
| <b>C</b> | <b>Graph Construction Details</b>             | <b>83</b> |
| C.1      | Candidate graph sizes . . . . .               | 83        |
| C.2      | Indexing and masking conventions . . . . .    | 83        |
| C.3      | Local actor graphs . . . . .                  | 84        |
| C.4      | Boundary verification . . . . .               | 84        |
| <b>D</b> | <b>Experiment Configurations</b>              | <b>85</b> |
| <b>E</b> | <b>Full-Test Checkpoint Registry</b>          | <b>88</b> |

# Chapter 1

## Thesis Plan

### 1.1 Working Title

**Graph Neural Networks for Decentralized Multi-Agent Reinforcement Learning in Power-Grid Topology Control**

### 1.2 Central Thesis Question

**How can graph neural networks help decentralized reinforcement learning agents learn reliable power-grid topology control policies that keep survival performance while scaling to larger environments?**

Three terms define what the thesis has to measure:

- **Reliable** is episodic survival on the chronics.
- **Decentralized** means each agent owns a fixed group of substations, observes a local subgraph, and acts on its own discrete topology actions without seeing what the other agents do. Only the critic is centralized.
- **Scaling** means larger grids, with more substations, loads and generators.

### 1.3 Research Questions

1. **How should the grid be represented for a local actor?** Is a graph representation useful for learning? If yes, how fine the nodes are, which relations to use, what to add at substation level, and how the graph is read out are all design choices. Chapter 5 defines them and Chapter 6 measures which ones change survival.
2. **Is there a transfer possible between a smaller and a larger grid?** Moving from bus14 to the 36-substation WCCI grid without changing the weights of a model tests whether a graph encoder transfers in practice and not only in parameter shape. (Chapters 8 and 9.)

### 1.4 Expected Contributions

- **A tuned decentralized MAPPO baseline on bus14.** The original framework trained unstably. The training and initialization changes that fix it are tested one at a time.
- **An ablation of sparse intervention mechanisms,** reported together on survival and intervention frequency. This links the learned behavior to what a control room would accept: act rarely, act

locally, and act when the grid is unsafe.

- **A shared GNN actor and a screened local graph design.** The GNN first matches the tuned MLP. Three representations are then compared. Encoder depth and width, message-passing operator, readout, and structural augmentations are screened.
- **A model for action scoring that does not depend on grid size or action space size.** A candidate-action head scores each action from the encoded nodes it changes.

# Chapter 2

## Introduction

### 2.1 Topology control

A transmission grid is not a fixed graph. Inside a substation, every connected element — each transmission line, generator, and load — is attached to one of several busbars. An operator can move an element from one busbar to another, or split a substation into two electrically separate nodes. Doing so changes which paths the power actually flows through, and therefore redistributes loading across the network. This is topology control.

Its appeal is economic. Congestion is usually relieved in one of two ways. Redispatching generation or curtailing renewable production costs money at every intervention; building new lines costs a great deal once and takes years. Topology control uses assets that already exist and switching equipment that is already installed, so the cost of one more action is close to zero [1]. The energy transition makes this a practical need and not just an elegant idea: variable renewable generation makes flows less predictable and pushes existing corridors closer to their thermal limits.

The catch is that topology control is hard to use well. Operators use it, but mostly through a small set of remedial actions built from experience and offline study. As a result, a large part of the available flexibility is never searched [2].

### 2.2 Why it has to be automated

Two main properties make manual search impractical.

First, the action space is combinatorial. A substation with  $d$  connected elements and two busbars has  $2^d$  possible assignments, and the number of joint configurations of a grid is the product over its substations. Even the small IEEE 14-bus benchmark used throughout this thesis has 178 distinct unitary topology actions. The 36-substation grid of Chapter 8 has more than 66 000, which is why it has to be reduced offline before an agent can be trained on it at all (Section 5.6).

Second, the decision is sequential and time-constrained. An action taken now changes which contingencies are survivable later. Cooldown periods forbid acting on the same substation twice in quick succession, and an operator has minutes to decide what to do.

These are the conditions under which reinforcement learning is a reasonable candidate: a sequential decision problem, a simulator that is expensive but available, no optimum that can be computed directly, and a clear success signal — the grid does not black out. The Learning to Run a Power Network (L2RPN) competition series was created by RTE to open exactly this problem to machine-learning research. The Grid2Op simulator provides the environment, the chronics of load and generation, and the operational rules that go with it [3], [4].

## 2.3 Why decentralized agents

Most published work treats the grid as a single agent controlling everything. Real grids are not operated that way. Transmission system operators divide the network into regions, each with its own control centre and its own operators, who act on their own substations with a detailed view of their own area and a much less detailed view of the rest [2]. A decentralized setup is therefore closer to how the job is actually done. It is also closer to how any automated assistant would eventually be deployed: as support inside one control room, not as a single controller over the whole network.

It is also the setup that scales. A single controller over the whole grid has to represent the joint action space, which multiplies with every extra substation. Splitting the grid into regions and giving each region its own policy replaces that product by a sum: each agent chooses among its own local actions only. This is the standard argument for decentralized multi-agent reinforcement learning, and it comes with the standard cost [5]. Because every agent learns while the others are also changing, each one faces an environment that is non-stationary from its own point of view. Each agent also observes only part of the grid, so agents can take individually sensible actions that are jointly harmful.

The usual answer is *centralized training with decentralized execution*: a critic is allowed to see the global state during training, while each agent sees only what it will have access to at deployment time [6], [7]. This thesis uses that setting throughout.

## 2.4 Reinforcement learning for topology control

Work on RL for topology control falls into four broad groups. A wider survey of graph-based RL for power systems is given by Hassouna et al. [8].

**Single-agent policies.** The first approaches train one agent to control the whole grid. Subramanian et al. [9] showed that deep RL over a hand-picked action set can beat a do-nothing policy on L2RPN benchmarks. SMAAC, winner of the L2RPN WCCI 2020 challenge, combines actor-critic learning over *afterstates* — configurations after an action but before the physics resolves — with a hierarchical choice of where and how to act [10]. These methods share one obstacle: the action space must be reduced, curated, or factorized before learning is feasible.

**Learned policies wrapped in heuristics.** A second group wraps learned policies in rules, most commonly querying the agent only when the maximum line loading  $\rho_{\max}$  indicates stress and doing nothing otherwise. PowRL combined an RL policy with an overload manager and reduced action set to win the L2RPN NeurIPS 2020 robustness track [11]. Lehna et al. [12] found similar survival for advanced rule-based and RL agents, with much cheaper execution for RL. Beating a well-engineered heuristic is therefore not automatic; such heuristics are now standard practice and are also used in this thesis.

**Hierarchical and centrally coordinated multi-agent methods.** A third group divides the decision but retains a coordinator. Manczak et al. [13] separate whether, where, and how to act; van der Sar et al. [14] place substation agents under a higher-level selector; and de Mol et al. [15] use a central agent to choose among regional proposals. These methods retain much of the scalability of factorization, but they still include a component that sees the whole grid and decides for everyone. They therefore do not answer whether purely local policies are enough.

**Decentralized multi-agent control.** Whether agents can operate a grid from local observations, without coordination or communication, remains less explored and is the focus of this thesis. The MARL2Grid-TR benchmark targets this gap because earlier work “focused on single-agent settings, neglecting the often decentralized, multi-agent nature of grid control” [16].

**Graph representations.** Across all four groups, the grid representation also matters. Flat vectors discard topology and fix the policy to one grid size. Graph neural networks preserve topology and have a parameter count independent of the number of substations, enabling transfer between grids [17], [18], [19].

Yet standard homogeneous graphs encode current connections but not those an action could create, an issue called *busbar information asymmetry*; heterogeneous representations address it [20]. Chapters 5 and 6 study this representation choice.

## 2.5 Starting point: MARL2Grid and MAPPO

This thesis builds on MARL2Grid [16], the multi-agent successor to RL2Grid [21]. It wraps Grid2Op in a multi-agent interface and supplies the region split, observation and action spaces, reward, and training loop modified in this work.

**How the grid is split.** The grid is divided in advance into one non-overlapping region per agent. In the IEEE 14-bus environment (**bus14**, 14 substations and 20 lines), three agents own substations  $\{0, 1, 2, 4\}$ ,  $\{3, 6, 7, 8\}$ , and  $\{5, 9, \dots, 13\}$ , with 52, 50, and 76 non-idle actions respectively. These are the 178 actions of the full grid, split rather than multiplied. Each agent observes only quantities belonging to its region, chooses among actions at its substations, and acts simultaneously without knowing the other agents' choices. Action 0 is always do-nothing.

**Reward.** A flat +1 per surviving step dominates the training reward, with an additional overload term, so the objective is essentially to keep the episode alive. The reported metric is episodic survival: the fraction of evaluation episodes completed without blackout.

**MAPPO.** Multi-agent PPO [7] extends PPO [22] to centralized training with decentralized execution. Each agent  $i$  has a stochastic policy  $\pi_{\theta_i}(a_i | o_i)$  over its discrete local action space, updated with the clipped surrogate objective

$$L^{\text{CLIP}}(\theta_i) = \hat{\mathbb{E}}_t \left[ \min \left( r_t(\theta_i) \hat{A}_t, \text{clip} \left( r_t(\theta_i), 1 - \epsilon, 1 + \epsilon \right) \hat{A}_t \right) \right], \quad r_t(\theta_i) = \frac{\pi_{\theta_i}(a_{i,t} | o_{i,t})}{\pi_{\theta_i^{\text{old}}}(a_{i,t} | o_{i,t})},$$

where  $\hat{A}_t$  is a generalized advantage estimate from a single centralized value function trained on the global state. That state is available in simulation but never given to the actors. Only the actors remain at execution time, making deployment decentralized.

**What this thesis starts from.** The starting baseline uses flat multilayer-perceptron actors and a critic trained on **bus14**. It is unstable at low survival out of the box. It is also tied to one grid: the MLP input width depends on the number of substations, so nothing it learns can be reused elsewhere. The following chapters address these limitations in order: stabilize training, make interventions more operator-like, then replace the fixed-width representation.

## 2.6 Outline

Chapter 3 tunes the flat baseline until it trains stably. Chapter 4 then changes the objective and asks whether the same agents can keep their survival while intervening rarely. Its early WCCI pilot shows the limits of keeping a fixed-width MLP actor. Chapter 5 therefore defines size-independent graph inputs: feature scaling, three grid representations, substation-level augmentations, the readout, and the candidate-action head. Chapter 6 first checks that a shared graph encoder remains competitive, then screens those design choices. Chapter 8 moves to the 36-substation WCCI grid and tests what actually transfers.

## Chapter 3

# Establishing a Baseline

This chapter builds a solid MLP actor baseline for the IEEE 14-bus task.

### 3.1 Evaluation Protocol

All experiments use an 80/20 train/test chronic split. The 80% training split is used for policy updates, while the 20% test split measures how well the policy generalizes. Evaluation rollouts run periodically during training using 10 episodes per split.

The main metric is test episodic survival (as a percentage): the fraction of evaluation timesteps where the agent keeps the grid alive. Results are reported as mean curves across seeds, with shaded regions showing the spread. Final evaluations report the checkpoint with the highest test survival.

### 3.2 IEEE 14-Bus Experiments

The initial MAPPO baseline from the MARL2Grid framework led to unstable policies and low survival rates on the IEEE 14-bus topology-control task. The experiments below test changes to the training and to the architecture, in order to build a strong baseline.

#### 3.2.1 MLP Baseline

**Question.** Which changes are needed to get a stable MLP MAPPO baseline on the 14-bus task?

**Baseline and changes.** We start from the MLP MAPPO baseline of the MARL2Grid framework and test several main changes meant to stabilize training:

- using reward normalization;
- changing the critic update so that the critic is updated once for the whole batch instead of once per agent;
- tuning the optimization hyperparameters to stabilize training, as summarized in Table [3.1](#).

Table 3.1: New hyperparameters for MAPPO training.

| Hyperparameter                       | New Value  | Paper Baseline |
|--------------------------------------|------------|----------------|
| <b>ENVIRONMENT</b>                   |            |                |
| n_envs (number parallel environment) | 20         | 10             |
| <b>PPO OPTIMIZATION</b>              |            |                |
| Total time step                      | 15,000,000 | 25,000,000     |
| Actor learning rate                  | 1e-4       | 3e-5           |
| Critic learning rate                 | 1e-4       | 3e-5           |
| Gamma                                | 0.9        | 0.99           |
| Update epochs                        | 10         | 80             |
| Minibatches                          | 16         | 4              |
| Maximum gradient norm                | 1.0        | 10.0           |

**Evidence.** Table 3.3 reports final and peak survival for the core ablations. Figure 3.1 overlays all core MLP baseline curves.

**Interpretation.** The **Optimized MLP Baseline**, which combines reward normalization and an optimized critic update, reaches the best performance and the fastest convergence, with the highest and most stable final survival ( $\sim 93\%$ ). Removing or reducing any of these three elements — using a smaller architecture, disabling reward normalization, or going back to a non-optimized critic update — lowers either learning speed or final survival. The original unoptimized baseline fails entirely to learn a competitive policy. Because the **Optimized MLP Baseline** consistently beats the ablations, it is used as the reference for all later comparisons.

Table 3.2: MLP baseline run configurations. The reference run is **Optimized MLP Baseline**.

| Run family                      | Actor/critic hidden layers    | Reward norm. | Initial action-0 prob. | Optimized critic update | Difference from <b>Optimized MLP Baseline</b> |
|---------------------------------|-------------------------------|--------------|------------------------|-------------------------|-----------------------------------------------|
| <b>Optimized MLP Baseline</b>   | $3 \times 256 / 3 \times 256$ | true         | 0.0                    | true                    | Reference baseline                            |
| Disable optimized critic update | $3 \times 256 / 3 \times 256$ | true         | 0.0                    | false                   | Disables optimized critic updates             |
| Smaller MLP actor/critic        | $2 \times 256 / 2 \times 256$ | true         | 0.0                    | true                    | Uses a smaller MLP actor and critic           |
| Disable reward normalization    | $3 \times 256 / 3 \times 256$ | false        | 0.0                    | true                    | Disables reward normalization                 |
| Initial Bias 50%                | $3 \times 256 / 3 \times 256$ | true         | 0.5                    | true                    | Uses action-0 bias 0.5                        |
| Initial Bias 70%                | $3 \times 256 / 3 \times 256$ | true         | 0.7                    | true                    | Uses action-0 bias 0.7                        |

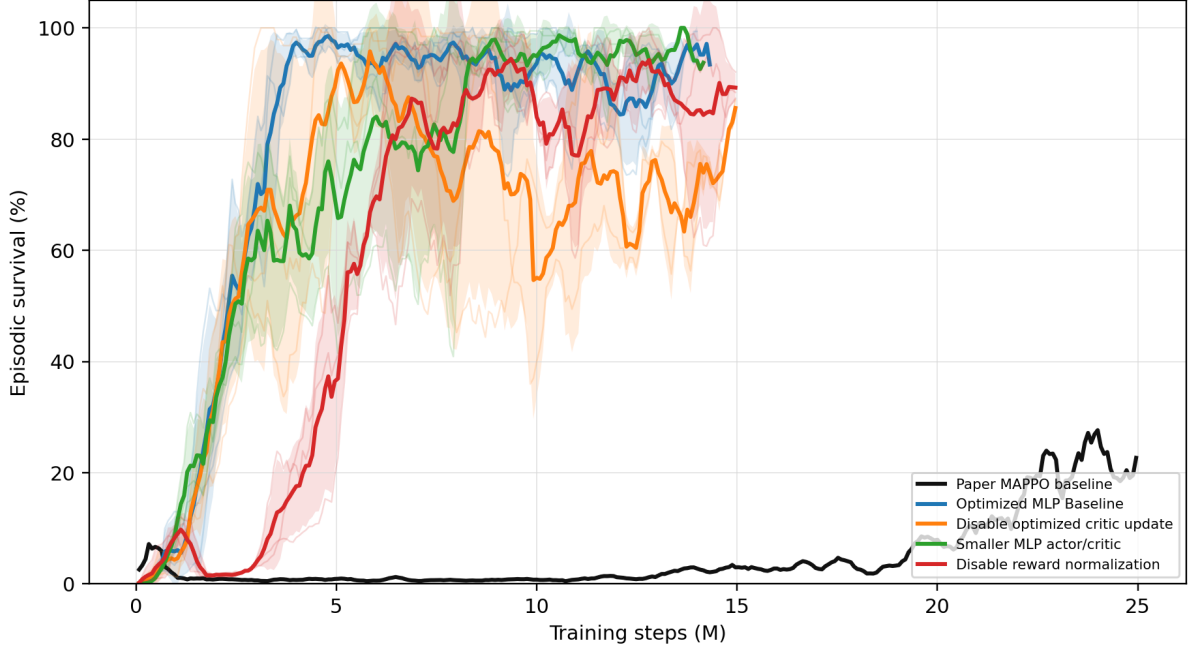


Figure 3.1: All MLP baseline core rerun survival curves shown on one axis.

Table 3.3: MLP baseline core rerun survival summary.

| Run                             | Intended change             | Final survival (%) | Peak survival (%) |
|---------------------------------|-----------------------------|--------------------|-------------------|
| Optimized MLP Baseline          | Optimized MLP baseline      | 93.4               | 98.6              |
| Disable optimized critic update | Non-optimized critic update | 93.1               | 95.8              |
| Smaller MLP actor/critic        | Smaller MLP architecture    | 93.7               | 100.0             |
| Disable reward normalization    | No reward normalization     | 81.8               | 97.1              |

### 3.2.2 Initial Do-Nothing Bias

**Question.** Does biasing the initial policy toward action 0, the do-nothing action, help or hurt learning? The idea makes sense: under normal conditions, human operators usually leave the grid alone and take no action.

**Baseline and changes.** Our baseline uses no initial do-nothing bias in this rerun set. The bias controls keep the same general protocol but change the initial probability mass assigned to action 0.

**Implementation detail.** The bias is applied only at initialization: it changes the starting action distribution before learning, without changing the reward, architecture, or PPO objective. It is implemented by raising the initial pre-softmax logits of the do-nothing action, so that the policy picks this safe action more often before any learning update happens. This makes the ablation a test of exploration bias rather than a test of model capacity.

**Evidence.** Table 3.4 and Figure 3.2 compare the zero-bias baseline with Initial Bias 50% and Initial Bias 70%.

**Interpretation.** The effect of the action-0 bias is large, and a stronger bias is not simply better or worse. The 0.5-bias curve remains unstable and well below the zero-bias baseline, while the 0.7-bias curve eventually reaches the highest final survival in this comparison. The current conclusion is that adding an initialization bias is generally harmful or very unstable for the models. Instead of helping exploration, a strong initial prior seems to stop the agent from reliably finding a wide range of useful actions. It should be treated with care as a main ablation variable.

Table 3.4: Initial do-nothing bias survival summary.

| Run                    | Final survival (%) | Peak survival (%) |
|------------------------|--------------------|-------------------|
| Optimized MLP Baseline | 93.4               | 98.6              |
| Initial Bias 50%       | 50.8               | 70.2              |
| Initial Bias 70%       | 99.3               | 100.0             |

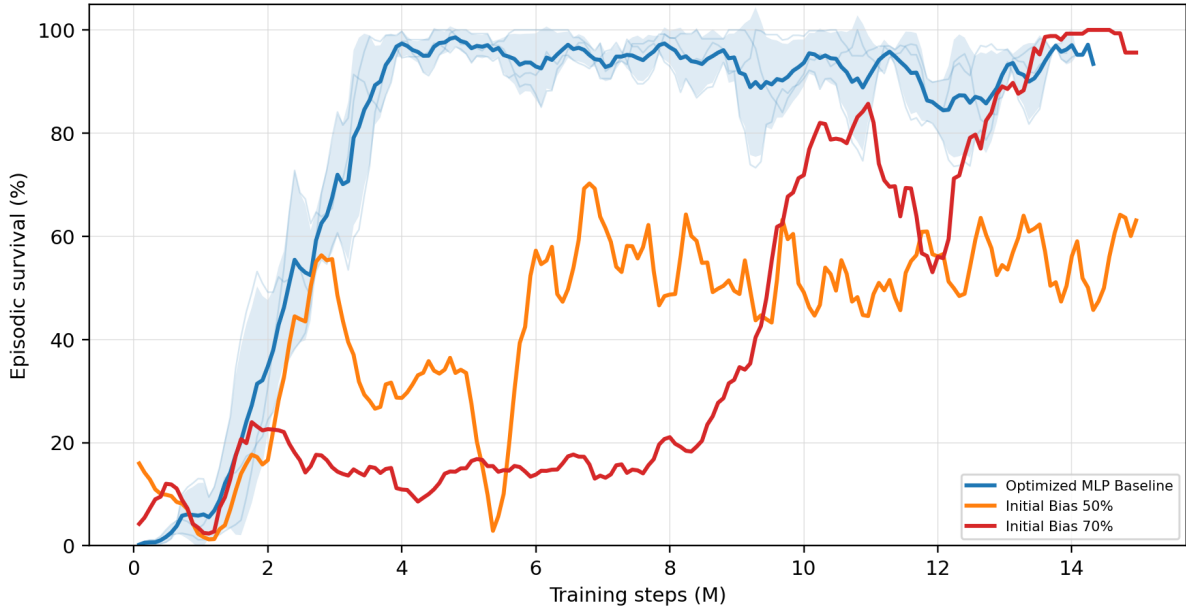


Figure 3.2: Optimized MLP baseline compared with initial do-nothing bias controls Initial Bias 50% and Initial Bias 70%.

### Chapter Summary

**Objective:** Establish a stable, high-performing flat MAPPO baseline.

#### Findings:

- **MLP optimization:** Hyperparameter tuning, reward normalization, and optimized critic updates raise the flat MLP baseline to roughly 93% survival.
- **Do-nothing initialization:** The effect is large but goes in both directions; it is not a reliable substitute for learning when to intervene.

**Conclusion:** The tuned flat actor is a strong survival baseline. The next step is to ask whether it can behave more like an operator by intervening less often without losing that survival. That is the subject of Chapter 4.

## Chapter 4

# Sparse Intervention Control

The previous chapter optimized survival on held-out chronics. This chapter adds a second objective: sparsity. A policy that keeps the grid alive by reconfiguring substations at every opportunity is impractical for a control room. Interventions carry operational costs, and an agent acting constantly is not trustworthy. The question is whether decentralized agents can keep high survival while intervening rarely, locally, and only when necessary.

In the literature, sparse activity is most often enforced by hard-coding a physical threshold that decides when the agent is allowed to act. Instead of relying on an external override, this chapter tries to make sparse intervention something the agent learns by itself.

Mechanisms are therefore measured on two competing axes: survival rate and intervention sparsity. A good mechanism must improve sparsity without hurting survival.

**Question.** Can the agents keep high survival while behaving more like operators: acting rarely and mainly when the grid state makes an intervention useful?

**Baseline and change.** This block starts from the flat decentralized MAPPO topology-control actor. For agent  $i$ , the baseline policy is one categorical distribution over the full local action space:

$$\pi_i(a_i \mid o_i), \quad a_i \in \{0, 1, \dots, |\mathcal{A}_i| - 1\}.$$

Action 0 is the no-op action. During training, actions are sampled from the policy, while deterministic evaluation uses the greedy action by default:

$$a_{i,t}^{eval} = \arg \max_a \pi_i(a \mid o_{i,t}).$$

This chapter tests five ways to reduce unnecessary topology changes: evaluation-time rho overrides, an intervention-gated actor, fixed non-idle action penalties, adaptive intervention budgets, and behavioral cloning from heuristics.

The intervention gate separates whether to act from which action to take, following the hierarchical splits used in topology control [13], [14]. The fixed and adaptive penalties instead treat non-idle interventions as a limited resource, following constrained and budgeted reinforcement learning [23], [24], [25].

### 4.1 Evaluation-Time Rho Heuristics

**Implementation detail.** The rho heuristic is an evaluation-only diagnostic. It does not change the training objective and it does not prove that the policy learned sparse behavior. The policy first proposes the deterministic action  $\bar{a}_{i,t}$ , then the evaluator may replace it with action 0.

The global version uses the pre-action maximum line loading

$$\rho_t^{global} = \max_{\ell \in \mathcal{L}} \rho_{\ell,t},$$

and executes

$$a_{i,t}^{exec} = \begin{cases} 0, & \rho_t^{global} < \rho_{safe}, \\ \bar{a}_{i,t}, & \text{otherwise,} \end{cases} \quad \rho_{safe} = 0.90.$$

The local version is decentralized. Agent  $i$  computes the maximum rho on the lines touching its regional substations,

$$\rho_{i,t}^{local} = \max_{\ell \in \mathcal{L}_i} \rho_{\ell,t},$$

and only that agent is forced to no-op when  $\rho_{i,t}^{local} < \rho_{safe}$ . A boundary line can appear in the local neighborhood of both adjacent agents, but no communication is required: both agents can observe the line through their own physical neighborhood.

**Interpretation.** The heuristic is useful to ask what happens if safe-state interventions are blocked, but it is not a learned method. Its action-0 rates must therefore be reported as final executed evaluation actions, not as evidence that the policy itself prefers action 0.

## 4.2 Intervention-Gated Actor

**Implementation detail.** The gated actor changes the actor factorization. Instead of one categorical head over all actions, each agent has a gate head and a non-idle action head:

$$\pi_i^{gate}(g_i | o_i), \quad g_i \in \{0, 1\},$$

where 0 means do nothing and 1 means intervene, and

$$\pi_i^{nonidle}(b_i | o_i), \quad b_i \in \{1, \dots, |\mathcal{A}_i| - 1\}.$$

During training, the gate is sampled first. If  $g_{i,t} = 0$ , then  $a_{i,t} = 0$ . If  $g_{i,t} = 1$ , the final action is sampled from the non-idle head. The PPO log-probability of the executed action is therefore

$$\log P(a_i = 0) = \log P(g_i = 0),$$

and, for  $a_i > 0$ ,

$$\log P(a_i) = \log P(g_i = 1) + \log P(b_i = a_i | g_i = 1).$$

Two deterministic decoders are used. The `final_action_map` decoder selects the most likely final executed action:

$$P(a_i = 0) = P(g_i = 0), \quad P(a_i = a > 0) = P(g_i = 1)P(b_i = a | g_i = 1).$$

The `hierarchical_greedy` decoder first chooses the most likely gate decision, then chooses the most likely non-idle action only if the gate selects intervention.

The original coupled entropy term is

$$H_i = H(g_i) + P(g_i = 1)H(b_i).$$

This can accidentally reward intervention because a larger  $P(g_i = 1)$  allows more non-idle entropy. The separated entropy variant is

$$H_i = \alpha_{gate}H(g_i) + \alpha_{nonidle}H(b_i),$$

and is used in the gate-plus-budget diagnostic runs.

**Interpretation.** The gate is learned and decentralized, but it can limit exploration more than a reward penalty does. If it collapses early to no-op, the non-idle head is rarely used and gets little learning signal. For this reason, the gate is treated as an important diagnostic rather than the main sparse control contribution.

### 4.3 Fixed Sparse-Action Penalty

**Implementation detail.** The fixed-penalty sweep adds a reward penalty to the final executed action of each agent:

$$r'_{i,t} = r_{i,t} - \lambda_{fixed} \mathbf{1}[a_{i,t} \neq 0],$$

with

$$\lambda_{fixed} \in \{0.0, 0.001, 0.003, 0.01\}.$$

The grid is

$$\text{actor} \in \{\text{flat}, \text{gated}\}, \quad \lambda_{fixed} \in \{0.0, 0.001, 0.003, 0.01\}, \quad \text{seed} \in \{0, 1, 2\},$$

which gives  $2 \times 4 \times 3 = 24$  runs. The state-dependent safety penalty is disabled in every condition (`safe_intervention_penalty = 0.0`), so the penalty is not rho-weighted.

**Interpretation.** This mechanism changes only the training reward. It does not override actions during evaluation and it does not block exploration during training. The policy can still sample non-idle actions, but each such action must be useful enough to pay its fixed cost. Comparing the flat and gated actors at  $\lambda_{fixed} = 0.010$  tests whether the gate still helps once a simple sparse objective already exists.

### 4.4 Adaptive Intervention Budget

**Implementation detail.** The adaptive intervention budget turns sparse topology control into a local constraint. For each agent, the cost is  $c_i(s_t, a_{i,t}) = \mathbf{1}[a_{i,t} \neq 0]w_i(s_t)$ , and the desired constraint is  $E_{\pi_i}[c_i(s_t, a_{i,t})] \leq d$ .

The implemented rollout reward removes the constant term that does not affect the PPO action gradient:

$$r'_{i,t} = r_{i,t} - \lambda_i c_i(s_t, a_{i,t}).$$

After each rollout, the local multiplier is updated by

$$\lambda_i \leftarrow \text{clip}\left(\lambda_i + \eta(\hat{C}_i - d), 0, \lambda_{\max}\right), \quad \hat{C}_i = \frac{1}{T} \sum_{t=1}^T c_i(s_t, a_{i,t}).$$

If the observed cost  $\hat{C}_i$  is above the budget  $d$ , future interventions become more expensive; if it is below the budget, the penalty relaxes. The default settings are `intervention_budget_lr = 0.02`, `intervention_budget_init_lambda = 0.0`, and `intervention_budget_max_lambda = 10.0`.

Two cost modes are used:

- **Non-idle cost:**  $w_i(s_t) = 1$ , so  $c_i(s_t, a_{i,t}) = \mathbf{1}[a_{i,t} \neq 0]$ . This is an adaptive version of a plain sparse-action penalty.
- **Local-safe cost:**  $w_i^{local}(s_t) = \sigma(k(\rho_{safe} - \rho_{i,t}^{local}))$ . This is the decentralized version: safe local states make interventions expensive, while stressed local states make them cheap.

In the AIB setting,  $\rho_{safe} = 0.90$  is not a hard action override. It is the center of a smooth sigmoid safety weight.

**AIB experiment grid.** The main local-budget ablation contains five conditions:

- flat actor, local-safe cost,  $d = 0.20$ ;
- flat actor, stricter local-safe budget,  $d = 0.10$ ;
- flat actor, looser local-safe budget,  $d = 0.35$ ;
- gated actor, hierarchical-greedy evaluation, separated entropy, local-safe cost,  $d = 0.20$ ;
- flat actor with adaptive plain non-idle cost,  $d = 0.20$ .

Comparing the local-safe and plain non-idle costs at  $d = 0.20$  isolates the value of local rho weighting.

## 4.5 Behavioral Cloning from Heuristics

**Implementation detail.** The evaluation-time rho heuristics (Section 4.1) produce highly sparse, safe behavior, but they require a manual override. To move this behavior into the policy weights, a teacher-student distillation approach is tested. The teacher is a MAPPO model running the local rho heuristic, and the student is trained via Behavioral Cloning (BC) to match the teacher’s executed actions. The student is then evaluated without the heuristic override, to see whether it learned to copy the sparsity on its own. Two balancing settings are tested to handle the strong class imbalance caused by the teacher’s high action-0 rate.

**Interpretation.** As reported in Appendix A.1, the student policies reach roughly 91–92% survival and 94–96% sparsity. This shows that sparse behavior can be partly distilled into the actor weights. Still, the resulting student is weaker than the AIB methods and gives up part of the teacher’s survival (97.1%). The distillation route is therefore less reliable than learning the sparse objective directly through the Adaptive Intervention Budget.

## 4.6 Results

Figure 4.1 summarizes survival and the mean action-0 fraction across agents. Exact values and per-agent action-0 fractions are reported in Appendix A.1.

Table 4.1: Sparse-control mechanisms and where they affect the policy.

| Method             | Training effect                             | Evaluation effect                       | Main risk                                                               |
|--------------------|---------------------------------------------|-----------------------------------------|-------------------------------------------------------------------------|
| Baseline           | Samples from the flat policy                | Greedy argmax by default                | Standard entropy-controlled exploration                                 |
| Rho heuristic      | No training change                          | Hard no-op override in safe states      | Evaluation is manually changed                                          |
| Intervention gate  | Changes action factorization and sampling   | Final-action MAP or hierarchical greedy | Gate collapse can starve non-idle exploration                           |
| Fixed penalty      | Fixed reward penalty on non-idle actions    | No action override                      | Penalty coefficient must be tuned                                       |
| AIB non-idle       | Adaptive reward penalty on non-idle actions | No action override                      | Multiplier can grow if target is too strict                             |
| AIB local-safe     | Adaptive state-dependent reward penalty     | No action override                      | Depends on the local rho                                                |
| Behavioral cloning | Distills heuristic into actor weights       | No action override                      | safety-weight design<br>Struggles with extreme action-0 class imbalance |

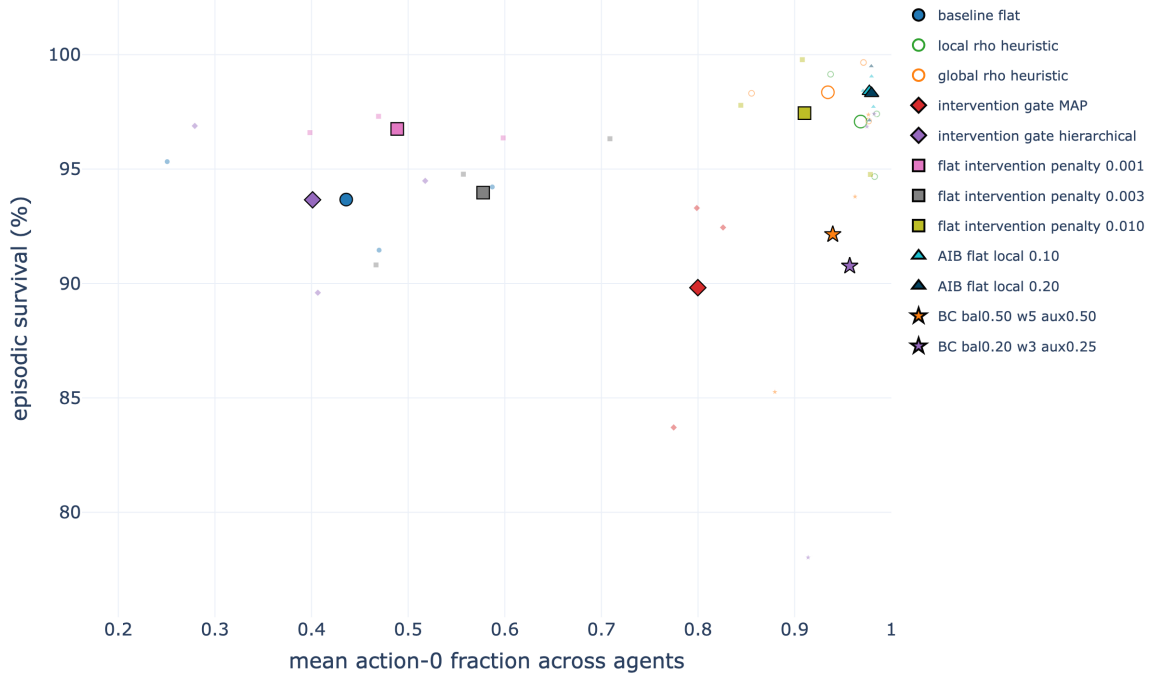


Figure 4.1: Joint action-0/survival comparison of the selected sparse-control mechanisms on **bus14**. Large markers show condition means and faint markers the individual seeds. The plot uses checkpoint-aligned action-0 rates and full-test survival, so it shows the chapter’s two objectives at once.

**Interpretation.** The current reading is that the gate is not the strongest contribution by itself. Sparse reward shaping already improves action-0 usage, the adaptive budget makes the sparse penalty self-tuning, and the local-safe cost makes the penalty more physically meaningful. In particular, the flat actor with  $\lambda_{fixed} = 0.010$  is a strong fixed-penalty baseline, while the flat local-safe AIB policy with  $d = 0.20$  is the best-justified learned sparse-control candidate. It keeps the original decentralized actor, does not rely on an evaluation-time override, learns the intervention penalty adaptively, and uses a local state-dependent safety weight. Figure 4.1 also makes the main failure mode visible: high action-0 usage is not enough when the gate collapses and survival falls.

## 4.7 Preliminary WCCI Stress Test

As an early scale test, representative mechanisms were retrained from scratch with an MLP actor on the maintenance-enabled WCCI grid. No **bus14** weights were transferred. All runs use the same reduced action space, three seeds, 60M training steps, and learning-rate schedule. The reported values are the final deterministic test evaluation, averaged over seeds and ten held-out episodes per seed. They are not full-test estimates.

Table 4.2: Preliminary WCCI stress test with scheduled maintenance at the final training-time test evaluation.

| Method                                        | Survival (%)      | Action-0 fraction |
|-----------------------------------------------|-------------------|-------------------|
| Matched MLP baseline                          | $23.75 \pm 7.70$  | 0.168             |
| Global rho heuristic, $\rho_{safe} = 0.90$    | $28.35 \pm 11.06$ | 0.974             |
| Local rho heuristic, $\rho_{safe} = 0.90$     | $9.46 \pm 3.54$   | 0.995             |
| Intervention gate, final-action MAP           | $25.92 \pm 10.69$ | 0.400             |
| Flat fixed penalty, $\lambda_{fixed} = 0.010$ | $27.00 \pm 12.13$ | 0.489             |
| Flat local-safe AIB, $d = 0.20$               | $21.65 \pm 2.21$  | 0.927             |

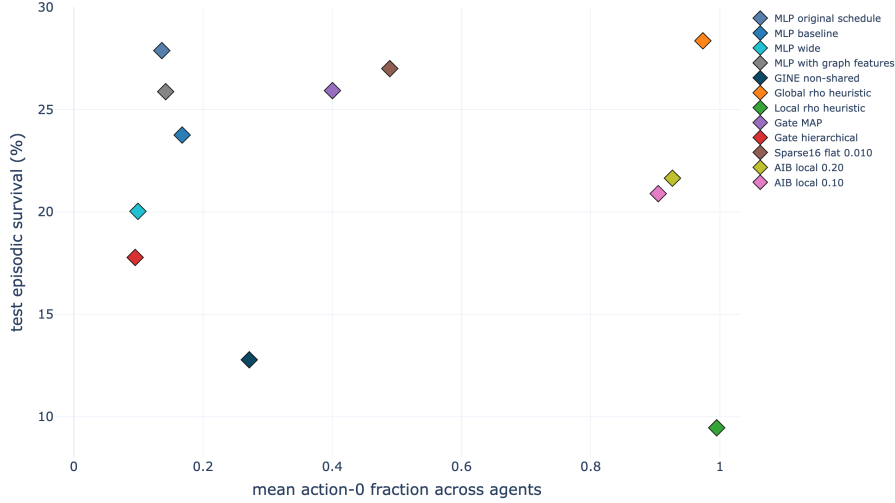


Figure 4.2: Action-0/survival trade-off for the early maintenance-enabled WCCI runs. Each point is a seed-aggregated run variant. The upper-right region is still the target, but no method combines high sparsity with strong survival on this larger grid.

The mechanisms change intervention frequency, but none reliably improves survival over the weak MLP baseline: the differences are small compared with the seed variation. The local rho heuristic and AIB are highly sparse but remain weak controllers, which suggests that sparsity transfers more easily than control quality. This failed pilot changes grid size, action space, and maintenance at once, so it cannot identify the cause. It still shows the limit of continuing with a fixed-width MLP actor on the larger grid. The next two chapters therefore introduce a size-independent graph representation and test its design choices. Chapter 8 later returns to WCCI and separates the factors that are mixed together here, before studying cross-grid transfer.

### Chapter Summary

**Objective:** Keep survival while reducing unnecessary topology interventions.

#### Findings:

- **Evaluation heuristics** reach 97–98% survival with action 0 in 94–97% of decisions, but sparsity is imposed after the policy acts.
- **A fixed penalty** already gives a strong learned controller:  $\lambda_{fixed} = 0.010$  reaches 97.44% survival and a 0.905 action-0 fraction without an evaluation override.
- **Local-safe AIB** gives the best learned trade-off: budgets  $d = 0.10$  and  $d = 0.20$  reach about 98.4% survival and 0.98 action-0 usage. The gated variants are less reliable and can collapse to almost permanent no-op behavior.

**Conclusion:** Sparse behavior can be learned on `bus14`, but the same MLP mechanisms remain weak on WCCI. This motivates the graph representation introduced next.

# Chapter 5

## Methods

Chapter 4 made the flat policy more operator-like on `bus14`, but its early WCCI stress test remained weak. Sparse intervention control changes when the policy acts; it does not remove the fixed-width MLP representation that ties the actor to one grid. This chapter therefore introduces the size-independent graph inputs and graph-actor variants used in the remaining experiments. This chapter separates four design choices: feature scaling, the base graph representation, substation-level augmentations, and graph pooling.

### 5.1 Physical Scaling and Voltage-Angle Representation

Two input transformations make observations more comparable across grids: a fixed physical scaling of the continuous features, and a reference-independent representation of voltage angles. Both act before the graph encoder and leave it unchanged.

#### 5.1.1 Physical scaling

Each scaled continuous feature is divided by a fixed reference of the same unit,  $x'_f = x_f/s_f$ . This changes the unit without changing the information and — importantly for the sparse graph features — keeps zero at zero, so a busbar carrying no asset still reads 0 on every grid. Two references carry the physical content: power is divided by the rating of the largest installed generator in the grid,  $s_P = \max_g P_g^{\max}$ , and angles by a fixed 180 degrees. The remaining discrete channels — the two cooldown counters, the overflow counter, and the maintenance timers — are divided by their Grid2Op parameter bounds, listed in Table A.13. Every reference is fixed at construction rather than estimated during training. The dimensionless loading ratio `rho` and all binary, mask, type, and relation indicators are left unchanged.

#### 5.1.2 Angle representation

An absolute voltage angle needs an arbitrary zero, normally fixed by the slack bus, so adding a constant to every bus angle describes the same physical state. The angle difference across a line does not depend on that choice:

$$\Delta\theta_\ell = \theta_\ell^{\text{or}} - \theta_\ell^{\text{ex}}.$$

This quantity is *gauge-invariant*: it does not depend on that choice of reference. Being defined per line, it is also dense, while an absolute angle exists only where a generator or load attaches. Enabling the drop removes the absolute-angle channels. Chapter 7 measures why this matters: the two grids differ by 5 to 7° in absolute angle, largely because they place their slack bus differently, and no fixed scaling can remove a constant offset.

The drop is used only with the disaggregated line-node representation of Section 5.2.3, where each physical line is itself a node and its attachment edges carry no measurements, so the drop is stored on the line node. This gives every local actor the drop for each line it observes, boundary lines included. A

disconnected line is assigned a zero drop, because the simulator may keep stale endpoint angles after an outage. The two schemas whose lines are edges would carry the drop on the edge instead, but no reported run uses that combination.

### 5.1.3 Use across the experiments

The screening experiments of Chapter 6 and the sparse-control experiments of Chapter 4 use raw features and absolute angles. The cross-grid transfer study of Chapter 9 pairs runs with both transformations enabled against runs with neither, on both grids, so the comparison isolates their effect. Running standardization is also implemented but no reported policy run uses it; Section 7.5.1 analyses it offline and finds that it increases the mismatch between grids.

## 5.2 Graph Representations

The graph observation can use one of three base representations, named here after how much equipment receives its own node. The *busbar* representation aggregates equipment values onto busbar nodes. The *disaggregated* representation gives each generator and load its own node while keeping physical lines as busbar-to-busbar edges. The *disaggregated line-node* representation goes one step further and turns each transmission line into a node as well.

Table 5.1 states what separates them. The node schema also determines how a boundary transmission line is represented in a local actor graph. As a result, the three representations do not carry the same information: they expose different amounts of context about the substation at the far end of a shared line. This difference is spelled out in Section 5.2.4.

Table 5.1: The three graph schemas. Each row follows from how much equipment receives its own node. In the node counts,  $S$  is the number of substations,  $B$  the busbars per substation,  $N_g$  and  $N_d$  the included generators and loads, and  $L$  the included transmission lines.

| Aspect                                 | busbar                                                        | disaggregated                                                       | disaggregated line-node                                           |
|----------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------|
| <b>gnn_graph_type</b>                  | <b>bus</b>                                                    | <b>heterogeneous</b>                                                | <b>heterogeneous_line</b>                                         |
| Node types                             | Busbar                                                        | Busbar, generator, load                                             | Busbar, generator, load, transmission line                        |
| Generator and load state               | Summed and averaged onto the busbar                           | One node per asset                                                  | One node per asset                                                |
| Physical line                          | Direct busbar-to-busbar edge                                  | Typed direct busbar-to-busbar edge                                  | Typed node between its endpoint busbars                           |
| Line state                             | Continuous edge attributes                                    | Continuous edge attributes                                          | Features of the line node                                         |
| Minimum useful depth                   | One layer for adjacent busbar exchange                        | One layer for adjacent busbars; two for asset-to-remote-busbar flow | Two layers for adjacent busbar exchange through a line node       |
| Hidden line state                      | None: the edge modifies a message                             | None: the edge modifies a message                                   | Yes: each line obtains and updates an embedding                   |
| Far-end context in a local actor graph | Live neighboring busbar, including aggregated power and angle | Live neighboring busbar, but no neighboring generator or load nodes | None; the shared line node replaces the need for a far-end busbar |
| Node count                             | $SB$                                                          | $SB + N_g + N_d$                                                    | $SB + N_g + N_d + L$                                              |

### 5.2.1 Busbar Representation

The busbar schema is the most compact of the three. Busbars are the only nodes, active transmission lines form the graph edges, and generator and load quantities are aggregated into their currently assigned busbars.

#### Nodes and features

With  $S$  substations and at most  $B$  busbars each, one node is created per substation-busbar pair, written  $i(s, b)$ , including busbars that currently hold no asset.

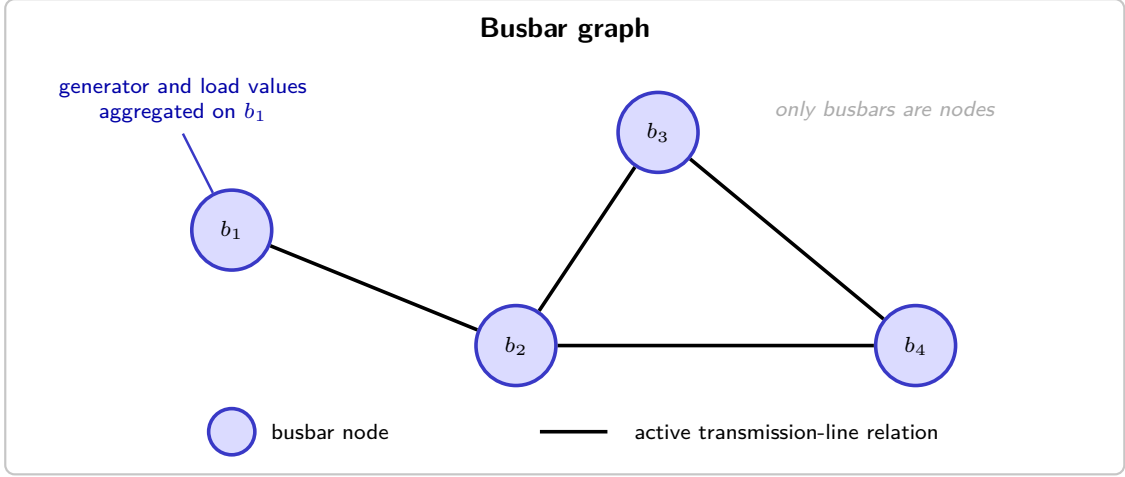


Figure 5.1: Busbar graph representation. Blue nodes are busbars and black connections are active transmission-line relations. Generator and load quantities are aggregated into the features of their associated busbars.

Table 5.2 lists the six node features. Each generator and load contributes to the busbar it is connected to: active powers are summed when several assets share a busbar, and their angles are averaged. Disconnected assets contribute nowhere. The substation cooldown is repeated on every busbar node of that substation.

Table 5.2: Features attached to each busbar node.

| Feature                               | Definition                                                                       |
|---------------------------------------|----------------------------------------------------------------------------------|
| <code>gen_p</code>                    | Sum of active generator power connected to the busbar.                           |
| <code>gen_theta</code>                | Mean voltage angle of generators connected to the busbar.                        |
| <code>load_p</code>                   | Sum of active load power connected to the busbar.                                |
| <code>load_theta</code>               | Mean voltage angle of loads connected to the busbar.                             |
| <code>time_before_cooldown_sub</code> | Remaining substation cooldown.                                                   |
| <code>domain_mask</code>              | One for a substation controlled by the agent and zero for a contextual neighbor. |

### Relations and edge features

The only physical relation in the busbar schema is the busbar-to-busbar transmission line. Table 5.3 gives the dynamic attributes copied onto each active line edge; the two maintenance features are present only for environments with scheduled maintenance.

Table 5.3: Features attached to physical-line edges.

| Feature                                | Definition                                              |
|----------------------------------------|---------------------------------------------------------|
| <code>line_status</code>               | Binary connection status of the physical line.          |
| <code>rho</code>                       | Line loading relative to its thermal limit.             |
| <code>timestep_overflow</code>         | Number of consecutive time steps in overload.           |
| <code>time_before_cooldown_line</code> | Remaining cooldown before another line action is legal. |
| <code>time_next_maintenance</code>     | Time until scheduled maintenance, when available.       |
| <code>duration_next_maintenance</code> | Duration of scheduled maintenance, when available.      |

### 5.2.2 Disaggregated Representation

The disaggregated schema keeps busbars as the backbone and adds one node per generator and load, joined to their busbar by typed directed relations. Transmission lines remain busbar-to-busbar edges, and generator and load states are kept separately.

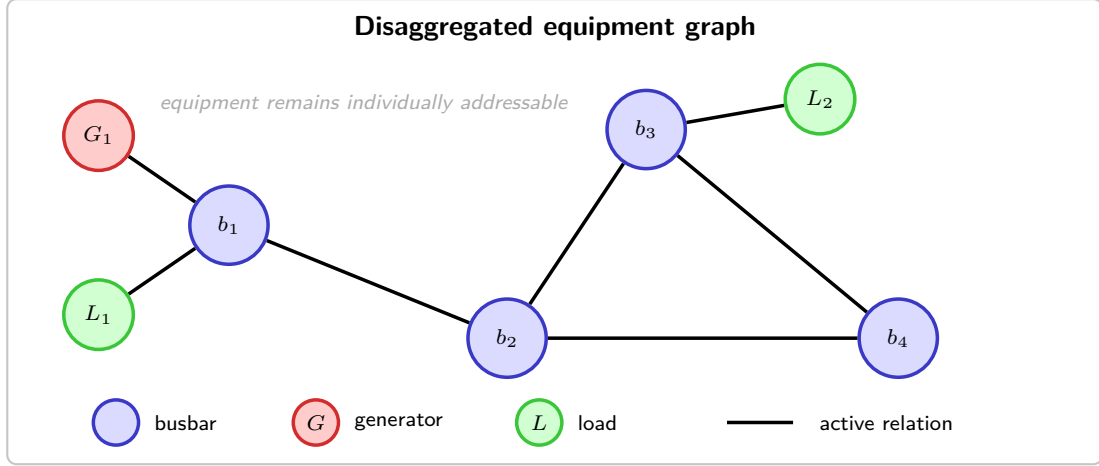


Figure 5.2: Disaggregated representation. Busbars are blue, generators are red, and loads are green. Physical transmission lines remain direct relations between busbar nodes.

### Nodes and features

Busbar nodes are created exactly as in Section 5.2.1. In addition, each generator and load whose substation is present in the graph receives its own node.

Every busbar, generator, and load node is represented by the same ordered seven-dimensional feature vector

$$\mathbf{x}_v = [\text{is\_busbar}, \text{is\_generator}, \text{is\_load}, p, \text{theta}, \text{time\_before\_cooldown\_sub}, \text{domain\_mask}]^T \in \mathbb{R}^7. \quad (5.1)$$

Table 5.4 shows the value placed in every shared column; a zero there is an explicitly zero-filled column, not an omitted feature.

Table 5.4: Values placed in the seven feature columns shared by every node in the disaggregated graph. Disconnected assets keep their node row but have zero power and angle.

| Shared feature column    | Busbar node                | Generator node                      | Load node                            |
|--------------------------|----------------------------|-------------------------------------|--------------------------------------|
| is_busbar                | 1                          | 0                                   | 0                                    |
| is_generator             | 0                          | 1                                   | 0                                    |
| is_load                  | 0                          | 0                                   | 1                                    |
| p                        | 0                          | gen_p if connected; 0 otherwise     | load_p if connected; 0 otherwise     |
| theta                    | 0                          | gen_theta if connected; 0 otherwise | load_theta if connected; 0 otherwise |
| time_before_cooldown_sub | Substation value           | 0                                   | 0                                    |
| domain_mask              | 1 controlled; 0 contextual | 1 controlled; 0 contextual          | 1 controlled; 0 contextual           |

### Relations and edge features

Every candidate directed edge is represented by the same ordered feature vector. Its physical block is

$$\mathbf{p}_{uv} = [\text{line\_status}, \text{rho}, \text{timestep\_overflow}, \text{time\_before\_cooldown\_line}, (\text{time\_next\_maintenance}, \text{duration\_next\_maintenance})_{\text{optional}}]^T. \quad (5.2)$$

The two maintenance columns are appended only when scheduled maintenance is available. The relation block is the same six-column vector for every edge:

$$\mathbf{r}_{uv} = [\text{relation\_physical\_line}, \text{relation\_same\_substation\_busbar}, \text{relation\_generator\_to\_busbar}, \text{relation\_busbar\_to\_generator}, \text{relation\_load\_to\_busbar}, \text{relation\_busbar\_to\_load}]^T. \quad (5.3)$$

The complete common edge vector is therefore

$$\mathbf{e}_{uv} = [\mathbf{p}_{uv}^T \mid \mathbf{r}_{uv}^T]^T \in \begin{cases} \mathbb{R}^{10}, & \text{without maintenance columns,} \\ \mathbb{R}^{12}, & \text{with maintenance columns.} \end{cases} \quad (5.4)$$

An active physical-line edge fills  $\mathbf{p}_{uv}$  with its measured line state. Every active structural or asset-attachment edge instead uses

$$\mathbf{p}_{\text{neutral}} = \mathbf{0}, \quad (5.5)$$

so a relation carrying no flow reports none. In every active edge row, exactly one column of  $\mathbf{r}_{uv}$  is one and the other five are zero.

Table 5.5 shows the values placed in these common blocks for every relation.

Table 5.5: Values placed in the common edge-feature vector shared by every active directed relation in the disaggregated graph. “Other five” refers to the remaining columns of the six-column relation block.

| Directed relation                                 | Active rule                                                     | Shared physical block         | Shared relation block                                            |
|---------------------------------------------------|-----------------------------------------------------------------|-------------------------------|------------------------------------------------------------------|
| $i_o \rightarrow i_e$ and reverse (physical line) | Line online and candidate busbars match the current endpoints   | Measured $\mathbf{p}_{uv}$    | <code>relation_physical_line= 1</code> ; other five = 0          |
| $i(s, b) \rightarrow i(s, b'), b \neq b'$         | Optional; every ordered same-substation busbar pair             | $\mathbf{p}_{\text{neutral}}$ | <code>relation_same_substation_busbar= 1</code> ; other five = 0 |
| $g \rightarrow i(s(g), b)$                        | Generator connected to candidate busbar $b$ ; direction enabled | $\mathbf{p}_{\text{neutral}}$ | <code>relation_generator_to_busbar= 1</code> ; other five = 0    |
| $i(s(g), b) \rightarrow g$                        | Same mask; reverse direction enabled                            | $\mathbf{p}_{\text{neutral}}$ | <code>relation_busbar_to_generator= 1</code> ; other five = 0    |
| $d \rightarrow i(s(d), b)$                        | Load connected to candidate busbar $b$ ; direction enabled      | $\mathbf{p}_{\text{neutral}}$ | <code>relation_load_to_busbar= 1</code> ; other five = 0         |
| $i(s(d), b) \rightarrow d$                        | Same mask; reverse direction enabled                            | $\mathbf{p}_{\text{neutral}}$ | <code>relation_busbar_to_load= 1</code> ; other five = 0         |

Generator and load directions are chosen independently as `bidirectional`, `asset_to_busbar`, or `busbar_to_asset`. The default is `bidirectional`. Physical-line and same-substation relations remain `bidirectional`.

### 5.2.3 Disaggregated Line-Node Representation

The disaggregated line-node schema replaces each direct physical-line edge with a transmission-line node between its endpoint busbars, so busbars, generators, loads, and lines are all nodes.

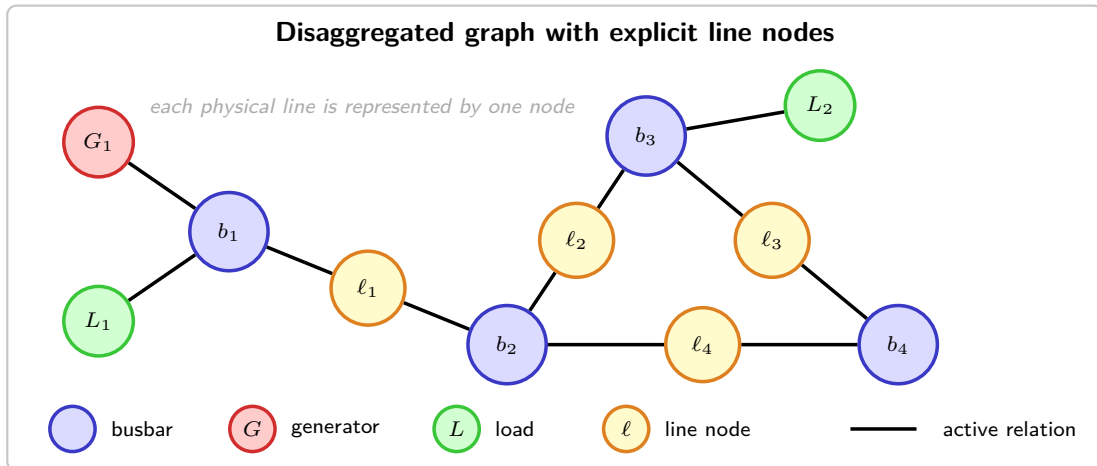


Figure 5.3: Disaggregated line-node representation. Yellow nodes represent physical lines and connect the blue busbars at their endpoints; generators and loads remain separate terminal nodes.

## Nodes and features

Every node type uses the same ordered feature vector:

$$\begin{aligned} \mathbf{x}_v = & [\text{is\_busbar}, \text{is\_generator}, \text{is\_load}, \text{is\_transmission\_line}, \\ & \text{p}, \text{theta}, \text{line\_status}, \text{rho}, \text{timestep\_overflow}, \\ & \text{time\_before\_cooldown\_line}, \\ & (\text{time\_next\_maintenance}, \text{duration\_next\_maintenance})_{\text{optional}}, \\ & \text{time\_before\_cooldown\_sub}, \text{domain\_mask}]^T, \\ \mathbf{x}_v \in & \begin{cases} \mathbb{R}^{12}, & \text{without maintenance columns,} \\ \mathbb{R}^{14}, & \text{with maintenance columns.} \end{cases} \end{aligned} \quad (5.6)$$

Table 5.6 lists every entry; 0 means an explicitly zero-filled column, not an omitted feature. For generators and loads, power and angle are set to zero when the asset is disconnected.

Table 5.6: Values placed in the common node-feature vector of the disaggregated line-node graph, when voltage angles are carried as absolute per-asset values. The two maintenance rows are included only when maintenance observations are enabled. When the angle is instead carried as the drop along each line, the angle row changes owner: the generator and load entries become 0 and the transmission line holds the drop across it, set to 0 while the line is out (Section 5.1.2).

| Shared feature column     | Busbar                        | Generator                                 | Load                                       | Transmission line                                  |
|---------------------------|-------------------------------|-------------------------------------------|--------------------------------------------|----------------------------------------------------|
| is_busbar                 | 1                             | 0                                         | 0                                          | 0                                                  |
| is_generator              | 0                             | 1                                         | 0                                          | 0                                                  |
| is_load                   | 0                             | 0                                         | 1                                          | 0                                                  |
| is_transmission_line      | 0                             | 0                                         | 0                                          | 1                                                  |
| p                         | 0                             | gen_p if connected;<br>0 otherwise        | load_p if connected;<br>0 otherwise        | 0                                                  |
| theta                     | 0                             | gen_theta if<br>connected; 0<br>otherwise | load_theta if<br>connected; 0<br>otherwise | 0                                                  |
| line_status               | 0                             | 0                                         | 0                                          | Observed line value                                |
| rho                       | 0                             | 0                                         | 0                                          | Observed $\rho_\ell$                               |
| timestep_overflow         | 0                             | 0                                         | 0                                          | Observed line value                                |
| time_before_cooldown_line | 0                             | 0                                         | 0                                          | Observed line value                                |
| time_next_maintenance     | 0                             | 0                                         | 0                                          | Observed line value                                |
| duration_next_maintenance | 0                             | 0                                         | 0                                          | Observed line value                                |
| time_before_cooldown_sub  | Substation<br>value           | 0                                         | 0                                          | 0                                                  |
| domain_mask               | 1 controlled;<br>0 contextual | 1 controlled; 0<br>contextual             | 1 controlled; 0<br>contextual              | 1 if either endpoint is<br>controlled; 0 otherwise |

## Relations and edge features

Every candidate directed edge uses the same ordered nine-dimensional feature vector:

$$\begin{aligned} \mathbf{e}_{uv} = & [\text{relation\_same\_substation\_busbar}, \\ & \text{relation\_busbar\_to\_line\_origin}, \text{relation\_line\_origin\_to\_busbar}, \\ & \text{relation\_busbar\_to\_line\_extremity}, \text{relation\_line\_extremity\_to\_busbar}, \\ & \text{relation\_generator\_to\_busbar}, \text{relation\_busbar\_to\_generator}, \\ & \text{relation\_load\_to\_busbar}, \text{relation\_busbar\_to\_load}]^T \in \mathbb{R}^9. \end{aligned} \quad (5.7)$$

These edges contain no continuous line-state attributes because the transmission-line node already contains rho, status, overflow, cooldown, and maintenance. For an active edge, exactly one relation column is one and the other eight are zero. An inactive candidate has all nine entries set to zero and is removed by edge\_mask=0 before message passing. Table 5.7 gives the active rule and the nonzero entries for every relation.

Table 5.7: Nonzero entries in the common 9-column edge-feature vector of the disaggregated line-node graph. All relation columns not named in a row are zero.

| Directed relation                         | Active rule                                              | Relation column set to 1                       |
|-------------------------------------------|----------------------------------------------------------|------------------------------------------------|
| $i(s, b) \rightarrow i(s, b'), b \neq b'$ | Optional; every ordered same-substation pair             | <code>relation_same_substation_busbar</code>   |
| $i_o \rightarrow \ell$                    | Online line, current origin busbar, direction enabled    | <code>relation_busbar_to_line_origin</code>    |
| $\ell \rightarrow i_o$                    | Same mask; reverse direction enabled                     | <code>relation_line_origin_to_busbar</code>    |
| $i_e \rightarrow \ell$                    | Online line, current extremity busbar, direction enabled | <code>relation_busbar_to_line_extremity</code> |
| $\ell \rightarrow i_e$                    | Same mask; reverse direction enabled                     | <code>relation_line_extremity_to_busbar</code> |
| $g \rightarrow i(s(g), b)$                | Generator attached to $b$ ; direction enabled            | <code>relation_generator_to_busbar</code>      |
| $i(s(g), b) \rightarrow g$                | Same mask; reverse direction enabled                     | <code>relation_busbar_to_generator</code>      |
| $d \rightarrow i(s(d), b)$                | Load attached to $b$ ; direction enabled                 | <code>relation_load_to_busbar</code>           |
| $i(s(d), b) \rightarrow d$                | Same mask; reverse direction enabled                     | <code>relation_busbar_to_load</code>           |

### 5.2.4 Local actor information boundary and the global state graph

Each agent  $a$  controls a set of substations  $\mathcal{C}_a$ . A boundary line connects  $\mathcal{C}_a$  to another agent’s domain, and its far-end busbar is the only possible neighboring busbar. The local graph exposes only the boundary information required by its representation.

**Why one hop, and why not more.** The one-hop boundary is a design choice: it keeps each actor regional while the centralized critic still observes the global state. Power flow has no natural one-hop cutoff, but the loading and angle drop of boundary lines give the immediate interaction with the outside without exposing neighboring asset states. One hop is therefore not claimed to be optimal.

#### Visibility and masking

A contextual busbar is visible only while a live boundary line connects it to the controlled region. When that connection is lost its feature row is zeroed, the relations touching it go inactive, and it is left out of the readout — including the denominator when the readout is a mean. It therefore reaches neither message passing nor the graph embedding, and a same-substation relation cannot restore it.

Controlled nodes stay visible when disconnected. In the disaggregated line-node schema a shared line node stays visible too, since it belongs to the agent’s own graph, though its attachment relations are inactive while the line is offline. Relations are active only when the line is online and the current busbar assignments match their endpoints. Neighboring generators and loads are never constructed.

#### Representation-dependent information

The representations do not carry the same information: Table 5.8 states what each one exposes of a neighboring domain.

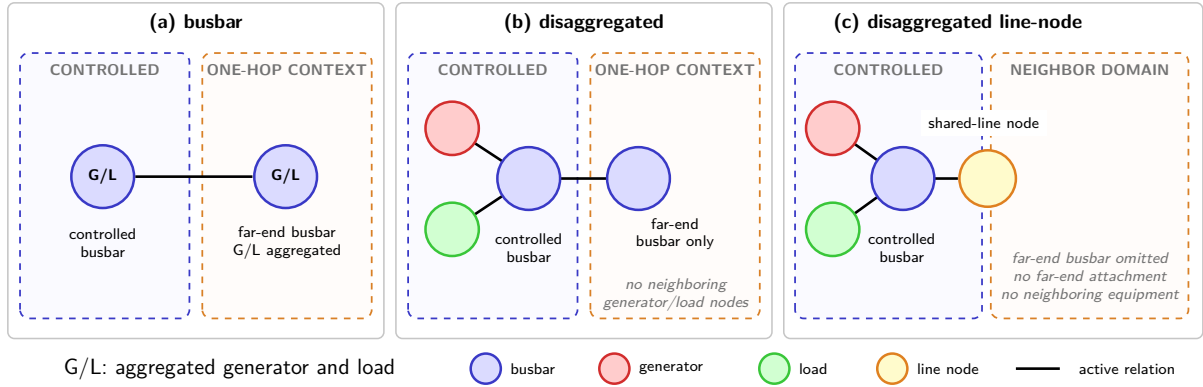


Figure 5.4: One live boundary line, seen under each representation. In every panel the blue dashed region is the agent’s controlled domain and the orange dashed region is what lies beyond it. The busbar graph (a) reaches a far-end busbar carrying the neighbor’s generation and load aggregated onto it. The disaggregated graph (b) reaches the same busbar, but no neighboring generator or load node is ever constructed, so that busbar carries no measurement of its own. The disaggregated line-node graph (c) reaches only the shared line node, and no far-end busbar exists at all.

Table 5.8: Information from another agent’s domain visible to a local actor.

| Representation          | Neighbor information available                                                                           |
|-------------------------|----------------------------------------------------------------------------------------------------------|
| busbar                  | Shared-line edge and attached far-end busbar, including aggregated generator/load power and angles       |
| disaggregated           | Shared-line edge and attached far-end busbar; no neighboring generator/load nodes or private-asset state |
| disaggregated line-node | Shared-line node only; no neighboring busbar, equipment, or far-end attachment                           |

Three consequences follow. In the busbar and disaggregated schemas the far-end busbar has to exist, because an edge needs both endpoints for the boundary line’s features to reach the controlled busbar through message passing. The remaining busbar candidates at that substation stay masked, and if the line disconnects none of them remains visible. In the disaggregated schema that contextual busbar carries only type and role indicators, since power and angle sit on asset nodes and neighboring assets are never constructed — all neighboring measurement therefore arrives through the boundary-line edge. The disaggregated line-node schema drops the contextual busbar entirely and puts the shared measurements on the line node. That node stays visible with `line_status=0` when the line disconnects, while its attachment edges go inactive.

These boundaries limit the actors only. The centralized critic reads the normalized flat state of the whole grid during training, while execution uses each actor’s local graph, as required by MAPPO’s centralized-training, decentralized-execution setup [7].

### 5.2.5 Graph actor and centralized critic

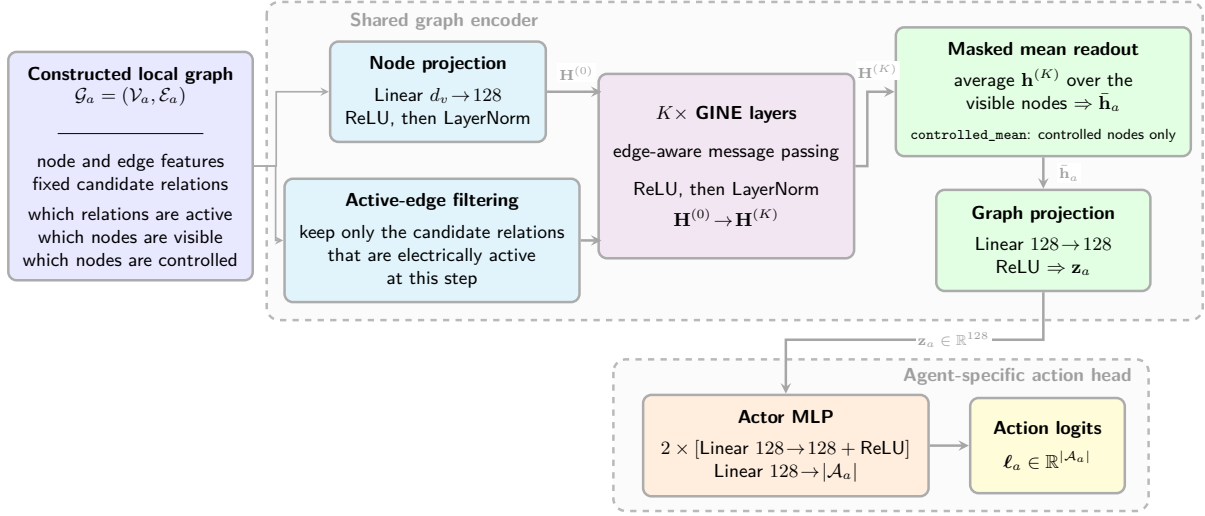


Figure 5.5: End-to-end graph-actor architecture. Candidate edges masked by the current topology are removed before message passing. The graph encoder and its output projection are shared across agents, while the final MLP has one output per action available to agent  $a$  and is therefore agent-specific.

In the reported configuration, the input node projection and the two message-passing layers have 128 features. LayerNorm is applied after the input projection and after each message-passing layer. The graph readout produces a 128-dimensional embedding, followed by an agent-specific action head with two 128-unit hidden layers. The graph encoder and output projection are shared across agents, while the action heads are separate because their action-set sizes may differ.

This is the default configuration. The candidate-scoring alternative of Section 5.5 replaces this head.

The centralized critic is a separate MLP over the normalized flat state. It does not reuse the actor graph embedding and is used only during centralized training. Its flat-state normalization is independent of the graph-input normalization. The exact GINE update, graph readout, output projection, and action-head equations are given in Appendix B.

**Purpose:** Define how the electrical state becomes the input of each actor and how that input is encoded.

- **Input processing:** Continuous physical quantities can be divided by fixed physical scales and optionally standardized from training statistics. The loading ratio `rho`, binary features, and masks remain unchanged. The graph-screening experiments use raw features; the cross-grid transfer study uses deterministic physical scaling.
- **Dynamic topology:** Nodes and candidate edges remain fixed, while time-dependent edge and node masks keep only the relations and contextual nodes that are electrically connected to the controlled region. An inactive contextual candidate has a zero feature row and participates in neither message passing nor readout.
- **Graph representations:** The busbar schema aggregates equipment on busbars; disaggregated gives generators and loads their own nodes; and disaggregated line-node also represents each transmission line as a node. This changes both where the features sit and the number of message-passing steps between physical objects.
- **Actor and critic:** Each actor receives its controlled region and only the schema-specific boundary context of Table 5.8. The reference architecture uses a two-layer, 128-feature GINE encoder shared across agents, followed by an agent-specific action head. The centralized critic remains a separate MLP over the normalized full-grid flat state.

**Takeaway:** The graph variants change both how state is represented and how much neighboring information is available. The MAPPO training structure and agent split remain unchanged, but the three representations must not be described as carrying the same information.

## 5.3 Substation Representation: Direct Edges and Summary Nodes

Two optional factors change how information is exchanged inside a substation. The factor  $e$  adds direct same-substation busbar edges. The factor  $n$  adds one learned substation-summary node per included substation. Both augmentations are part of the constructed graph; their nodes and edges therefore enter the same node projection, edge filtering, and message-passing pipeline as the base graph.

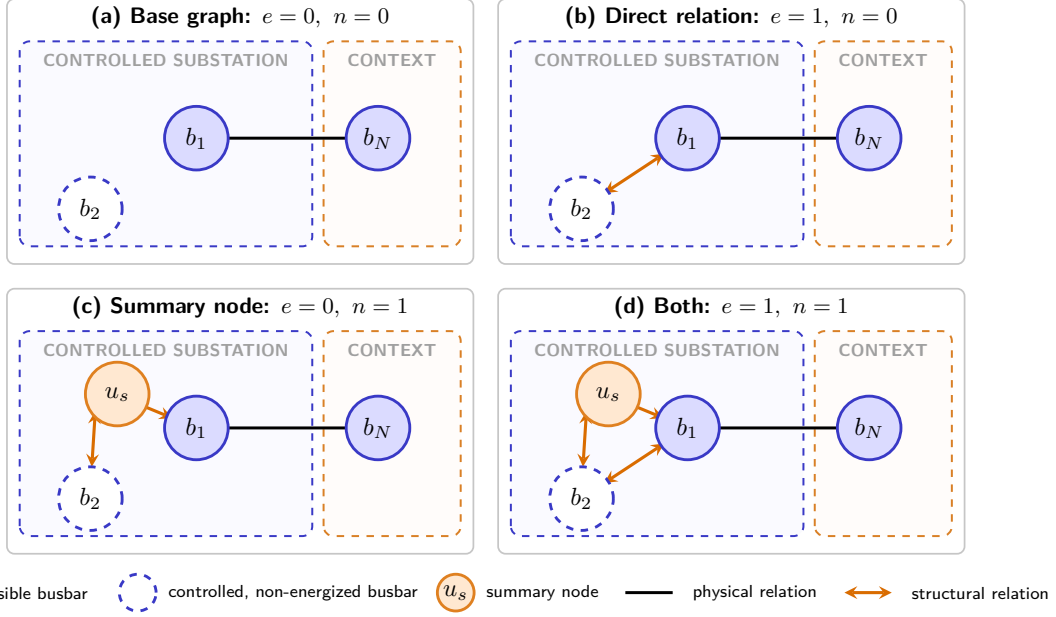


Figure 5.6: The four combinations of the independent substation factors  $e$  and  $n$ . The hollow node  $b_2$  is controlled but non-energized. Factor  $e$  adds a bidirectional relation between the controlled busbars, while factor  $n$  connects a summary node  $u_s$  to both. Neither augmentation extends to the contextual busbar  $b_N$ , which retains only its physical relation.

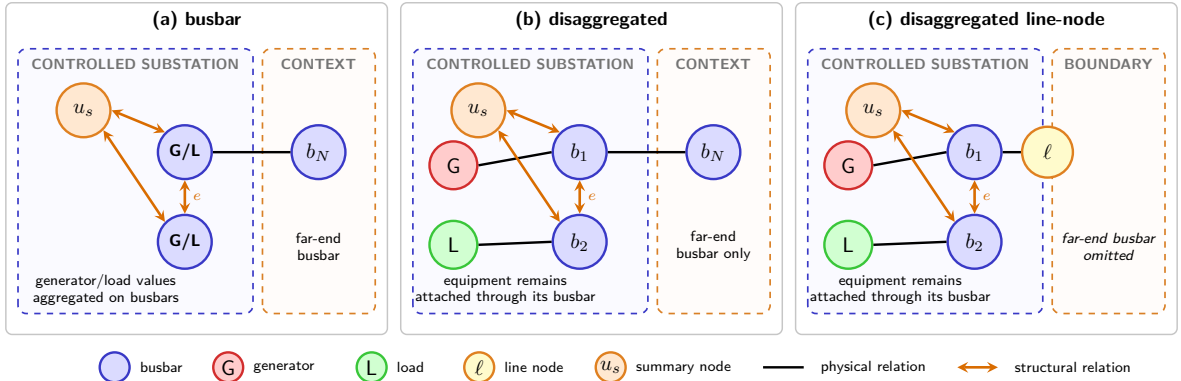


Figure 5.7: Placement of the two augmentations when  $e = n = 1$ . In every graph representation, the direct relation and summary node attach only to controlled busbars. Equipment and line nodes retain their ordinary busbar relations, and no augmentation is added in the contextual domain.

### 5.3.1 Direct same-substation edges (e)

In addition to physical-line relations, the augmentation inserts  $i(s, b) \rightarrow i(s, b')$  for every ordered pair  $b \neq b'$  of busbars in the same *controlled* substation. The edges are therefore bidirectional between each pair of busbars, remain active throughout the episode, and do not represent an electrical connection. As a result, a non-energized controlled busbar remains connected to the other busbar of its substation through this structural relation, without being considered electrically energized. Figure 5.7 shows this relation across the three graph representations.

Table 5.9 gives the exact feature values of the added edges. Every physical attribute is zero: a relation that carries no flow reports none.

Table 5.9: Node and edge features introduced by the direct same-substation augmentation. “None” means that the augmentation neither adds nodes nor changes the features of existing nodes.

| Graph schema            | Added node features | Feature values on each added directed edge                                                                                                                                                       |
|-------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Busbar                  | None                | Every entry is 0: this schema has no relation columns. An active physical line always carries <code>line_status</code> = 1, so an all-zero row is itself the signature of a structural relation. |
| Disaggregated           | None                | <code>relation_same_substation_busbar</code> = 1; every other entry is 0.                                                                                                                        |
| Disaggregated line-node | None                | <code>relation_same_substation_busbar</code> = 1; every other entry is 0.                                                                                                                        |

**Why controlled substations only.** The relation represents the actor’s ability to move elements between busbars, not an electrical connection. Adding it inside a neighboring substation would encode another agent’s action space and create a nonphysical message path.

### 5.3.2 Substation summary nodes (n)

Augmentation  $n$  adds one learned summary node  $u_s$  per controlled substation and connects it in both directions to every visible busbar of that substation, adding one node and  $2B$  directed edges each. Two busbars of the same substation then communicate in two steps,  $i(s, b) \rightarrow u_s \rightarrow i(s, b')$ . This is the same role inside the substation as the direct relation of Section 5.3.1, one message-passing step slower, and limited to the controlled domain for the same reason. Only busbars attach to  $u_s$ , and only visible ones, so the summary carries no contextual state of its own. Figure 5.7 shows both augmentations across the three representations.

The model initializes  $u_s$  from a *structural descriptor* rather than from the substation’s identity. Let  $\phi_s \in \mathbb{R}^4$  hold its numbers of busbars, connected lines, attached generators, and attached loads. The last three are divided by a fixed 4, the same on every grid, so a count means the same thing whatever network it comes from. The busbar count is divided by the grid’s busbars per substation, and is therefore 1 throughout both grids studied here, so the descriptor varies only through its other three entries:

$$u_s^{(0)} = \mathbf{W}_{\text{sub}} \phi_s + \mathbf{b}_{\text{sub}}, \quad \mathbf{W}_{\text{sub}} \in \mathbb{R}^{d_h \times 4}, \quad \mathbf{b}_{\text{sub}} \in \mathbb{R}^{d_h}. \quad (5.8)$$

The summary node therefore holds no grid measurement of its own; messages from its busbars are what give it an observation-dependent state.

An embedding table indexed by substation identity would have shape  $S \times d_h$  and could not transfer between grids with different  $S$ . Equation 5.8 applies one shared projection to four fixed-scale counts instead, so the parameter shapes do not depend on grid size while substations of different structure still start from different states. This keeps the augmentation compatible with the transfer study of Chapter 9 if necessary.

The augmentation behaves the same way in all three schemas: every  $i \leftrightarrow u_s$  edge carries an all-zero attribute vector, since no schema defines a summary relation column. Each  $u_s$  counts as one node in a mean readout. The two augmentations may be enabled together, and either may be combined with the virtual-node readout of Section 5.4.2.

## 5.4 Pooling and Graph Readout

The final graph embedding is obtained either by pooling node embeddings directly or by reading out a learned virtual node. The output projection and action head are the same in both cases.

### 5.4.1 Mean and maximum pooling

A final readout is two independent choices: *which* nodes are aggregated, and *how*. Write  $w_v \in \{0, 1\}$  for the resulting weight of node  $v$ . It is the visibility mask  $m_v$  of Section 5.2.4, optionally intersected with the static controlled-node mask  $c_v$ , and optionally with an energized mask  $e_v$  that requires the node to touch an active electrical relation. Self and same-substation relations do not count as electrical. The reducer is an average or a per-channel maximum,

$$\bar{\mathbf{h}}_G^{\text{mean}} = \frac{\sum_v w_v \mathbf{h}_v^{(K)}}{\max(1, \sum_v w_v)}, \quad \left(\bar{\mathbf{h}}_G^{\text{max}}\right)_c = \max_{v: w_v=1} \left(\mathbf{h}_v^{(K)}\right)_c. \quad (5.9)$$

An unnormalized sum is not a candidate. Its size grows with the number of pooled nodes, which differs across agents, representations, and grids, so the scale of the embedding would change with the graph it came from, and change again under the transfer of Chapter 9. A single readout is taken over all node types at once; there is no separate aggregation per type. A node with  $w_v = 0$  contributes neither to a numerator nor to a denominator, so masking enforces the construction boundary rather than averaging in zero placeholders. The two restrictions combine freely, which gives the eight names of Table 5.10.

Table 5.10: The eight final readouts, named [controlled\_] [energized\_] {mean, max}. Rows choose the pooled population, columns the reducer. The virtual-node readout of Section 5.4.2 replaces final pooling entirely.

| Pooled nodes ( $w_v$ )                  | mean                             | max                             |
|-----------------------------------------|----------------------------------|---------------------------------|
| All visible, $m_v$                      | <b>mean</b>                      | <b>max</b>                      |
| Controlled, $m_v c_v$                   | <b>controlled_mean</b>           | <b>controlled_max</b>           |
| Energized, $m_v e_v$                    | <b>energized_mean</b>            | <b>energized_max</b>            |
| Controlled and energized, $m_v c_v e_v$ | <b>controlled_energized_mean</b> | <b>controlled_energized_max</b> |

The restrictions act on the readout only, never on message passing, so a contextual node excluded from  $w_v$  still reaches the controlled nodes through active relations. They are therefore pooling ablations, not information-boundary fixes.

**What each axis trades.** The two axes answer different questions, so their trade-offs are separate (Table 5.11). Until the ablation of Section 6.4 is done, the safest default is **controlled\_mean**: stable in population, aligned with the action domain, and keeping the empty busbars that actions may target. **controlled\_max** is the risk-sensitive alternative when a decision is expected to depend on one critical node. Without one-hop context the controlled and visible sets are the same, and in the disaggregated line-node schema the context option changes little because boundary line nodes are already present.

Table 5.11: Trade-offs along each axis of Table 5.10.

| Choice                   | Main advantage                                                                                          | Main limitation                                                                               |
|--------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <i>Reducer</i>           |                                                                                                         |                                                                                               |
| <b>mean</b>              | Smooth regional summary; gradients reach every pooled node.                                             | A rare but critical activation is diluted.                                                    |
| <b>max</b>               | Keeps the strongest activation and ignores how many nodes there are.                                    | Sparse gradients, and channel maxima may come from unrelated nodes.                           |
| <i>Pooled population</i> |                                                                                                         |                                                                                               |
| All visible              | Boundary conditions enter the readout directly.                                                         | Context can dilute a mean or dominate a max, and the population moves with boundary topology. |
| Controlled               | Aligned with the action domain and fixed for an episode; context still arrives through message passing. | Relies on message passing to carry every useful boundary signal.                              |
| Energized                | Focuses on the network currently carrying power.                                                        | Drops available action targets, and the population changes after each switching action.       |

### 5.4.2 Virtual-node graph readout

A virtual-node readout replaces final pooling with one learned graph-summary node  $v_*$ , and works with any of the three schemas. The graph gains that node plus one directed edge into it from every member of a summary set  $\mathcal{V}_{\text{sum}}$ : the controlled substation nodes when the augmentation of Section 5.3.2 is enabled, and the visible controlled busbars otherwise. Every added edge points *toward*  $v_*$  in all reported runs, so it aggregates node states and never influences them. It has no direct edge to contextual busbars, generators, loads, or line nodes; those reach it only through ordinary message-passing paths into the controlled region.

Its initial state  $\mathbf{h}_*^{(0)}$  is learned, and the same message-passing layers that update the physical nodes update it. After  $K$  layers,  $\mathbf{h}_*^{(K)}$  enters the usual output projection in place of the pooled vector, so the embedding dimension is unchanged. The added edges are structural. Without substation nodes the augmentation adds one edge per visible controlled busbar; with them, one edge per controlled substation.

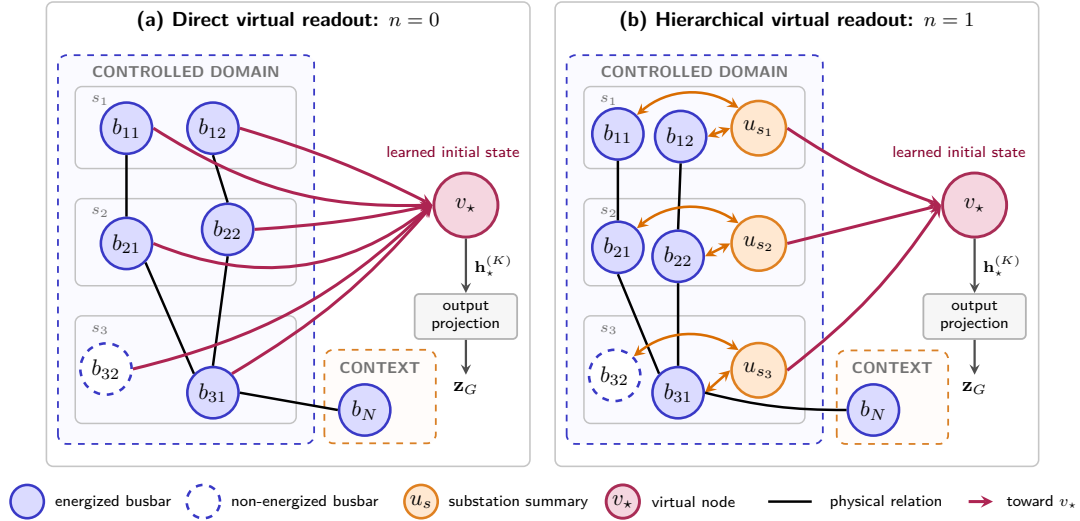


Figure 5.8: Virtual-node readout in the reported one-way setting. Without substation nodes, every controlled busbar—including non-energized  $b_{32}$ —sends directly to  $v_*$ . With augmentation  $n$ , the same busbars communicate through  $u_s$ , and only the substation nodes send to  $v_*$ . The final state  $\mathbf{h}_*^{(K)}$  replaces final pooling.

#### Chapter Summary (2/3): Augmentations and Graph Readout

**Purpose:** Separate changes to information exchange inside the graph from changes to the final graph-level representation.

- **Direct substation edges ( $e$ ):** Add structural relations between busbars of the same substation, without adding nodes or representing a physical transmission line.
- **Substation nodes ( $n$ ):** Add one learned node per *controlled* substation. It has no raw electrical measurement, is initialized from a structural descriptor, and is updated from its busbars.
- **Final pooling:** A readout is a pooled population — all visible nodes, or the controlled and/or energized subsets — reduced by a mean or a per-channel maximum. Limiting the population changes the readout only: excluded context still reaches the controlled nodes through message passing.
- **Virtual readout ( $v$ ):** Replace final pooling with one learned grid-summary node. Messages travel only toward this node, from the busbars or from the added substation nodes, and its final state becomes the graph embedding.

**Takeaway:** Factors  $e$  and  $n$  change message passing, while pooling and factor  $v$  determine how the encoded local graph is reduced to one vector for action selection.

## 5.5 Candidate-Action Scoring from the Disaggregated Line-Node Graph

The default action head produces one output column per discrete action. That column is tied to an action index and knows nothing about the physical objects the action changes. This section describes an alternative head that scores each action from the graph nodes it touches. It is available only for the disaggregated line-node schema of Section 5.2.3; the default remains the action-index head of Appendix B.4. The PPO objective, the categorical policy, sampling, and deterministic evaluation are unchanged.

### 5.5.1 From action indices to candidate scoring

Throughout this section,  $\mathbf{z}_a$  is the graph embedding of agent  $a$ : the node states of its local graph pooled as in Section 5.4 and passed through the output projection (Equation B.7). It is one 128-dimensional vector per agent per step, and it is the only thing the actor has ever seen of the graph.

With the default head, agent  $a$  produces its logit vector in one step from  $\mathbf{z}_a$ :

$$\ell_a = \text{MLP}_a(\mathbf{z}_a), \quad \ell_a \in \mathbb{R}^{|\mathcal{A}_a|}. \quad (5.10)$$

The last layer has one row per action index, so what makes an action good must be learned separately for every index, and the head grows with  $|\mathcal{A}_a|$ .

Candidate scoring replaces that by one scalar-valued network shared across all non-idle actions. The idle action may use the same network or a dedicated one:

$$\ell_{a,j} = \text{score} \left( \underbrace{\mathbf{z}_a}_{\text{whole graph}}, \underbrace{\mathbf{c}_j}_{\text{nodes action } j \text{ touches}}, \underbrace{\phi_j}_{\text{optional action descriptor}} \right), \quad j \geq 1. \quad (5.11)$$

The same parameters score all non-idle actions, so the head no longer grows with  $|\mathcal{A}_a|$ . Two actions are ranked by the difference between their touched-node contexts rather than by two independent output rows. This is what makes the line nodes usable: an action next to a loaded line is scored from that line’s own embedding.

Three choices combine independently: how the touched nodes are pooled into  $\mathbf{c}_j$ , whether the static descriptor  $\phi_j$  is appended, and whether the idle action uses the shared scorer or its own. The two uniform pooling rules therefore give eight variants, and learned attention adds a third pooling family. Figure 5.9 shows how the three enter one candidate score; the following subsections define them.

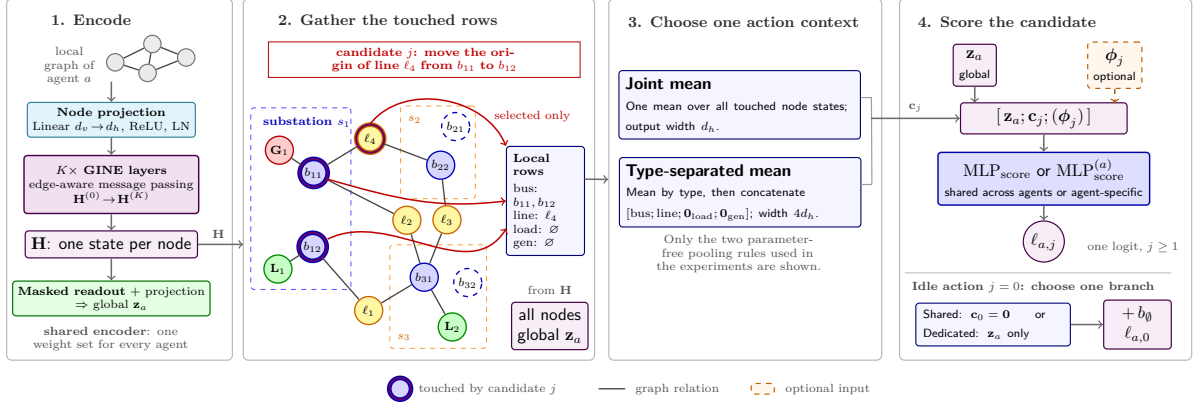


Figure 5.9: Candidate-action scoring for an endpoint-move candidate. Stage 1 is the shared GNN encoder, given in full in Figure 5.5: it maps the local graph to node states  $\mathbf{H}$ . The same stack gives the global embedding  $\mathbf{z}_a$  by masked readout and projection, so the two inputs to the scorer come from one set of weights rather than two. In stage 2 the violet outlines mark the only rows gathered from  $\mathbf{H}$  for the local context: both busbars of the modified substation and the touched line node. One pooling construction then produces  $\mathbf{c}_j$ . The static descriptor  $\phi_j$  may be appended before the shared non-idle scorer. The idle action uses either zero local context in that scorer or a dedicated scorer.

### 5.5.2 What each action touches

The first ingredient is one lookup table per agent, with one row per action, answering a single question: which nodes of the local graph does this action physically touch? It is built once, when the environment is created, and never recomputed during training.

A Grid2Op topology action states which entries of the topology vector it changes, and each entry belongs to a known object: a line endpoint, a generator, or a load. Reading those entries gives the objects the action changes, and the graph builder’s row maps turn each object into the row of the graph that holds its embedding. Table 5.12 follows one action through this.

Table 5.12: One action, from Grid2Op to node rows: moving the origin end of line 4 onto the other busbar of substation 6. Row indices are illustrative.

| Step                       | Result for this action                                                             |
|----------------------------|------------------------------------------------------------------------------------|
| Objects the action changes | Substation 6, line 4                                                               |
| Their rows in the graph    | Both busbars of substation 6, here 12 and 13, and the line node of line 4, here 22 |
| Stored for scoring         | Busbar [12, 13], line [22], generator and load empty                               |

Two decoding rules are worth stating. A touched substation contributes *all* of its busbar rows, not only the busbar in use, because the action changes which of them the equipment sits on. A boundary line contributes its line-node row and the controlled endpoint rows that are present in the local graph. Scoring does not add the far-end busbar or any neighboring private asset. The rows are stored as fixed-size arrays with a mask, padded to the widest action, so that a single gather serves the whole action space.

Write  $\mathcal{N}_{\text{bus}}(j)$ ,  $\mathcal{N}_{\text{line}}(j)$ ,  $\mathcal{N}_{\text{load}}(j)$ , and  $\mathcal{N}_{\text{gen}}(j)$  for the four sets of node rows that action  $j$  touches. Because a topology change moves line endpoints, the line node must be present whenever the line touches the agent’s controlled domain. The leakage-free construction ensures this without building the non-controlled endpoint.

Each action also carries a static feature vector  $\phi_j \in \mathbb{R}^{12}$ , listed in Table 5.13. It describes what the action does independently of the current grid state, so the scorer can tell a single endpoint move from an action that reconfigures a whole substation even when both touch the same nodes.

Table 5.13: The static action-feature vector  $\phi_j$ . The five indicators are binary; the seven counts are divided by their largest value over that agent’s action space, so every entry lies in  $[0, 1]$  and does not depend on the grid size.

| Entry                                | Meaning                                |
|--------------------------------------|----------------------------------------|
| <i>Indicators</i>                    |                                        |
| <code>is_do_nothing</code>           | The action is the idle action          |
| <code>has_busbar_context</code>      | It modifies at least one substation    |
| <code>has_line</code>                | It touches at least one line           |
| <code>has_load</code>                | It touches at least one load           |
| <code>has_generator</code>           | It touches at least one generator      |
| <i>Scaled counts</i>                 |                                        |
| <code>n_substations</code>           | Substations modified                   |
| <code>n_topology_changes</code>      | Topology-vector entries changed        |
| <code>n_line_status_changes</code>   | Lines connected or disconnected        |
| <code>n_line_endpoint_changes</code> | Line endpoints moved to another busbar |
| <code>n_generator_changes</code>     | Generators moved to another busbar     |
| <code>n_load_changes</code>          | Loads moved to another busbar          |
| <code>n_other_changes</code>         | Changed objects of any other type      |

The descriptor is optional in the final scorer. Including it tells the model what operation is proposed and can tell apart actions that touch similar nodes. Omitting it isolates what can be learned from the graph state and the touched-node context alone. This choice is independent of the pooling rule and of how the idle action is handled.

### 5.5.3 Pooling the touched nodes

The second ingredient turns the touched rows into  $\mathbf{c}_j$ . In  $\mathbf{z}_a$  the nodes have already been averaged away, so the encoder also returns the node states themselves:

$$(\mathbf{z}_a, \mathbf{H}) = \text{encoder}_{\text{nodes}}(\mathcal{G}_t^{(a)}), \quad \mathbf{H} \in \mathbb{R}^{|\mathcal{V}| \times d_h}, \quad (5.12)$$

where row  $v$  of  $\mathbf{H}$  is the state  $\mathbf{h}_v^{(K)}$  of node  $v$  after the last message-passing layer. These are exactly the vectors that mean pooling averages into  $\mathbf{z}_a$ ; nothing new is computed, and a run with the default head is unaffected. Only the physical nodes appear in  $\mathbf{H}$ : substation-summary and virtual nodes are excluded, since an action touches equipment and not a learned hub.

Write  $\mathcal{N}_t(j)$  for the rows of type  $t \in \{\text{bus, line, load, gen}\}$  that action  $j$  touches. The masked mean over any such set is

$$\mathbf{c}_j^{(t)} = \frac{1}{\max(1, |\mathcal{N}_t(j)|)} \sum_{v \in \mathcal{N}_t(j)} \mathbf{h}_v^{(K)}, \quad (5.13)$$

where the clamped denominator makes an untouched type give the zero vector instead of a division by zero. The joint rule applies this once to all touched rows regardless of type, giving width  $d_h$ . The type-separated rule applies it per type and concatenates the four results, giving width  $4d_h$  and keeping node roles (Table 5.14).

Table 5.14: Implemented constructions of the action-specific node context.

| Construction                     | Eligible rows                     | Aggregation                                                      | Width  |
|----------------------------------|-----------------------------------|------------------------------------------------------------------|--------|
| Joint uniform mean               | All nodes touched by the action   | One equal-weight mean; node roles are mixed                      | $d_h$  |
| Type-separated uniform mean      | All nodes touched by the action   | One equal-weight mean per type, then concatenation               | $4d_h$ |
| Type-separated learned attention | Determined by the attention scope | One action-conditioned weighted sum per type, then concatenation | $4d_h$ |

With learned attention, an action-specific query assigns normalized weights within each node type:

$$\mathbf{c}_j^{(t)} = \sum_{v \in \mathcal{E}_t(j)} \alpha_{jv}^{(t)} \mathbf{v} \mathbf{h}_v^{(K)}, \quad \sum_{v \in \mathcal{E}_t(j)} \alpha_{jv}^{(t)} = \mathbb{I}[|\mathcal{E}_t(j)| > 0]. \quad (5.14)$$

Here  $\mathcal{E}_t(j)$  is the eligible set; an empty set returns the zero vector. The implemented choices are shown in Table 5.15. The action query is formed from the global graph state and the static action descriptor in the reported attention comparisons. This is separate from whether the descriptor is also appended to the final scorer.

Table 5.15: Possible eligibility rules for learned action-specific attention.

| Eligibility rule            | Nodes allowed to receive weight                                         | Purpose                                                                                |
|-----------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Fixed physical support      | Only nodes identified as touched by the action                          | Learn their relative importance without changing the physical mapping.                 |
| Physical support as a prior | Every local node of the same type, with a score bonus for touched nodes | Allow attention to extend the physical mapping while favoring it initially.            |
| Fully learned support       | Every local node of the same type, without a score bonus                | Test whether the model can discover the relevant nodes without the mapping as a prior. |

The two uniform means are parameter-free controls. Learned attention can keep a critical node that a mean would dilute, but adds parameters and must also learn a sensible weighting.

#### 5.5.4 Scoring and the idle action

All non-idle actions are scored by one network with a single output unit. Its input depends on whether the static action descriptor is included:

$$\mathbf{x}_{a,j} = \begin{cases} [\mathbf{z}_a; \mathbf{c}_j; \phi_j], & \text{with the descriptor,} \\ [\mathbf{z}_a; \mathbf{c}_j], & \text{without it,} \end{cases} \quad \ell_{a,j} = \text{MLP}_{\text{score}}(\mathbf{x}_{a,j}), \quad j \geq 1. \quad (5.15)$$

The idle action touches no node, so its local context is the zero vector. It either reuses the same scorer or gets a dedicated one:

$$\ell_{a,0} = \begin{cases} \text{MLP}_{\text{score}}(\mathbf{x}_{a,0}) + b_{\emptyset}, & \text{shared, with } \mathbf{c}_0 = \mathbf{0}, \\ \text{MLP}_{\emptyset}(\mathbf{z}_a) + b_{\emptyset}, & \text{dedicated.} \end{cases} \quad (5.16)$$

In both cases  $b_{\emptyset}$  is a learned scalar applied only to the idle action. Sharing the scorer keeps parameter sharing across the full action set, but gives it an empty context that never occurs for non-idle actions. A dedicated scorer instead treats “do nothing” as a global safety decision, at the cost of extra parameters and a different scoring mechanism.

#### 5.5.5 Global-maximum and action-delta extension

The Gmax-delta variant keeps the candidate scorer above but changes two inputs. First,  $\mathbf{z}_a$  is obtained by a component-wise maximum over energized node states instead of a mean. Second, each exact modification in action  $j$  is encoded from the modified object, the mean state of its substation busbars, the target busbar, their difference, and static operation flags. The modification tokens are combined into  $\delta_j$  and appended to the scorer input. Figure 5.10 summarizes the resulting architecture. It uses the current graph embeddings only: it does not construct the post-action graph or simulate the candidate.

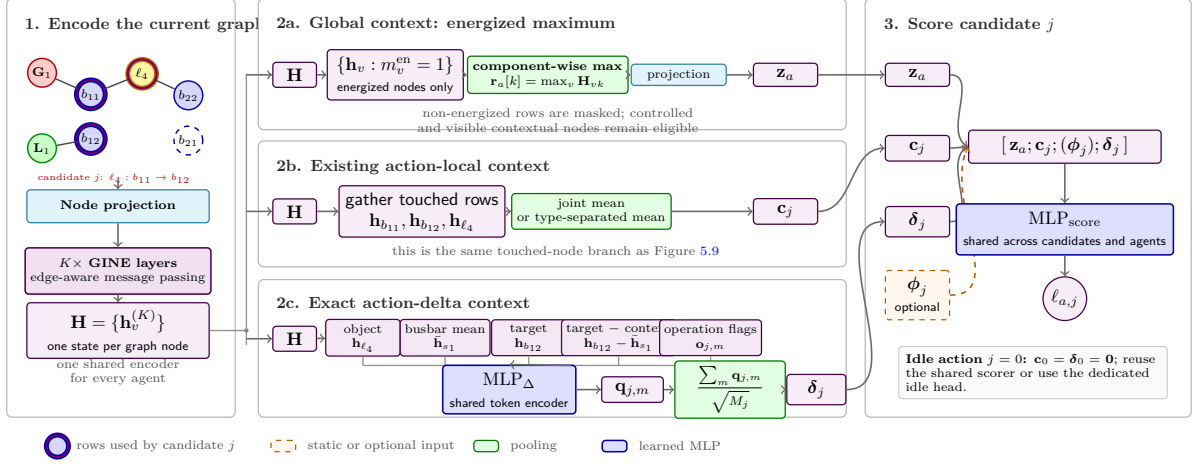


Figure 5.10: Gmax-delta extension of Figure 5.9. The shared GINE encoder returns node states  $\mathbf{H}$ . The upper branch builds the global vector by maximum pooling over energized nodes; the middle branch keeps the original touched-node context  $\mathbf{c}_j$ ; and the lower branch encodes each exact object modification and aggregates the resulting tokens into  $\delta_j$ . The shared scorer receives all three state-dependent contexts and, optionally, the static descriptor  $\phi_j$ . No post-action graph or power flow is computed.

### 5.5.6 Cost and current limits

The candidate contexts form a tensor of shape  $[\text{batch}, |\mathcal{A}_a|, d_c]$ , so memory grows with the size of the action space. This is the reason to start on **bus14** and on reduced action spaces before the full WCCI one.

The compared variants do not have the same parameter counts: separating node types widens the scorer input, static descriptors add input columns, learned attention adds projections, and a dedicated idle scorer adds another network. These counts should therefore be reported. When the dedicated idle scorer is used, its different mechanism also makes the action-0 frequency an explicit comparison metric rather than something that can be assumed comparable.

The head requires the disaggregated line-node graph and does not combine with the intervention gate, which is refused at construction rather than silently ignored. One-hop context makes no difference here: the neighbor-inclusion setting adds no far-end equipment to this schema, because the line node already supplies the shared-line state.

## 5.6 Action-Space Reduction for Large Grids

The representations above change what an agent sees. This section changes what it can do, and is what makes the larger grids workable at all. The WCCI bus36 setting has a far larger discrete topology-action space than **bus14**, so before training MAPPO on it the action space is reduced offline from simulated action outcomes. During rollouts, states are collected when the grid is stressed,

$$\rho_{\max} > 0.90.$$

For each collected state  $s$ , candidate actions are simulated and compared against do nothing:

$$\Delta(s, a) = \rho_{\text{after}}(s, a) - \rho_{\text{after}}(s, 0).$$

An action is counted as useful when it lowers the post-action loading compared with do nothing, i.e.  $\Delta(s, a) < -\epsilon$ . Each action is then scored by its empirical improvement rate,

$$\text{score}(a) = \frac{\#\{\text{tested states where } a \text{ improves the grid}\}}{\#\{\text{tested states where } a \text{ is evaluated}\}},$$

with a minimum-count filter to avoid selecting rarely sampled actions. The final reduced action space keeps the  $k$  best-scoring actions per agent, while always keeping action 0; an agent with fewer than  $k$

actions keeps all of them.

Choosing  $k$  is an empirical question rather than a design one, so it is decided where the reduced space is used. The candidate sets are sanity-checked with a one-step simulator-greedy controller that, at each risky state, simulates the available reduced actions and executes the best improving one. That controller also sets the size of the reduction, since it shows how much survival the action set can still reach once cut. Section 8.1 reports the resulting sizes and the value of  $k$  used for the transfer experiments.

### Chapter Summary (3/3): Action Scoring and Large-Grid Reduction

**Purpose:** Define how graph embeddings become action logits and how the large WCCI action space is made workable.

- **Default action head:** An agent-specific MLP maps the graph embedding directly to one logit per discrete action.
- **Candidate-action head:** The optional disaggregated line-node variant scores each non-idle action from the encoded nodes it changes and from static action descriptors. The idle action has a separate score. The PPO objective and categorical policy are unchanged.
- **Current scope:** Candidate scoring requires the disaggregated line-node graph, and its memory cost still grows with the number of candidate actions. One-hop context is optional, since the line node already carries the shared measurements.
- **WCCI reduction:** Actions are ranked offline by how often they reduce loading in simulated stressed states. The best  $k$  per agent are kept and action 0 is always kept; a one-step simulator-greedy controller checks the reduced sets and fixes  $k$  before policy training.

**Takeaway:** Graph representation, action scoring, and action-space size are separate design choices. This chapter defines them; the following chapters evaluate the resulting policies.

## Chapter 6

# Graph-Design Screening

This chapter begins by verifying that a shared GNN actor can match the tuned MLP baseline, then reports five graph-design screens (Screens A–E). All screens evaluate seed-0 factorials. They ask whether a specific choice in the actor of Chapter 5 changes survival on held-out chronics. Screens A, B, and C evaluate busbar graphs, while Screens D and E evaluate disaggregated graphs. All screens form a unified, numerically comparable sequence.

### 6.1 Preliminary GNN Encoder Viability

Before screening individual graph-design choices, an exploratory study checks that a graph actor can match the tuned MLP baseline. This is a prerequisite for transfer: the agents must be able to share one size-independent encoder without sacrificing survival.

**Question.** Can a shared graph encoder perform as well as the MLP? If each agent requires a separate encoder, or if the GNN cannot train stably, its structural transferability is lost.

**Baseline and changes.** The study compares several GINE variants. It varies encoder size (light or heavy), actor sharing (shared or unshared encoder), critic architecture (MLP or GNN), and the training optimizations established for the MLP baseline, including optimized critic updates and reward normalization.

**Protocol.** All GINE runs use `gnn_type = gine`, 15,000,000 total timesteps, evaluation every 80,000 timesteps, 2,000 rollout steps, deterministic evaluation, and the same 80/20 chronic split used by the MLP experiments. GINE is used because it incorporates edge features and is expressive over graph structures [19]. The complete graph-actor pipeline is described in Appendix B; Appendix D gives the full configurations in Tables D.1 and D.2.

**Evidence.** Table 6.1 and Figure 6.1 compare the main architectural improvements with the MLP Baseline. Figure 6.2 shows the broader search over isolated changes.

Table 6.1: GINE rerun survival summary.

| Run                             | Intended change                      | Final survival (%) | Peak survival (%) |
|---------------------------------|--------------------------------------|--------------------|-------------------|
| Heavy GINE Baseline             | Historical GINE baseline             | 59.2               | 93.9              |
| Light GINE (Concat, MLP Critic) | Light GINE with MLP critic           | 78.9               | 90.4              |
| No Bias, Opt Critic             | Optimized critic and bias 0.0        | 82.5               | 85.7              |
| Optimized Light GINE            | Light optimized GINE with MLP critic | 95.7               | 97.4              |

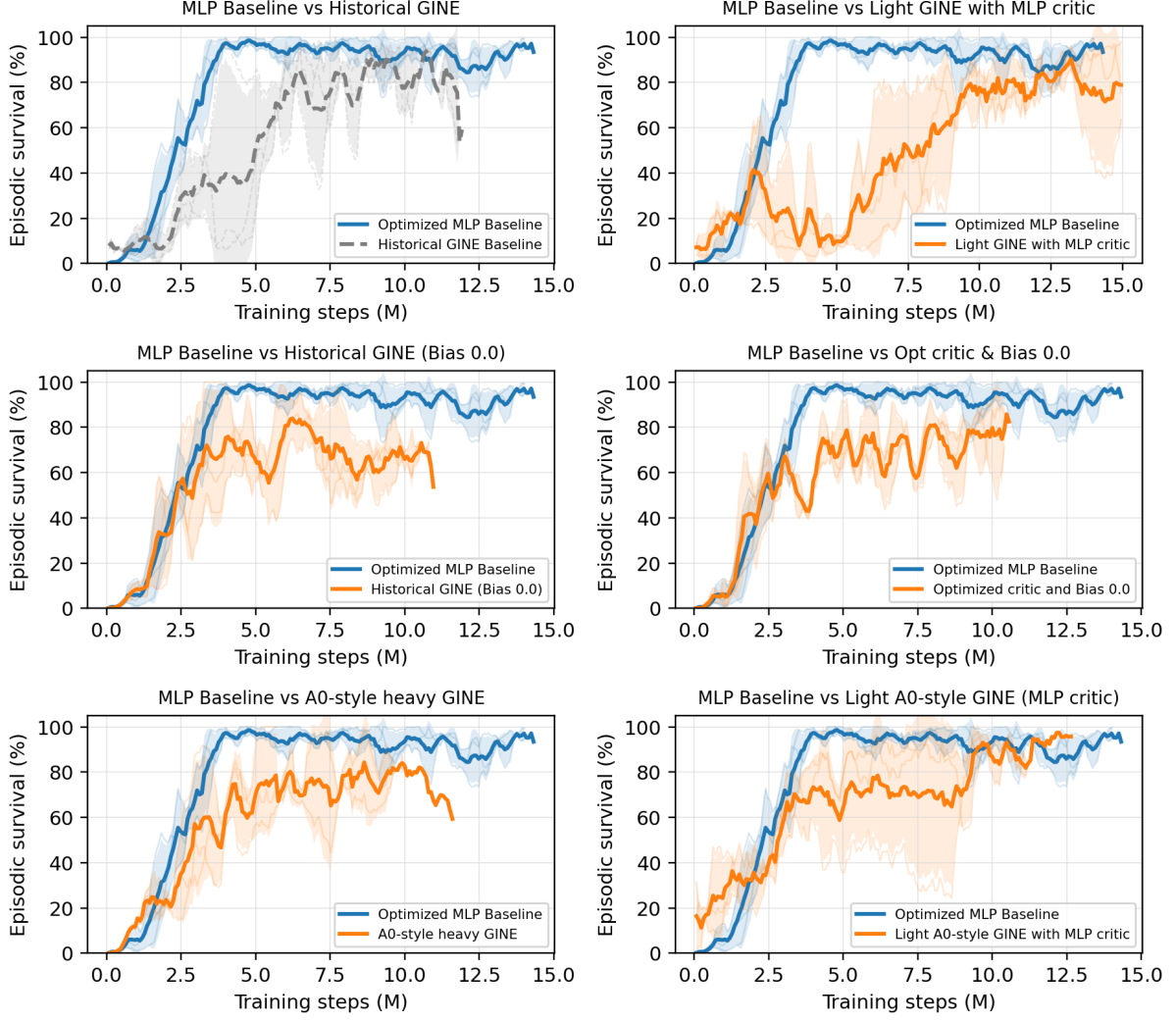


Figure 6.1: Seed-aggregated test episodic survival comparing the MLP Baseline with high-performing GINE architectural variants.

**Interpretation.** The broad search gives three conclusions:

- **Shared encoder viability.** The unshared-actor variant does not outperform the shared variants, so one encoder can serve all agents without becoming the performance bottleneck.
- **Critic architecture.** A GNN centralized critic is unstable and degrades performance. The strongest variants pair the shared GNN actor with the standard MLP critic.
- **Optimization dependence.** Optimized critic updates, reward normalization, and removal of the initial do-nothing bias are also important for stable GNN training.

The strongest result is the **Optimized Light GINE**, which combines the shared light GINE actor, the MLP critic, and the baseline optimizations to reach 95.7% final survival. A shared GNN is therefore competitive on `bus14`, justifying the representation screens that follow.

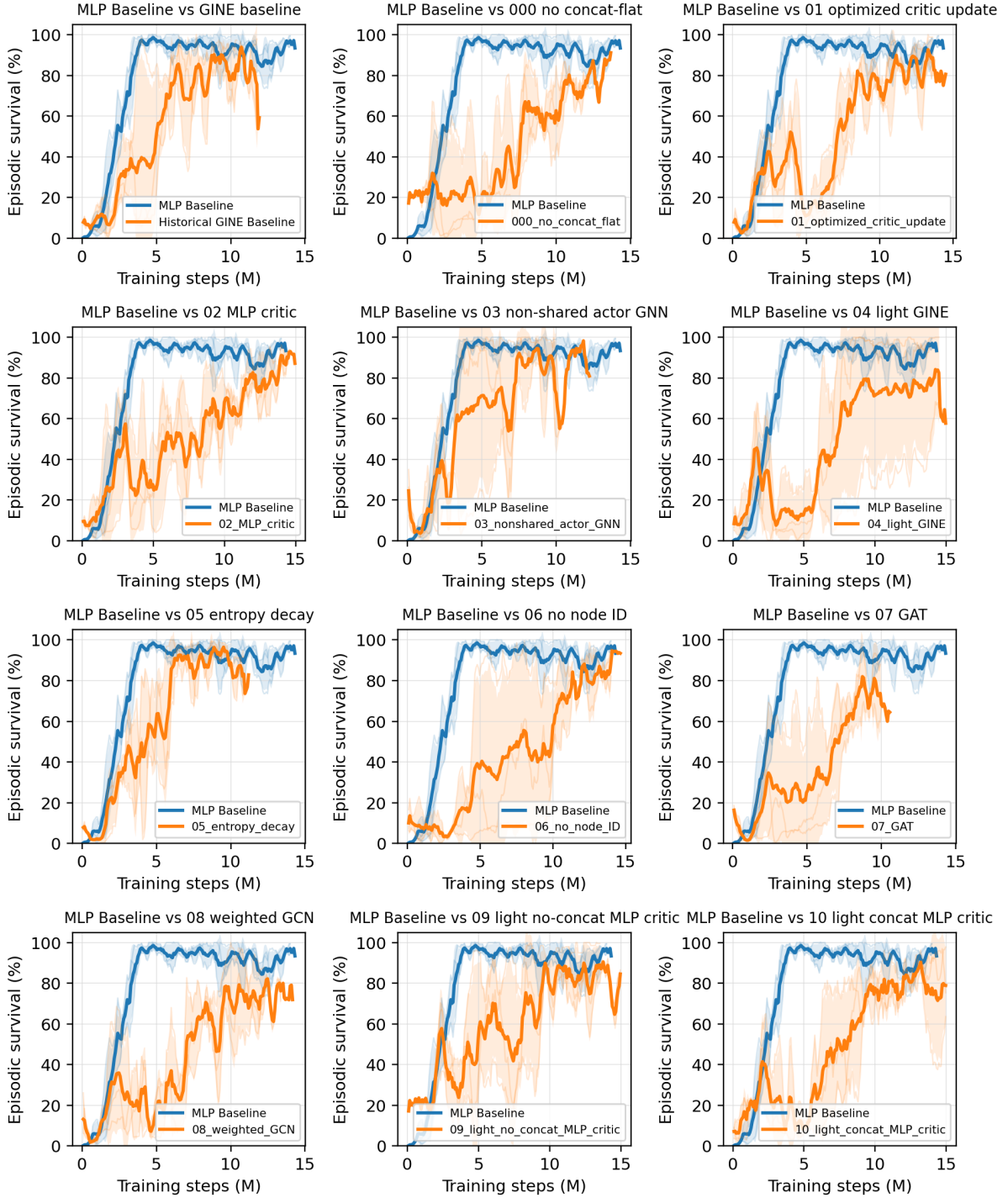


Figure 6.2: GINE survival comparisons. Each subplot compares the MLP Baseline with one variant from `configs/gine_s0_s1_s2`. Full hyperparameters and architectural details are given in Appendices B and D.

## 6.2 Shared Protocol

Table 6.2 separates the common optimization protocol from the boundary and seed settings that differ by campaign. Within a screen, every setting not named as a screened factor is fixed, so each screen is a comparison against its own baseline cell.

Table 6.2: Common optimization settings and campaign-specific boundary and seed settings.

| Component             | Fixed setting                                                                                                                            |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Environment</b>    | bus14 (difficulty 0), topology actions, decentralized agents, normalized observations/rewards, 80/20 train/test chronic split.           |
| <b>Actor</b>          | Shared GNN encoder. LayerNorm, node pre-encoder, two 128-unit action-head layers.                                                        |
| <b>Critic</b>         | Centralized MLP with three 256-unit hidden layers.                                                                                       |
| <b>Preprocessing</b>  | None.                                                                                                                                    |
| <b>Budget</b>         | 15,000,000 environment steps (72 parallel envs, 576 rollout steps).                                                                      |
| <b>MAPPO</b>          | 10 epochs, 16 minibatches. LR $10^{-4}$ (linear decay). $\gamma = 0.9$ , $\lambda = 0.95$ , clip 0.2, target KL 0.02, entropy coef 0.01. |
| <b>Exploration</b>    | No initial do-nothing prior; no action-0 logit bonus.                                                                                    |
| <b>Sparse control</b> | Disabled.                                                                                                                                |
| <b>Evaluation</b>     | Deterministic, every 82,944 env steps, 10 episodes per split.                                                                            |
| <b>Seeds</b>          | 0 (for Screens A, B, C, D, and E).                                                                                                       |

### 6.2.1 Endpoint statistics

Survival is the percentage of an episode’s maximum length the agent keeps the grid alive (Section 3.1). To reduce each run to a single number, the checkpoint with the highest test survival seen during training is deterministically re-evaluated over all 201 held-out chronics. Every run reaches the 15M-step budget, and all screens report exactly one seed per cell.

### 6.2.2 Compute

The campaigns were executed on JED (CPU) and IZAR (GPU). Hardware and execution setup were held fixed within each factorial to ensure fair comparisons. Since the screens evaluate exactly one training seed per cell, performance differences are descriptive rather than estimates of between-seed effects.

Training throughput is dominated by stepping the 72 Grid2Op environments rather than the model’s forward and backward passes. For example, in Screen A (Section 6.3), tripling the actor parameter count from 83.6k to 244.9k has a negligible effect on wall-clock time. Parameter count is therefore not the bottleneck when selecting architectures.

## 6.3 Screen A: Encoder Depth and Width

### Question

This screen measures the effect of message-passing depth  $K$  and encoder width  $d_h$ . On the busbar graph, depth controls reach: one layer crosses one transmission-line edge, while two layers let each busbar combine information across paths of two physical edges. The graph-level readout then pools all valid node embeddings, so  $K$  need not equal the diameter of the local graph. Width controls how much information the node and pooled embeddings can retain.

The result applies directly only to the busbar schema; using the same pair for graphs with explicit asset or line nodes remains an assumption.

## Design

A complete  $3 \times 4$  grid at seed 0, yielding 12 runs. This screen is executed entirely on the busbar graph representation. Table 6.3 lists the screened parameters and their levels. The graph output dimension is matched to the hidden dimension, so a cell varies one width rather than two. Every other setting is inherited from the reference configuration of Section 6.2.

Table 6.3: Screen A. Screened parameters and the values tested. All other settings are held at Table 6.2.

| Parameter                  | Setting                     | Values tested   |
|----------------------------|-----------------------------|-----------------|
| Message-passing layers $K$ | <code>gnn_layers</code>     | 1, 2, 3         |
| Encoder width $d_h$        | <code>gnn_hidden_dim</code> | 16, 32, 64, 128 |

## Result

The complete endpoint and marginal tables, the training curves, and the parameter-count diagnostic are reported in Appendix A.2.

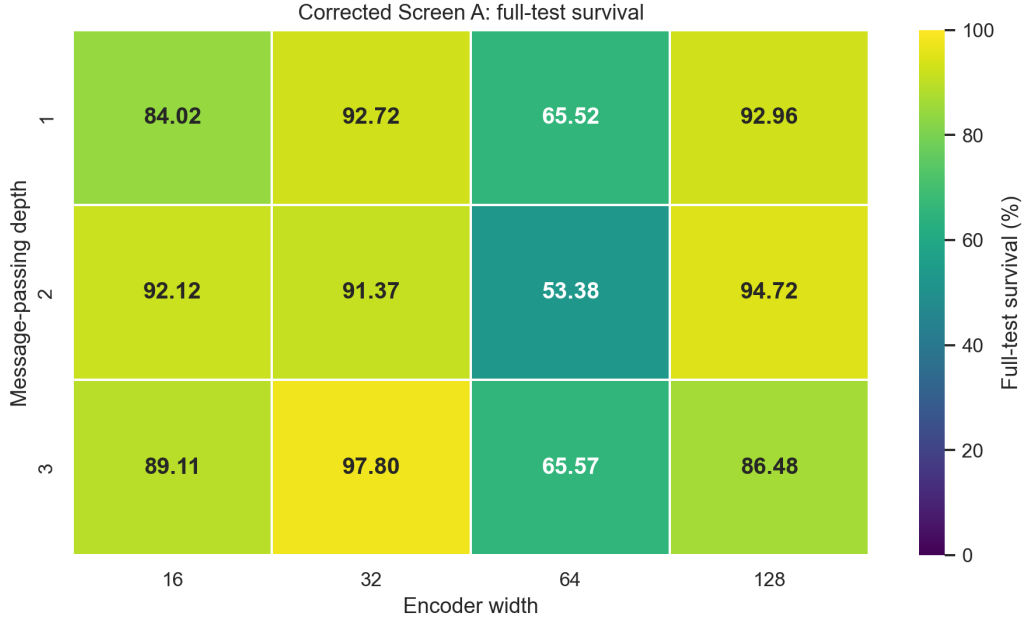


Figure 6.3: Screen A. The same twelve cells as a heatmap of the full-test survival of each run’s best checkpoint at seed 0.

## Reading

**The endpoint leader is mp3\_h32.** Its best checkpoint reaches 97.80% full-test survival, 3.08 points above mp2\_h128 at 94.72%.

**Depth does not separate cleanly.** The row means are 83.80%, 82.90%, and 84.74% for  $K = 1, 2, 3$ . Their ranges overlap heavily, and the best and second-worst cells both occur at  $K = 3$ . These four single-seed cells per row do not establish a reliable depth ordering.

**Width is strongly non-monotonic.** Averaged across depths,  $d_h = 32$  leads at 93.97%, followed by 128 at 91.39% and 16 at 88.42%; every  $d_h = 64$  cell is weak, yielding a 61.49% marginal mean.

**The factorial does not yield a simple architecture rule.** The depth marginals differ by less than two points, width is non-monotonic, and the ranking changes across depth–width combinations.

The grid is therefore better used to select a robust operating point than to claim a monotonic depth or capacity effect.

Each endpoint covers all 201 test chronics, but it still reflects a checkpoint selected from roughly 180 ten-episode evaluations and only one training seed. Table 6.4 therefore complements it with metrics computed from the complete training curves.

Table 6.4: Screen A. Convergence measured from the training curves rather than from a single checkpoint. Steps to 90% is the first evaluation at which the smoothed test survival reaches 90%; the curve mean is the time-average of the smoothed curve over training. The final-window mean averages the raw last 20 periodic evaluations, spanning approximately the final 1.58M environment steps. This is the campaign’s late-stability window: it reduces sensitivity to the last few evaluations while remaining local to the converged regime. Dashes mark architectures that never reached the threshold.

| Architecture | Steps to 90% (M) | Curve mean (%) | Final-window mean (%) |
|--------------|------------------|----------------|-----------------------|
| mp2_h128     | <b>6.22</b>      | <b>71.40</b>   | <b>94.04</b>          |
| mp1_h16      | 6.88             | 70.92          | 84.93                 |
| mp3_h32      | 7.88             | 69.63          | 91.28                 |
| mp2_h16      | 8.79             | 69.16          | 88.21                 |
| mp1_h128     | 11.03            | 68.50          | 91.11                 |
| mp1_h32      | 6.64             | 63.13          | 93.77                 |
| mp3_h128     | —                | 58.58          | 80.20                 |
| mp1_h64      | —                | 51.60          | 63.37                 |
| mp3_h64      | —                | 49.80          | 67.58                 |
| mp3_h16      | —                | 48.82          | 60.00                 |
| mp2_h32      | 13.69            | 46.51          | 91.08                 |
| mp2_h64      | —                | 45.00          | 53.41                 |

**Why carry mp2\_h128 forward.** Physically,  $K = 2$  is a sensible choice because in a small grid like **bus14**, most substations are within two hops of each other. Empirically, **mp2\_h128** dominates all convergence and stability metrics (Table 6.4): it crosses 90% survival first, maintains the highest average performance during training, and is significantly more stable in the late-training window. While its absolute best checkpoint is slightly behind **mp3\_h32** (94.72% vs 97.80%), this gap is well within the noise variance of a single seed. We therefore select **mp2\_h128** as the robust default architecture for the remaining screens.

## 6.4 Screen B: Graph Readout Scope and Reducer

### Question

The local graph may contain nodes that provide message-passing context without being equally useful in the final graph summary. Should readout pool every visible node, only controlled nodes, only energized nodes, or the intersection of controlled and energized nodes? For each scope, should the valid embeddings be averaged or reduced by a per-channel maximum?

The scope mask is applied only at graph readout. Context nodes remain in the message-passing graph, so this screen changes which embeddings form the policy summary without changing what information can propagate to a controlled node.

### Design

A  $4 \times 2$  factorial crosses four readout scopes—all visible, controlled, energized, and controlled  $\cap$  energized—with mean and max reduction at seed 0. All cells use GINE with  $K = 2$  and  $d_h = 128$  on the busbar graph. Seven cells are new; the all-visible/mean cell reuses **mp2\_h128** from Screen A because it is exactly the same configuration.

## Result

Table 6.5: Screen B. Full-test survival by readout scope and reducer at seed 0. The all-visible/mean cell is inherited from Screen A.

| Readout scope               | Reducer            |              |
|-----------------------------|--------------------|--------------|
|                             | Mean               | Max          |
| All visible                 | 94.72 <sup>†</sup> | 96.31        |
| Controlled                  | 94.60              | 94.79        |
| Energized                   | 98.93              | <b>99.39</b> |
| Controlled $\cap$ energized | 91.45              | 96.47        |

<sup>†</sup>Screen-A mp2\_h128 reference; not a new run.

Table 6.6: Screen B training-curve diagnostics at seed 0, ranked by curve mean. “To 90%” is the first crossing of 90% by the five-evaluation trailing mean; curve mean is its time-weighted mean over training. Final-20 statistics use the raw final 20 periodic evaluations (approximately 1.58M environment steps).

| Scope/reducer                    | To 90% (M)  | Curve mean (%) | Final-20 mean (%) | Final-20 std (pp) |
|----------------------------------|-------------|----------------|-------------------|-------------------|
| Energized/max                    | <b>3.40</b> | <b>79.26</b>   | 97.79             | 4.12              |
| Energized/mean                   | 8.71        | 73.59          | <b>98.01</b>      | <b>2.90</b>       |
| Controlled $\cap$ energized/mean | 5.97        | 72.97          | 91.18             | 8.59              |
| All visible/mean <sup>†</sup>    | 6.22        | 71.40          | 94.04             | 5.29              |
| Controlled/mean                  | 6.47        | 70.94          | 91.21             | 6.08              |
| Controlled $\cap$ energized/max  | 8.88        | 68.65          | 86.31             | 8.80              |
| All visible/max                  | 4.73        | 68.38          | 88.57             | 8.08              |
| Controlled/max                   | 10.87       | 68.27          | 90.67             | 6.81              |

The corresponding training curves are reported in Appendix A.3.

## Reading

**Energized-node readout leads with both reducers.** Its mean-pooling cell reaches 98.93% and its max-pooling cell reaches 99.39%, only 0.46 points apart. Relative to pooling all visible nodes, energized-only pooling improves survival by 4.21 points under mean and 3.08 points under max.

**Max is directionally consistent across scopes.** For every matched scope, max pooling exceeds mean pooling on the full test. The gain ranges from 0.19 points (controlled nodes) to 5.01 points (controlled-and-energized intersection). The magnitude depends strongly on scope, but the positive direction does not.

**Why carry energized/max forward.** It combines the best scope with the reducer that wins every within-scope full-test comparison. Table 6.6 also identifies it as the strongest overall training curve: it is the earliest to cross 90% (3.40M steps) and has the highest time-weighted curve mean (79.26%). While energized/mean has slightly better late-window variance, **energized\_max** has the strongest full-test endpoint (99.39%) and is the most consistently supported operating point overall. We fix it for Screens C and D to prevent readout from becoming an additional confounding factor.

## 6.5 Screen C: Structural Augmentations of the Busbar Graph

### Question

Three optional augmentations add computational structure to the busbar graph without adding any physical measurement: same-substation edges ( $e$ ), which let the busbars of one substation exchange

messages directly; substation-summary nodes ( $n$ ), which insert one learned hub per substation; and virtual-node readout ( $v$ ), which replaces mean pooling with a learned graph-summary node. Each is motivated by the same intuition, shorten the path between related busbars, so the question is whether any of them earns its cost, and whether they compose.

## Design

A complete  $2^3$  factorial at seed 0, 8 runs. The graph type is busbar throughout and the baseline cell is the unaugmented **e0n0v0**. It has the same  $K = 2$ , width-128, GINE/**energized\_max** design as the reference from Screen B. Cells are named by their three switches, so **e1n0v1** means same-substation edges on, summary nodes off, virtual-node readout on.

## Result

The detailed result tables and training curves are reported in Appendix A.4.

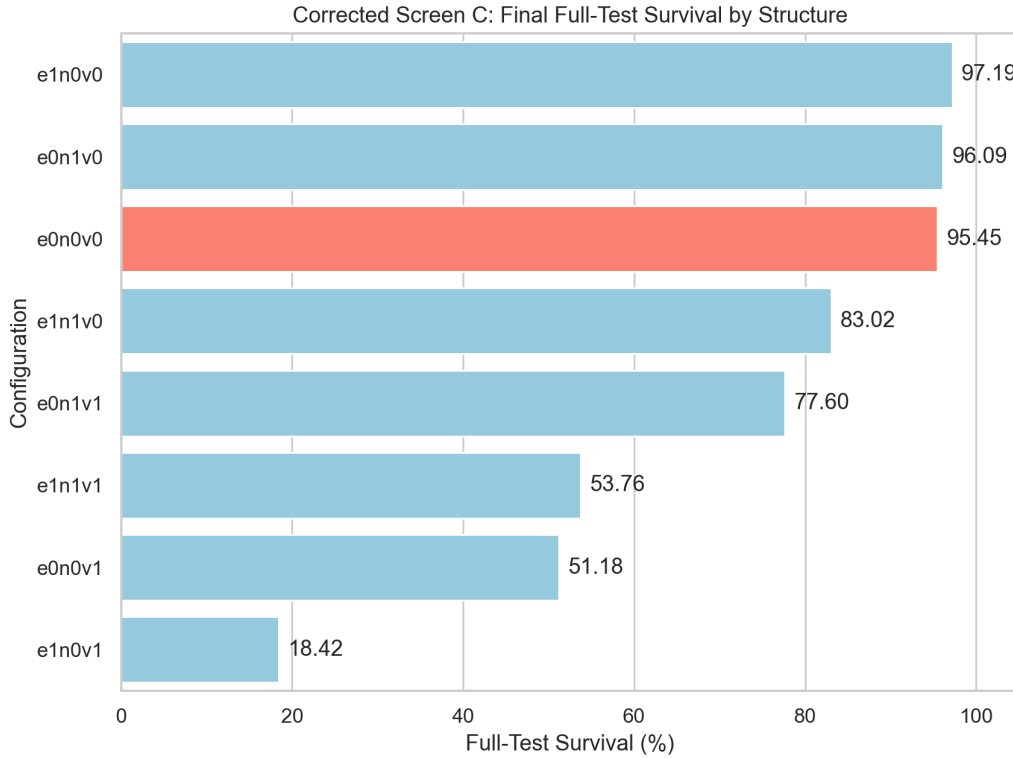


Figure 6.4: Screen C. The factorial results as a bar plot, with the baseline **e0n0v0** highlighted.

## Reading

The same-substation edges provide a slight improvement. The **e1n0v0** cell is the best at 97.19%, outperforming the unaugmented **e0n0v0** baseline which achieved 95.45%<sup>1</sup>. Figure A.5 shows that the cells separate well before the training budget ends.

The hierarchy factors have mixed effects. Summary nodes alone (**e0n1v0**) slightly improve the baseline to 96.09%, but the virtual-node readout (**v1**) is highly destructive, with cells containing it dropping

<sup>1</sup>This **e0n0v0** baseline is mathematically identical to the best-performing Energized/Max cell from Screen B, which achieved 99.39%. The discrepancy in their final survival rates is due to the different hardware used for training (Screen B on Izar GPUs, Screen C on Jed CPUs), which affects the non-deterministic environment stepping and hardware-specific operations, leading to divergent trajectories.

to between 18% and 77%. When summary nodes and same-substation edges are combined (**e1n1v0**), performance degrades to 83.02%.

The severe drop from virtual-node readout likely stems from redundancy: both mechanisms add a learned aggregation node, and together they stack two hubs between the busbars and the readout (Figure 5.8). With only two message-passing layers ( $K = 2$ ), most physical nodes cannot reach that readout.

Same-substation edges are milder and more consistent, successfully boosting survival by about 1.74 points in isolation. However, when mixed with the hierarchy factors, their effect becomes highly erratic and often negative.

The subsequent stages therefore keep the plain busbar graph with **energized\_max** readout, as the modest gain from edges does not justify the additional graph density. Any future hierarchy should be tested with a deeper encoder.

## 6.6 Screen D: Generator and Load Message Direction

### Question

In the disaggregated graph, each generator and load is a node attached to its busbar. Section 5.2.2 leaves the direction of that attachment open: the asset may send messages to the busbar, receive them from it, or both. The two directions are not symmetric in effect, because the **energized\_max** readout pools asset nodes as well as busbars. Does the choice matter, and do the generator and load relations behave independently?

### Design

A complete  $3 \times 3$  factorial over the generator and load directions at seed 0: 9 runs. The graph type is disaggregated throughout; the line and summary directions stay **bidirectional** and are inert for this schema. The encoder matches the reference configuration: GINE, two layers, 128 features, **energized\_max** readout, no augmentations, so the only screened factors are the two attachment directions. The baseline cell is bidirectional on both relations. Directions are abbreviated **bi** (bidirectional), **a2b** (asset to busbar), and **b2a** (busbar to asset).

### Result

The training curves and complete endpoint ranking are reported in Appendix A.5.

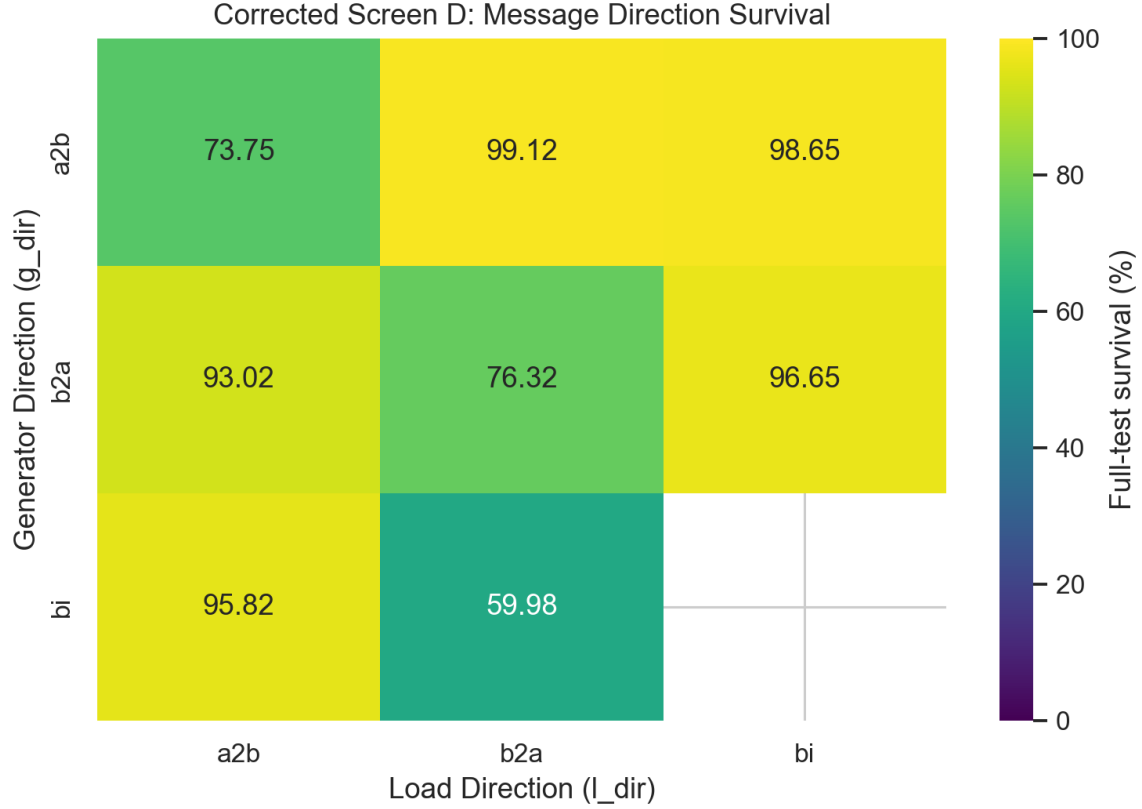


Figure 6.5: Screen D. Full-test survival of the best checkpoint by generator direction (rows) and load direction (columns). The baseline cell is `bi/bi`.

## Reading

**Asymmetric message passing is optimal.** The best configuration uses `a2b` (asset-to-busbar) for generators and `b2a` (busbar-to-asset) for loads, achieving 99.12% survival. Replacing the load direction with bidirectional (`bi`) yields a close second at 98.65%. Conversely, symmetric or reversed configurations fail heavily: `b2a/b2a` drops to 76.32%, `a2b/a2b` falls to 73.75%, and `bi/b2a` collapses completely to 59.98%. Figure A.7 shows that these failure modes separate from the top configurations early in training.

**Why `gen_a2b/load_b2a` makes physical sense.** A plausible mechanism relies on the fact that `energized_max` pooling includes generator and load nodes. By pushing generator information inward to the busbars (`a2b`) and pulling busbar information outward to the loads (`b2a`), the network correctly routes production capacity into the core topology while allowing load nodes to contextualize their demand before the global readout. This asymmetric flow prevents the dilution of the pooled graph embedding that occurs when edges incorrectly blur these distinct physical roles.

## 6.7 Screen E: Candidate-Action Scoring

### Question

The candidate-action head of Section 5.5 scores every action from the encoded nodes it modifies rather than from a fixed-width layer over an action index. Three choices in that head are unresolved: how the touched nodes are pooled, whether the static action descriptor is appended to the score, and whether the idle action gets its own head. This screen varies all three.

## Design

A  $2^3$  factorial on `bus14`, seed 0, eight runs. Pooling is `mean` or `typed_mean`; the action descriptor is appended (`f1`) or withheld (`f0`); the idle action is scored by a dedicated head (`a0h1`) or by the shared scorer (`a0h0`). Every run uses the disaggregated line-node graph without one-hop context, a two-layer GINE encoder of width 128, and the 15M-step budget of Section 6.2. All eight reached 15M steps.

## Result

Table 6.7 reports the full-test survival of each best checkpoint over the 201 held-out chronics. The corresponding training curves and endpoint ranking are reported in Appendix A.6.

Table 6.7: Screen E, full-test survival of the best checkpoint on 201 held-out chronics. One seed per cell.

| Pooling                 | Descriptor      | Idle head         | Survival (%) | Best checkpoint |
|-------------------------|-----------------|-------------------|--------------|-----------------|
| <code>mean</code>       | <code>f0</code> | <code>a0h1</code> | 99.52        | 14,929,920      |
| <code>typed_mean</code> | <code>f1</code> | <code>a0h0</code> | 99.50        | 14,929,920      |
| <code>mean</code>       | <code>f0</code> | <code>a0h0</code> | 99.18        | 14,929,920      |
| <code>mean</code>       | <code>f1</code> | <code>a0h1</code> | 98.66        | 14,929,920      |
| <code>mean</code>       | <code>f1</code> | <code>a0h0</code> | 98.32        | 14,846,976      |
| <code>typed_mean</code> | <code>f0</code> | <code>a0h1</code> | 98.17        | 14,764,032      |
| <code>typed_mean</code> | <code>f0</code> | <code>a0h0</code> | 92.50        | 12,358,656      |
| <code>typed_mean</code> | <code>f1</code> | <code>a0h1</code> | 57.84        | 11,612,160      |

## Reading

**The candidate-action head is a viable actor.** Six of the eight cells reach 98.2% or better and three exceed 99%. No single factor strictly separates the performance. The apparent marginal gaps—such as `mean` at 98.9% against `typed_mean` at 87.0%, and `f0` at 97.3% against `f1` at 88.6%—are artifacts of two low-performing cells; removing the single failure at 57.84% brings the pooling gap to about a point. At one seed, the defensible statement is that the head works and that the three choices do not cleanly order themselves.

**Withholding the descriptor (`f0`) is safer.** This is the one asymmetry worth recording: the best cell withholds the static action descriptor and both failing cells include it, which runs counter to a prior that favours the more elaborate feature. It is not redundant information, since every action still reaches the scorer with its own pooled context—agent 0’s 57 actions produce 57 distinct row-sets, because an action toggles one to three of a substation’s lines rather than a single busbar. Withholding it removes a static label and safely leaves the learned topological description intact.

### 6.7.1 Why the descriptor hurt: a normalization defect

The `f1` result has a concrete cause, and it lies in how the descriptor was built rather than in the information it carries. Its count columns were divided by the largest value in that agent’s *own* action set.

Table 6.8: Divisors applied to the action count descriptors under the previous per-agent rule, `bus14`.

| Agent | Actions | Topology changes | Line endpoints |
|-------|---------|------------------|----------------|
| 0     | 57      | 5                | 3              |
| 1     | 51      | 5                | 4              |
| 2     | 83      | 6                | 3              |

An action moving three elements therefore reached the shared scorer as 0.60 for agent 0 and 0.50 for agent 2, and every agent’s busiest action reached it as exactly 1.0 whether it moved five elements or six. Because one scorer serves all agents, it was shown mutually inconsistent encodings of the same quantity,

and on another grid the implicit divisors would shift again.

The correction is to divide these counts by a global constant (`candidate_action_feature_scaling`), under which the busiest actions read 1.25, 1.25 and 1.5 and keep their true relative scale across agents and grids. Chapter 7 finds the same per-population normalization defect in the node features, where the populations are grids rather than agents.

## 6.7.2 Replication with Improved Graph Features

At seed 0, the eight cells were repeated with improved graph features (NLS): deterministic physical scaling and line-angle differences, without running standardization. The GNN remains shared, but each agent retains its own scoring MLP and optional idle head; this tests preprocessing, not scorer sharing.

Table 6.9: Full-test survival over 201 held-out `bus14` chronics with baseline (NL) and improved (NLS) graph features. The final column is NLS minus NL in percentage points. One seed per cell.

| Pooling    | Descriptor | Idle head | NL (%) | NLS (%)       | $\Delta$ |
|------------|------------|-----------|--------|---------------|----------|
| mean       | f1         | a0h0      | 98.32  | <b>100.00</b> | +1.68    |
| typed_mean | f0         | a0h0      | 92.50  | 99.84         | +7.33    |
| mean       | f0         | a0h1      | 99.52  | 99.46         | -0.07    |
| mean       | f0         | a0h0      | 99.18  | 99.43         | +0.25    |
| typed_mean | f1         | a0h1      | 57.84  | 90.71         | +32.87   |
| mean       | f1         | a0h1      | 98.66  | 83.81         | -14.85   |
| typed_mean | f1         | a0h0      | 99.50  | 71.67         | -27.82   |
| typed_mean | f0         | a0h1      | 98.17  | 69.67         | -28.50   |

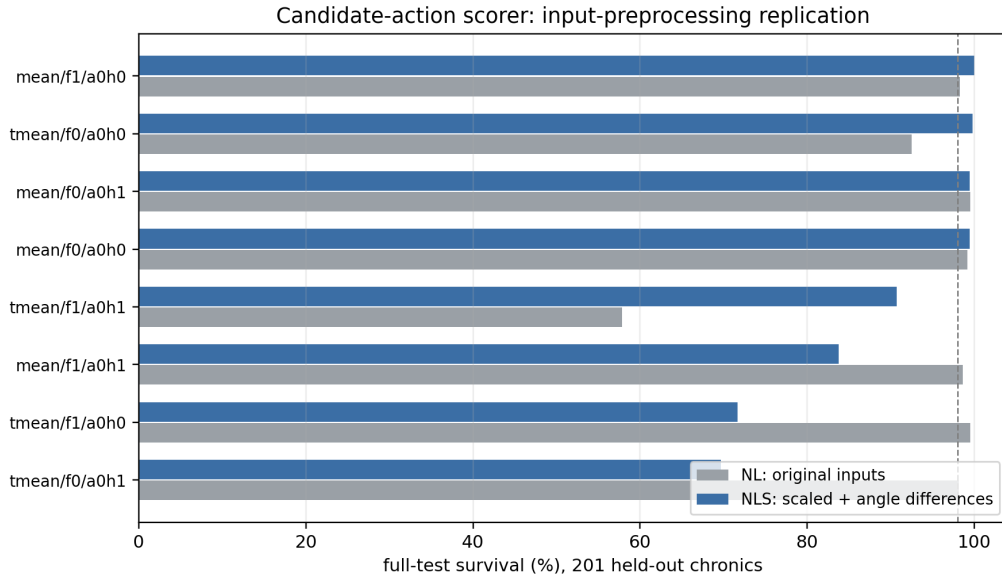


Figure 6.6: The paired comparison of Table 6.9. NLS changes both power scaling and angle representation; the differences cannot be attributed to scaling alone.

## Reading

**The preprocessing effect is highly volatile at one seed.** Four NLS cells exceed 98%, but three remain below 91%, with changes above 25 points in either direction. At one seed, this indicates an interaction with the candidate head rather than a stable preprocessing effect. Only `mean/f0` is robust: both idle-head choices exceed 99.1% with NL and 99.4% with NLS. Chapter 9 separately studies transfer with a scorer shared across agents.

## Chapter 6 Summary: Experimental Screens

**Purpose:** To systematically navigate the high-dimensional architecture space of GNN-based agents for power grid control, identifying the most effective graph structures, readout mechanisms, and message-passing directions on the **bus14** environment.

**Content:** We executed five targeted screens (A–E). We evaluated encoder depth and width (Screen A), discovering that  $K = 2$  and  $d_h = 128$  offers an ideal balance of reach and stability. Screen B confirmed that pooling only energized nodes with a max-reducer (**energized\_max**) strictly dominates other readout scopes. Screen C tested structural augmentations on the busbar graph, finding that virtual readouts cause redundancy issues while same-substation edges or summary nodes offer minor benefits. Screen D demonstrated that asymmetric message passing (**gen\_a2b/load\_b2a**) is optimal for the disaggregated graph. Finally, Screen E confirmed the viability of the candidate-action head, with **mean** pooling and no descriptor (**f0**) emerging as the most robust choice.

**Takeaway & Transfer to WCCI:** The screens successfully distilled the architecture space into a subset of high-performing, robust models. To evaluate their generalization capabilities on the larger and more complex WCCI L2RPN environment in the subsequent chapters, we carry forward the following top-performing configurations:

- Busbar graph: **e0n0v0** (unaugmented baseline)
- Busbar graph: **e1n0v0** (same-substation edges)
- Busbar graph: **e0n1v0** (summary nodes)
- Heterogeneous graph: **gen\_a2b/load\_b2a** (optimal asymmetric)
- Heterogeneous graph: **gen\_bi/load\_bi** (bidirectional baseline)
- The candidate-action scoring (**cas\_h1**) transferable runs.

# Chapter 7

## What the Shared Encoder Reads

One encoder serves every agent, and the same encoder is what a transfer study moves between grids. Its inputs are raw physical quantities, so before either form of sharing can be interpreted the inputs themselves have to be measured. This chapter establishes what the encoder consumes, how far those distributions differ between two grids, which of the differences a network absorbs and which it does not, and what corrections follow. Without those measurements, a deficit under transfer could not be told apart from a representation reading miscalibrated inputs. Chapter 8 applies the corrections and Chapter 9 runs the transfer.

The actor and the critic do not face the same problem. The critic consumes the normalized flat observation and is grid-specific in any case; the encoder reads raw graph features, with physical scaling and running standardization initially disabled, so a transferred `bus14` encoder meets WCCI's raw physical quantities directly. Only that second distribution is at issue here.

### 7.1 Measurement protocol

Both environments were rolled out under a do-nothing policy for 6,000 environment steps, with a cap of 300 steps per chronic so the sample spreads over roughly twenty scenarios per grid rather than concentrating in one. At each step the per-agent local graphs were rebuilt and every visible node recorded, pooling agents together because a single encoder is shared across them. This yields 426,000 node observations for `bus14` and 1,236,000 for WCCI.

Only nodes are measured, because on this schema only nodes carry measurements: the nine edge channels are relation indicators, exactly one of which is set on any active edge, with no continuous content to compare. The transfer runs also set `gnn_include_neighbors = false`, so every node in a local graph is controlled and `domain_mask` is constant at one throughout.

Four statistics summarize each feature. Writing  $F_A$  and  $F_B$  for the two empirical distributions with means  $\mu_A, \mu_B$  and standard deviations  $\sigma_A, \sigma_B$ , and  $\sigma_p = \sqrt{(\sigma_A^2 + \sigma_B^2)/2}$  for their pooled spread:

$$\text{offset} = \frac{\mu_B - \mu_A}{\sigma_p}, \quad \text{scale ratio} = \frac{\sigma_B}{\sigma_A}, \quad \text{KS} = \sup_x |F_A(x) - F_B(x)|. \quad (7.1)$$

**Conventions.**  $A$  is always `bus14` and  $B$  always WCCI, so an offset is positive when WCCI sits higher and a scale ratio above one means WCCI is the wider of the two.  $F_A$  and  $F_B$  are the empirical CDFs built from the samples of Section 7.1, and the Kolmogorov–Smirnov statistic is the largest gap between them: it lies in  $[0, 1]$  and is drawn in the last panel of Figure 7.1.

The pairing is deliberate: the offset is blind to a change in spread and the scale ratio is blind to a displacement, so a channel is only unproblematic when both sit near their neutral values, and KS summarises whatever disagreement remains. Section 7.4 shows why the two failures need separating.

## 7.2 What the encoder reads

The measurements are taken on the disaggregated line-node graph, the schema the transfer study uses. Its twelve node channels are specified in Table 5.6: four type indicators, the shared `p` and `theta` columns, four line-state columns, the substation cooldown, and `domain_mask`.

Every node type shares one feature vector, so a channel is filled only on the type that owns it and is zero-filled elsewhere. That makes the sparsity structural and severe: `p` and `theta` carry a value only on generator and load nodes, which are 24.0 % of rows on `bus14` and 28.7 % on WCCI, while `rho` and the other line-state channels are filled only on transmission-line nodes, 36.6 % and 36.4 % of rows. A statistic taken over all rows therefore mixes a measurement with the zeros of three other node types, which is why every channel below is reported twice.

Three channels are constant or near-constant in this sample rather than sparse: the two cooldown columns, which a do-nothing policy never triggers, and `timestep_overflow`. One channel is dimensionless by construction: `rho` divides each line’s flow by that line’s own thermal limit, while every other physical channel carries the units of the grid it came from.

## 7.3 Results

Table 7.1 reports the shift statistics twice: over all sampled rows, and over only the node type that carries the channel. The two readings differ for the reason given above — a shared column mixes a point mass at zero with the measurement of the type that owns it.

Table 7.1: Distribution shift between the two grids on the disaggregated line-node graph, ordered by KS. Offset is in pooled standard deviations and the scale ratio is WCCI over `bus14`. The right-hand block restricts each measured channel to the node type that owns it: generator and load nodes for `p` and `theta`, transmission-line nodes for the line state. *The type indicators and `domain_mask` own no type, so the restriction does not apply; `constant` marks a channel with no variation in this sample, whose ratio is undefined.*

| Channel                                | Zeros (%)          |      | All sampled rows |       |       | Owning node type |             |              |
|----------------------------------------|--------------------|------|------------------|-------|-------|------------------|-------------|--------------|
|                                        | <code>bus14</code> | WCCI | offset           | ratio | KS    | offset           | ratio       | KS           |
| <code>theta</code>                     | 77.5               | 73.8 | +0.653           | 0.83  | 0.157 | +1.864           | 1.43        | <b>0.678</b> |
| <code>p</code>                         | 77.8               | 76.7 | −0.037           | 2.14  | 0.042 | −0.158           | <b>2.54</b> | 0.201        |
| <code>rho</code>                       | 63.4               | 63.6 | −0.088           | 0.92  | 0.053 | −0.260           | 1.05        | 0.141        |
| <code>line_status</code>               | 63.4               | 63.6 | −0.005           | 1.00  | 0.002 | constant at one  |             |              |
| <code>timestep_overflow</code>         | 100                | 100  | −0.002           | 0.70  | 0.000 | −0.004           | 0.70        | 0.000        |
| <code>time_before_cooldown_line</code> | 100                | 100  | —                | —     | 0.000 | constant         |             |              |
| <code>time_before_cooldown_sub</code>  | 100                | 100  | —                | —     | 0.000 | constant         |             |              |
| <code>is_busbar</code>                 | 60.6               | 65.0 | −0.093           | 0.98  | 0.045 | —                |             |              |
| <code>is_generator</code>              | 91.5               | 89.3 | +0.076           | 1.11  | 0.022 | —                |             |              |
| <code>is_load</code>                   | 84.5               | 82.0 | +0.066           | 1.06  | 0.025 | —                |             |              |
| <code>is_transmission_line</code>      | 63.4               | 63.6 | −0.004           | 1.00  | 0.002 | —                |             |              |
| <code>domain_mask</code>               | 0                  | 0    | —                | —     | 0.000 | constant at one  |             |              |

Read on all rows the table looks harmless: no channel reaches  $KS = 0.2$ . That is the type mixing, not agreement. Restricted to the owning type, `theta` reaches 0.678 with +1.86 standard deviations of displacement, and `p` widens by a factor of 2.54. The four type indicators differ only slightly, which says the two grids have a similar mix of busbars, assets, and lines per local graph rather than saying anything about their physics.

The extremes make the same point concretely. `p` spans  $[0, 97]$  MW on `bus14` and  $[-281, 399]$  MW on WCCI, where 7.8 % of asset rows are negative and `bus14` has none. `theta` spans  $[-11.6, 1.5]$  degrees against  $[-7.1, 16.8]$ : not a wider version of the same distribution but a displaced one.

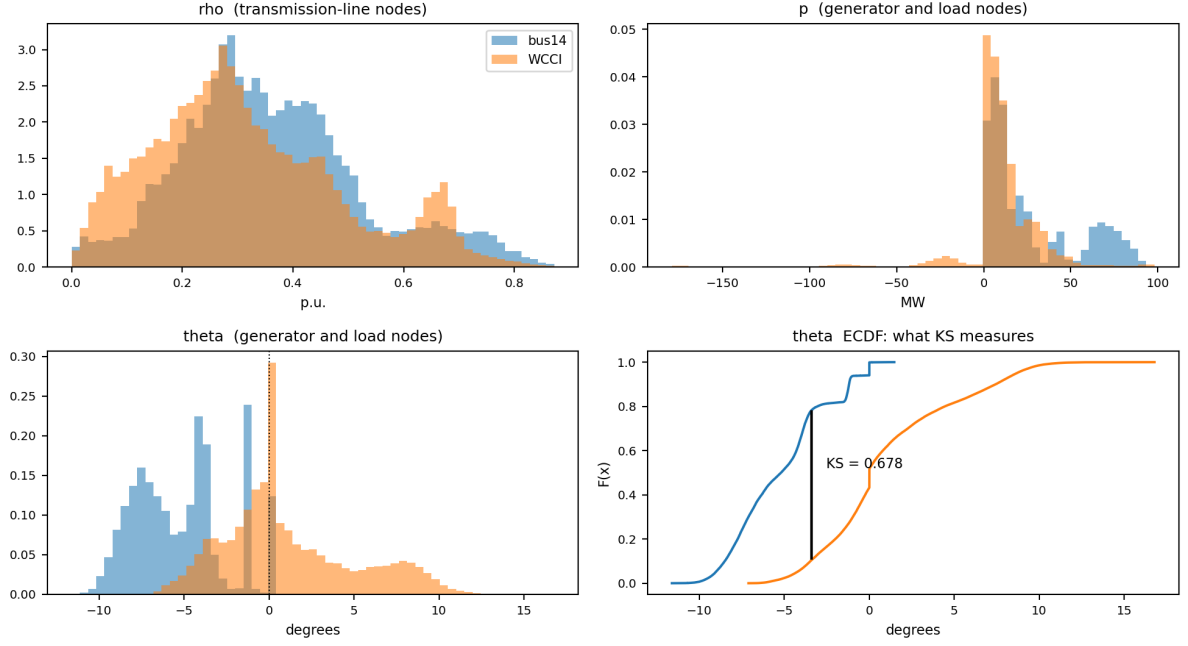


Figure 7.1: The measured channels on the node type that owns them. `rho` lands almost on top of itself, as expected from a quantity already normalized by each grid’s own thermal limits. `p` shares a centre but not a width, and WCCI alone produces negative values. `theta` is displaced rather than rescaled: `bus14` sits almost entirely on the negative side. The last panel reads that displacement as the KS statistic. Full ranges on log density are in Figure A.10.

## 7.4 Two failure modes

The left panel of Figure 7.2 separates the channels along the two axes; the right panel repeats the diagnostic after the corrections and is read in Section 7.5.4. The distinction is not cosmetic, because the architecture responds to the two axes differently.

**Reading the axes.** Each point is one channel, placed by the two statistics of Equation (7.1) over all sampled entries: horizontally the offset magnitude  $|\mu_B - \mu_A|/\sigma_p$ , vertically the scale ratio  $\sigma_B/\sigma_A$  on a log axis so that halving and doubling sit equidistant from 1. A channel needing no correction sits at the bottom left, so distance from that corner is distance from comparability and its direction names which of the two failures is at work.

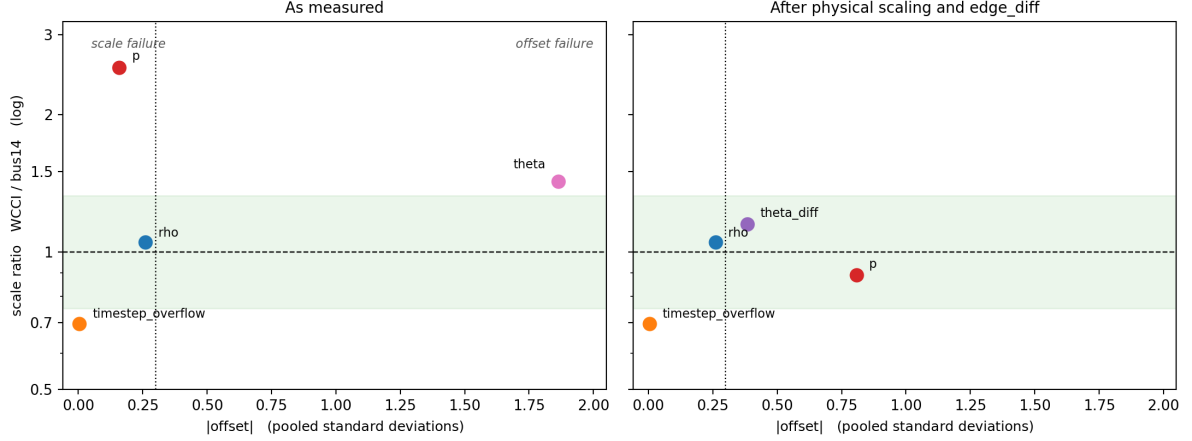


Figure 7.2: Absolute offset against scale ratio, over all sampled entries. The shaded band and the dashed line mark a scale ratio within a third of unity; the dotted vertical line marks an offset of 0.3 pooled standard deviations. *Left*: as measured. Channels outside those bounds fail in one of two distinct ways — the power channels keep their centre and widen, the angles keep their width and move. *Right*: the same diagnostic after the two corrections of Section 7.5, discussed in Section 7.5.4. Points are placed on the owning node type. Every channel now sits inside the scale band and the worst displacement falls from 1.86 to 0.81, but *p* acquires an offset it did not have.

The actor’s node pre-encoder is a linear map followed by ReLU and LayerNorm. LayerNorm effectively absorbs a uniform change in *scale*, as multiplying inputs by a constant leaves the normalized activations largely unchanged. However, it cannot absorb an *offset*: displacing a channel moves the pre-activations across the ReLU threshold, fundamentally altering which units fire. Post-hoc normalization cannot undo this destroyed activation pattern.

This makes the angle, not the power, the critical failure point. The power channel *p* differs in width by a factor of 2.54, which LayerNorm partially self-corrects (though WCCI’s deep negative region remains unseen territory). Conversely, *theta* is displaced by 1.86 standard deviations, an offset the architecture cannot correct.

*rho*, meanwhile, needs no correction at all: scale ratio 1.050, offset  $-0.26$  — the same values the busbar schema gives, since it is the same set of lines carrying the same loadings. The single most decision-relevant channel is already comparable across grids because it was defined in per-unit terms, an argument for expressing features in grid-relative units wherever the physics permits it.

## 7.5 Correcting the inputs

Two preprocessing mechanisms exist in the codebase. Deterministic physical scaling divides each channel by a per-environment physical constant. Running standardization maintains per-environment, per-node-type statistics over *p* and *theta*, excluding *rho*, which the measurements above confirm needs no help. On the diagnosis alone the second looks like the match, being the only one that can move a displaced channel. It is not.

### 7.5.1 Running standardization makes the mismatch worse

Standardizing each environment with its own statistics aligns the aggregate moments exactly: every channel comes out at offset 0.00 and scale ratio 1.00. It also moves both channels it touches an order of magnitude further apart (Table 7.2).

Table 7.2: KS between the two grids before and after per-environment standardization, over all sampled rows. The last column is the number a node that does *not* own the channel presents to the encoder once standardized. Raw, such a row reads 0 on both grids; standardizing sends it to  $-\mu/\sigma$ , a property of the grid.  $\rho$  is shown for contrast and is excluded from the running-statistics feature set.

| Channel               | KS raw | KS standardized | “Not this type” encoding<br>bus14 vs WCCI |
|-----------------------|--------|-----------------|-------------------------------------------|
| <b>p</b>              | 0.042  | <b>0.773</b>    | −0.371 vs −0.145                          |
| <b>theta</b>          | 0.157  | <b>0.673</b>    | +0.481 vs −0.143                          |
| <b>rho</b> (excluded) | 0.053  | 0.634           | −0.658 vs −0.622                          |

The cause is the shared-column structure of Section 7.2. Subtracting a per-grid mean moves each channel’s point mass at zero to a grid-dependent location, and since three quarters of all rows are zeros for **p** and **theta**, the encoder’s single most frequent input stops meaning the same thing on the two grids — a larger mismatch than the displacement being corrected. On **theta** a row that does not own the channel reads +0.481 on bus14 and −0.143 on WCCI, so the same absence lands on opposite sides of zero.

The same defect, with agents rather than grids as the populations, affected the static action descriptor; it is diagnosed alongside Screen E in Section 6.7.1.

Two further properties rule the mechanism out independently of these numbers. Standardization is scale-invariant, so it erases any scaling applied before it — dividing by a constant and then standardizing reproduces standardizing alone to four decimals — meaning the two mechanisms cannot be combined. And running statistics begin empty and converge during training, so a transferred encoder would spend early target training reading a distribution that is itself still moving, where a deterministic scaling is fixed at construction and identical on both grids from the first step.

### 7.5.2 Physical scaling for the power channels

Physical scaling, defined in Section 5.1, divides the power channels by each environment’s own maximum generator capacity. Two properties answer the failure diagnosed above: being multiplicative it leaves zeros at zero, so the point mass stays aligned on both grids, and being fixed at construction it does not drift the way estimated moments do. The divisor is a single generator’s nameplate rating, a crude proxy for grid scale, so it is worth checking against the alternatives (Table 7.3); it is the closest of the four to the divisor that would equalize the two spreads exactly.

Table 7.3: Candidate power normalizers and the p scale ratio each would produce on asset nodes. A ratio of 1.00 would be exact; the raw ratio is 2.54.

| Divisor                                     | bus14 | WCCI  | Resulting ratio |
|---------------------------------------------|-------|-------|-----------------|
| Exact                                       | —     | —     | 1.00            |
| <b>Maximum <math>P_{\max}</math> (used)</b> | 140   | 400   | <b>0.89</b>     |
| Total $P_{\max}$                            | 540   | 2,450 | 0.56            |
| Generator count                             | 6     | 22    | 0.69            |
| Median $P_{\max}$                           | 85    | 71    | 3.04            |

### 7.5.3 The angle displacement is a gauge artifact

Scaling cannot repair the angle displacement. Absolute voltage angles are referenced to the slack bus, and because the two grids place their slack buses differently, the displacement observed in Figure 7.1 is a gauge artifact rather than a physical difference. No rescaling or recentering can fix a reference-dependent quantity; it must be replaced entirely.

The solution is to use the angle drop along each line (Section 5.1.2). As a difference, it is gauge-invariant. On the disaggregated graph, this feature lives on the transmission-line node (the most populous

type), gracefully trading a feature owned by a quarter of the rows for one owned by a third. We therefore replace absolute angles with line-angle differences in both environments.

#### 7.5.4 The corrected distribution

Table 7.4 measures both changes together, under the same protocol as Section 7.1, and the right panel of Figure 7.2 places the result back on the offset–scale diagnostic. The angle channel lands in the band occupied by  $\rho$ , and the power spread comes within 11 % of parity.

Table 7.4: Distribution shift with `gnn_angle_representation = edge_diff` and `gnn_physical_scaling = true`, against the uncorrected baseline, on the node type owning each channel. The offset column is included because scaling moves it: dividing each grid by its own constant is not offset-preserving when the two constants stand in a different relation to typical operation.

| Channel           | Uncorrected |       |       | Corrected |             |              |
|-------------------|-------------|-------|-------|-----------|-------------|--------------|
|                   | offset      | ratio | KS    | offset    | ratio       | KS           |
| <b>theta</b>      | +1.864      | 1.43  | 0.678 | replaced  |             |              |
| <b>theta_diff</b> | —           | —     | —     | −0.383    | <b>1.15</b> | <b>0.230</b> |
| <b>p</b>          | −0.158      | 2.54  | 0.201 | −0.808    | <b>0.89</b> | 0.511        |
| <b>rho</b>        | −0.260      | 1.05  | 0.141 | −0.260    | 1.05        | 0.141        |

The angle correction is unambiguous: replacing a reference-dependent quantity by a gauge-invariant one takes the worst channel in the chapter from 0.678 to 0.230, and its scale ratio stays near one throughout.

The power correction is a genuine exchange rather than a repair. Dividing by each grid’s largest generator rating brings the spread from 2.54 to 0.89, but the raw means, nearly equal at  $-0.158$ , are not preserved: `bus14` runs its generators at 39 % of its fleet maximum against WCCI’s 20 %, so a divisor taken from the largest unit lands the two grids at different normalized levels and the displacement grows to  $-0.808$ . KS rises with it, from 0.201 to 0.511. Section 7.4 argued that offsets, not scale errors, are what the architecture cannot absorb, so this moves `p` onto the axis that binds.

The exchange is still favourable: the worst displacement across all channels falls from 1.864 to 0.808, every scale ratio ends inside the band, and no channel remains outside both bounds at once. But it is an exchange, not an elimination — a divisor reflecting typical rather than peak generation would plausibly do better on both axes, and selecting one against the offset as well as the scale ratio is left open. Chapter 9 is the test of whether the residual displacement matters in practice, and Section 9.1 runs both input conditions precisely so that question can be answered.

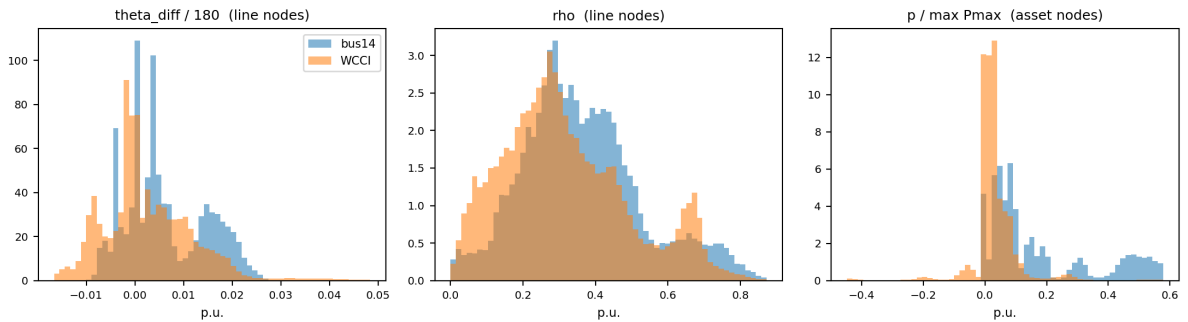


Figure 7.3: Corrected distributions, for comparison with Figure 7.1. The angle drop is centred on both grids, and the scaled power channels overlap over most of their range instead of differing by a factor of 2.5 in width.

## Chapter 7 Summary: What the Shared Encoder Reads

**Purpose:** To measure what the shared encoder actually reads on both grids, so that a deficit under transfer can be told apart from a representation reading miscalibrated inputs.

**Content:** Do-nothing rollouts on both grids (426,000 and 1,236,000 node observations) compared channel by channel on offset, scale ratio and KS. The pre-encoder’s LayerNorm absorbs a change in *scale* but not an *offset*, which moves pre-activations across the ReLU threshold — so the two axes are not equally serious.

### Findings:

- **rho** needs no correction (ratio 1.05, offset  $-0.26$ ): it is already in per-unit terms.
- **theta** is displaced by  $1.864\sigma$ , a gauge artifact of each grid’s slack bus. The gauge-invariant angle drop takes it from  $KS = 0.678$  to  $0.230$ .
- Physical scaling of **p** is an exchange, not a repair: spread  $2.54 \rightarrow 0.89$ , but offset  $-0.158 \rightarrow -0.808$ . The worst displacement overall still falls from  $1.864$  to  $0.808$ .
- Running standardization is rejected: it aligns the moments exactly yet triples KS on **p** ( $0.042 \rightarrow 0.773$ ), because a per-grid mean displaces the point mass at zero that three quarters of all rows carry.
- The action descriptor was normalized per agent, so the same action read  $0.60$  for one agent and  $0.50$  for another — which explains why the descriptor-free **f0** cells were robust in Screen E.

**Takeaway:** Deterministic physical scaling and the edge-difference angle are adopted, running standardization is not. Since the power correction trades a scale error for a displacement, both input conditions are carried forward for Chapter 9 to test.

## Chapter 8

# WCCI36: Target Environment and Controls

The experiments now move from **bus14** to the larger WCCI 2020 grid. This chapter defines the target setting and checks which models can learn on it before testing transfer.

### 8.1 Target environment

WCCI has more substations, lines and agents than **bus14** (Table 8.1). The agents still control separate areas, but their areas are not equal in size.

Table 8.1: Source and target environments. Both use topology actions, decentralized agents and the same train/test split rule.

|                | <b>bus14</b>      | <b>bus36_wcci_nomaint</b> |
|----------------|-------------------|---------------------------|
| Substations    | 14                | 36                        |
| Lines          | 20                | 59                        |
| Agents         | 3                 | 4                         |
| Chronics       | 1,004             | 2,880                     |
| Episode length | up to 8,064 steps | up to 8,062 steps         |
| Maintenance    | none              | disabled (Section 8.1.1)  |

#### 8.1.1 Removing maintenance

The WCCI environment normally includes scheduled line maintenance, while **bus14** does not. Maintenance is disabled here so that a transferred policy is not tested on a hazard absent from its source environment. This also keeps the observation spaces aligned.

On the full WCCI test split, removing maintenance raises do-nothing mean survival from 27.3% to 57.3% (Table 8.2). The change is large enough that maintenance-on and maintenance-off results should not be compared directly.

Table 8.2: Do-nothing performance on the full WCCI test split. The 576 chronics are the same in both rows.

| Setting              | Mean survival | Median survival | Complete episodes |
|----------------------|---------------|-----------------|-------------------|
| Maintenance enabled  | 27.3%         | 12.1%           | 39/576 (6.8%)     |
| Maintenance disabled | 57.3%         | 100.0%          | 303/576 (52.6%)   |

### 8.1.2 Reduced action spaces

The full decentralized action space contains almost 67,000 actions and is strongly unbalanced across agents. The candidate-action models are therefore evaluated with reduced spaces of  $k \in \{32, 64, 128, 256\}$ . Here,  $k$  is a cap: agents with fewer valid actions keep their complete list.

Table 8.3: Per-agent action counts on `bus36_wcci_nomaint`.

| Agent                | Substations | Full list | At $k = 256$ |
|----------------------|-------------|-----------|--------------|
| <code>agent_0</code> | 5           | 77        | 77           |
| <code>agent_1</code> | 5           | 65,642    | 256          |
| <code>agent_2</code> | 9           | 127       | 127          |
| <code>agent_3</code> | 17          | 1,119     | 256          |
| Total                | 36          | 66,965    | 716          |

A one-step greedy policy is used only as a reference for what each reduced space contains. Its difficult-cohort survival rises from 33.1% at  $k = 32$  to 94.9% at  $k = 256$ . A larger action list therefore gives more useful choices, but it also gives the learned policy a harder selection problem. This trade-off is measured in Chapter 9.

## 8.2 Evaluation protocol

All reported transfer evaluations use the same 50 WCCI chronics. Do nothing finishes 26 of them. These 26 form the *easy* cohort; the remaining 24 form the *difficult* cohort. Do-nothing mean survival is 56.0% overall and 8.34% on the difficult cohort.

Reporting both values separates two goals: preserving cases that already need no action and improving cases where inaction fails. The number of difficult chronics completed is reported as *rescues*. The one-step greedy policy is a strong reference, not an upper bound, because it queries the simulator at every step.

## 8.3 Inputs used for transfer

Chapter 7 found two input changes that make the graph features more comparable across grids. Power is divided by the largest installed generator on each grid, and voltage angle is represented by the difference across each line. Running standardization is not used. The study keeps both the original inputs (NL) and these corrected inputs (NLS) so that the effect of preprocessing can be measured.

## 8.4 Controls trained on WCCI

Figure 8.1 compares three controls trained directly on WCCI: a flat MLP, fixed-list GNN actors from the earlier screens, and the candidate-action scorer. Final checkpoints are used and the local heuristic is disabled.

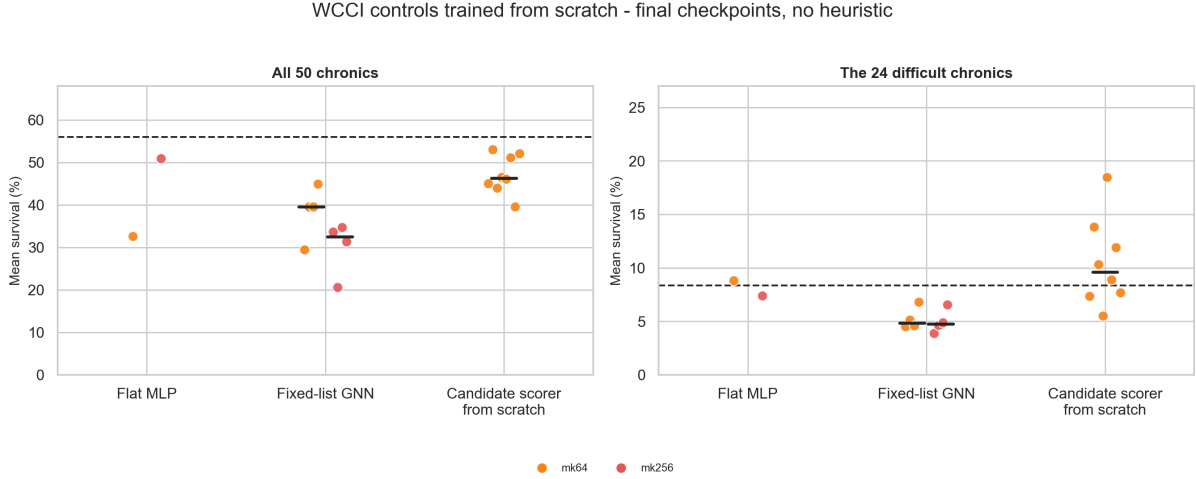


Figure 8.1: WCCI controls trained from scratch. Each point is one model; colours show the reduced action-space size. Short black lines show medians when several models share a setting. The dashed line is do nothing.

The flat MLP and fixed-list GNNs remain below do nothing overall. The fixed-list GNNs also stay below it on the difficult cohort and produce no rescues. The candidate scorer is more promising: at  $k = 64$ , its median is 9.61% on the difficult cohort and its best cell reaches 18.46%. However, its overall score stays below do nothing because it loses some easy chronics.

The candidate scorer is also the only graph actor here whose complete policy can cross grids. Its shared scorer is applied to each candidate, so its output size does not depend on the number of agents or actions. A fixed-list head has one output per action index and cannot be reused unchanged on WCCI.

Exact flat-MLP and fixed-list GNN results are given in Appendix A.9.

#### Chapter 8 Summary

WCCI is evaluated without maintenance and with reduced action spaces. Results are split into easy and difficult chronics so that preserving safe cases is not confused with solving hard ones. Among the graph actors, only candidate-action scoring is both useful on the difficult cohort and fully portable across grids. Chapter 9 therefore studies this actor.

## Chapter 9

# Zero-Shot Transfer and Target Adaptation

This chapter tests whether the shared candidate-action actor learned on **bus14** can control WCCI. It first evaluates the actor without target training, then studies small policy changes and fine-tuning.

### 9.1 Comparison rules

The actor varies along four binary choices: original or corrected inputs, mean or typed-mean candidate pooling, action descriptor off or on, and a shared or dedicated idle-action head. Table 9.1 defines the three training routes.

Table 9.1: Training routes used in the transfer study.

| Route        | Initial actor             | WCCI training         |
|--------------|---------------------------|-----------------------|
| Zero-shot    | Shared <b>bus14</b> actor | None                  |
| From scratch | Random weights            | 15M target steps      |
| Fine-tuned   | Shared <b>bus14</b> actor | About 5M target steps |

Final checkpoints are used for actors trained on WCCI. Zero-shot evaluation uses the source checkpoint selected on **bus14**. The optional heuristic blocks an agent’s action when its local maximum line loading is below 0.95. Every comparison uses the same 50 chronics and seed 0.

### 9.2 What controls zero-shot transfer?

Figure 9.1 separates three questions. The top panel gives mean survival over all 50 chronics. The lower-left panel restricts the score to the 24 difficult chronics. The lower-right panel then adjusts that difficult-cohort score to the greedy reference available at the same  $k$ :

$$R_{k,\pi} = 100 \frac{S_\pi - S_{\text{DN}}}{S_{\text{greedy},k} - S_{\text{DN}}}. \quad (9.1)$$

Here 0% means do-nothing performance and 100% means the greedy result at the same action count. A negative value is worse than do nothing. This avoids crediting a policy simply because it receives a larger candidate list.

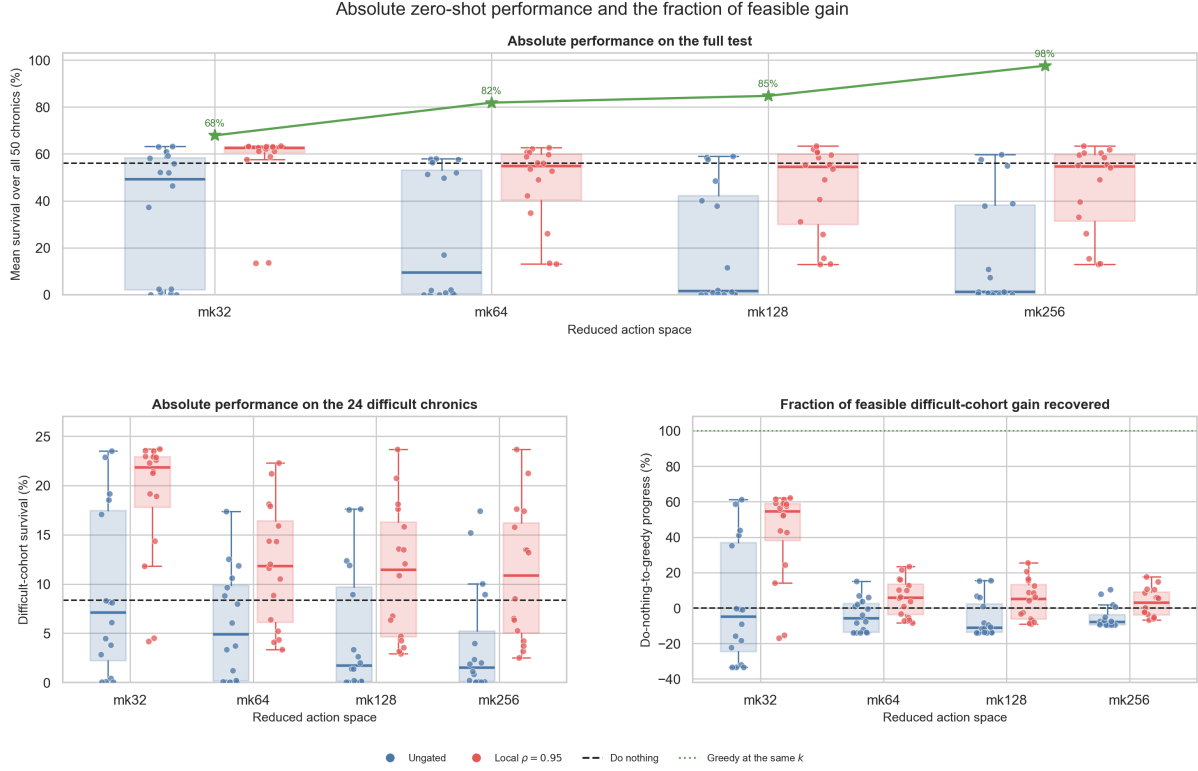


Figure 9.1: Absolute and feasible-relative zero-shot performance. Each box contains all 16 NL/NLS architectures. Green stars give the greedy full-test score and dashed lines give do nothing. The lower-right score measures progress from do nothing to the greedy reference at the same action-space size.

The distributions show that absolute and feasible-relative performance answer different questions. With the heuristic, median full-test survival is 62.5% at  $k = 32$  and about 54.5% for the larger action spaces; median difficult-cohort survival falls from 21.9% to roughly 11%. Relative to the same- $k$  greedy reference, the median recovers 54.6% of the available gain at  $k = 32$ , but only 5.8%, 5.2% and 2.9% at  $k = 64, 128, 256$ . The gate helps, but the actors exploit a smaller fraction of what becomes feasible as the action set grows. Ungated median performance remains below do nothing at every  $k$ .

Figure 9.2 then separates the architecture effects by gate condition. Typed-mean pooling is the clearest positive effect: +5.4 points ungated and +8.7 points gated. Physical scaling adds about one point in both cases. The descriptor is small on average, while the dedicated idle head is negative ungated and nearly neutral gated.

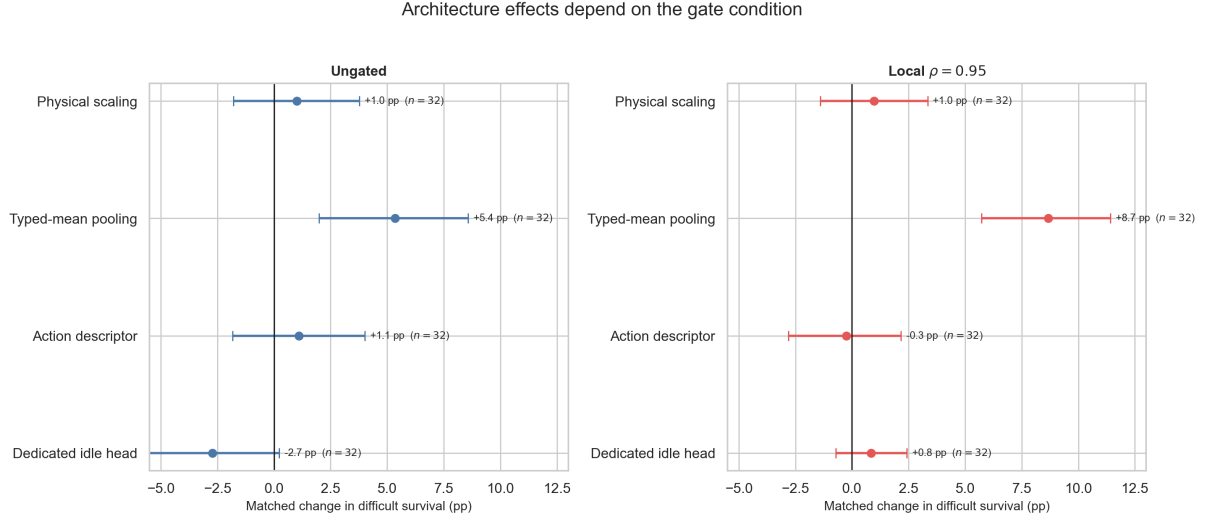


Figure 9.2: Matched architecture effects on the difficult cohort, shown separately for ungated actors and the local- $\rho = 0.95$  heuristic. Positive values improve survival. Error bars are bootstrap intervals over matched architecture cells;  $n$  is the number of available pairs.

These comparisons use one seed. They identify useful directions but do not establish a final architecture ranking.

### 9.3 Policy changes and the local heuristic

Two changes are tested around the zero-shot actor. The first uses the global-maximum readout and exact action-delta encoder of Figure 5.10. The second uses an adaptive intervention budget. Figure 9.3 reports their matched effect and the effect of the local- $\rho$  heuristic.

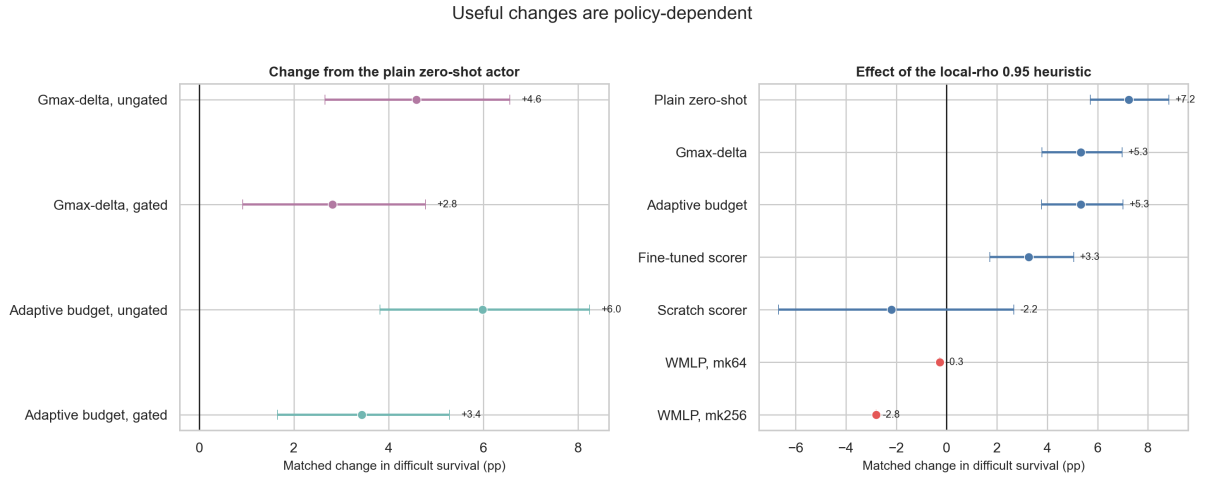


Figure 9.3: Matched changes on the difficult cohort. Positive values are better. The Gmax-delta and adaptive-budget blocks are compared with the same plain zero-shot architectures.

Without the heuristic, Gmax-delta adds 4.6 points and the adaptive budget adds 6.0 points. Across gate conditions, the heuristic adds 7.2 points to the plain zero-shot actor and 5.3 points to both modified actors. It is not a universal safety layer: it reduces the scratch scorer by 2.2 points and the flat MLP by up to 2.8 points. The gate must therefore be checked with each policy.

The plain block contains all 64 matched architecture- $k$  pairs. Gmax-delta and the adaptive budget each contain 60 pairs because their complete designs use 15 architectures rather than 16.

## 9.4 Does target training help?

Figure 9.4 compares the same eight NLS architectures at  $k = 64$ . The left panel tracks each architecture across zero-shot, scratch and fine-tuned routes. The right panel shows the main trade-off: survival on the difficult cohort against the number of easy chronics kept.

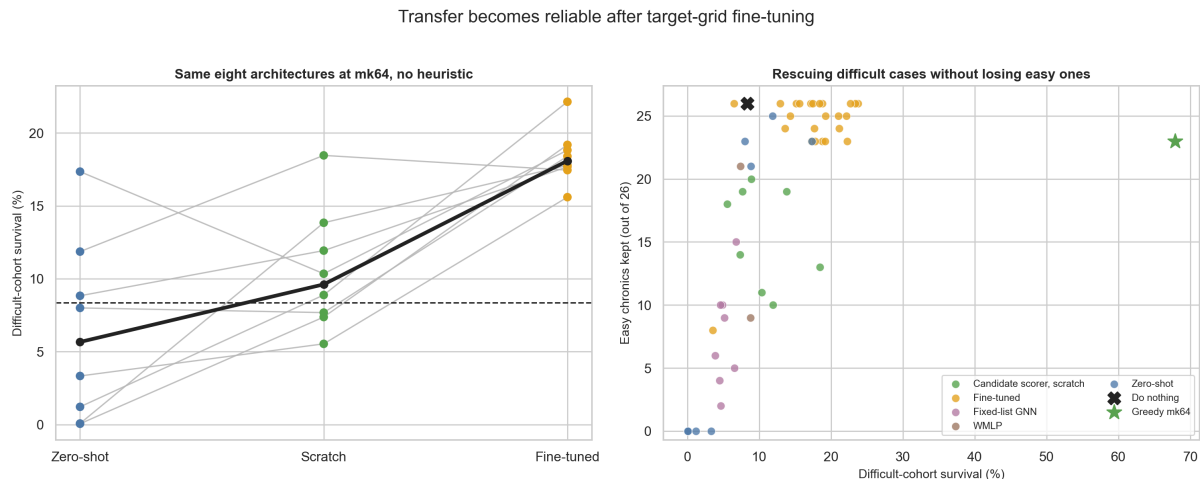


Figure 9.4: Effect of target-grid training. Grey lines join the same architecture in the left panel. The black cross and green star in the right panel are do nothing and the greedy  $k = 64$  reference.

Fine-tuning is the most consistent route. It beats training from scratch on seven of the eight difficult-cohort comparisons, with a mean gain of 7.9 points. It also beats zero-shot transfer in all eight cells, by 12.0 points on average. The median difficult-cohort survival rises from 5.7% zero-shot to 9.6% from scratch and 18.1% after fine-tuning.

The right panel adds an important qualification. Some actors improve difficult chronics while losing cases that do nothing already solves. The useful region is therefore the upper-right corner, not simply the highest difficult-cohort score. The target-trained controls remain far below the greedy policy, showing that much more performance is available in the action space.

## 9.5 Limits of the study

The study uses one training seed and 50 evaluation chronics. Bootstrap intervals describe variation across architecture cells, not across independent training runs. Some campaigns also use different target-training budgets. The results support comparisons inside the matched blocks, but small gaps between unrelated blocks should not be overinterpreted. Full matrices and exact control tables are provided in Appendix A.9.

### Chapter 9 Summary

Zero-shot transfer works best with a small candidate set and typed-mean pooling, but it is not yet reliable by itself. In the complete plain matrix, the local heuristic improves 58 of 64 cells, although its benefit falls as the action set grows and remains policy-dependent. The strongest result is target fine-tuning: it improves the transferred actor in all eight matched comparisons against zero-shot and usually beats the same architecture trained from scratch.

# Appendix A

## Supplementary Results

This appendix collects detailed result tables and diagnostic figures that support the experiment and transfer chapters. The main chapters retain the compact comparisons needed for the discussion.

### A.1 Sparse Intervention Control

The joint survival/action-0 comparison is retained in Figure 4.1; the tables below provide the exact aggregate and per-agent values.

The following tables report average full-test episodic survival and action-0 fractions over the three evaluated seeds. The all-agent value is the arithmetic mean of the three per-agent fractions. The do-nothing row is a reference policy that always selects action 0.

Table A.1: Full-test summary for evaluation-time rho heuristics.

| Run type                                   | Avg. episodic survival (%) | Avg. action-0 all agents | Avg. action-0 agent 0 | Avg. action-0 agent 1 | Avg. action-0 agent 2 |
|--------------------------------------------|----------------------------|--------------------------|-----------------------|-----------------------|-----------------------|
| Do-nothing policy                          | 20.25                      | 1.000                    | 1.000                 | 1.000                 | 1.000                 |
| Baseline flat policy                       | 93.67                      | 0.434                    | 0.554                 | 0.385                 | 0.363                 |
| Global rho heuristic, $\rho_{safe} = 0.90$ | 98.36                      | 0.936                    | 0.933                 | 0.936                 | 0.940                 |
| Local rho heuristic, $\rho_{safe} = 0.90$  | 97.07                      | 0.968                    | 0.992                 | 0.985                 | 0.928                 |

Table A.2: Full-test summary for intervention-gated actors.

| Run type                               | Avg. episodic survival (%) | Avg. action-0 all agents | Avg. action-0 agent 0 | Avg. action-0 agent 1 | Avg. action-0 agent 2 |
|----------------------------------------|----------------------------|--------------------------|-----------------------|-----------------------|-----------------------|
| Do-nothing policy                      | 20.25                      | 1.000                    | 1.000                 | 1.000                 | 1.000                 |
| Baseline flat policy                   | 93.67                      | 0.434                    | 0.554                 | 0.385                 | 0.363                 |
| Intervention gate, final-action MAP    | 89.82                      | 0.802                    | 0.826                 | 0.889                 | 0.690                 |
| Intervention gate, hierarchical greedy | 93.66                      | 0.395                    | 0.066                 | 0.844                 | 0.276                 |

Table A.3: Full-test summary for flat-policy fixed-penalty runs.

| Run type                               | Avg. episodic survival (%) | Avg. action-0 all agents | Avg. action-0 agent 0 | Avg. action-0 agent 1 | Avg. action-0 agent 2 |
|----------------------------------------|----------------------------|--------------------------|-----------------------|-----------------------|-----------------------|
| Do-nothing policy                      | 20.25                      | 1.000                    | 1.000                 | 1.000                 | 1.000                 |
| Baseline flat policy                   | 93.67                      | 0.434                    | 0.554                 | 0.385                 | 0.363                 |
| Flat policy, $\lambda_{fixed} = 0.000$ | 98.23                      | 0.494                    | 0.524                 | 0.463                 | 0.496                 |
| Flat policy, $\lambda_{fixed} = 0.001$ | 96.75                      | 0.486                    | 0.642                 | 0.401                 | 0.416                 |
| Flat policy, $\lambda_{fixed} = 0.003$ | 93.97                      | 0.575                    | 0.691                 | 0.346                 | 0.688                 |
| Flat policy, $\lambda_{fixed} = 0.010$ | 97.44                      | 0.905                    | 0.907                 | 0.954                 | 0.853                 |

Table A.4: Full-test summary for gated-policy fixed-penalty runs.

| Run type                                | Avg. episodic survival (%) | Avg. action-0 all agents | Avg. action-0 agent 0 | Avg. action-0 agent 1 | Avg. action-0 agent 2 |
|-----------------------------------------|----------------------------|--------------------------|-----------------------|-----------------------|-----------------------|
| Do-nothing policy                       | 20.25                      | 1.000                    | 1.000                 | 1.000                 | 1.000                 |
| Baseline flat policy                    | 93.67                      | 0.434                    | 0.554                 | 0.385                 | 0.363                 |
| Gated policy, $\lambda_{fixed} = 0.000$ | 78.51                      | 0.787                    | 0.747                 | 0.949                 | 0.665                 |
| Gated policy, $\lambda_{fixed} = 0.001$ | 73.82                      | 0.862                    | 0.956                 | 0.832                 | 0.799                 |
| Gated policy, $\lambda_{fixed} = 0.003$ | 94.59                      | 0.887                    | 0.934                 | 0.929                 | 0.799                 |
| Gated policy, $\lambda_{fixed} = 0.010$ | 85.69                      | 0.974                    | 0.990                 | 0.984                 | 0.947                 |

Table A.5: Full-test summary for adaptive intervention budget runs.

| Run type                                                  | Avg. episodic survival (%) | Avg. action-0 all agents | Avg. action-0 agent 0 | Avg. action-0 agent 1 | Avg. action-0 agent 2 |
|-----------------------------------------------------------|----------------------------|--------------------------|-----------------------|-----------------------|-----------------------|
| Do-nothing policy                                         | 20.25                      | 1.000                    | 1.000                 | 1.000                 | 1.000                 |
| Baseline flat policy                                      | 93.67                      | 0.434                    | 0.554                 | 0.385                 | 0.363                 |
| Flat local-safe AIB, $d = 0.20$                           | 98.31                      | 0.978                    | 0.980                 | 0.989                 | 0.966                 |
| Flat local-safe AIB, $d = 0.10$                           | 98.39                      | 0.979                    | 0.992                 | 0.995                 | 0.949                 |
| Flat local-safe AIB, $d = 0.35$                           | 96.61                      | 0.934                    | 0.959                 | 0.947                 | 0.896                 |
| Gated local-safe AIB, $d = 0.20$ ,<br>hierarchical greedy | 33.87                      | 0.999                    | 1.000                 | 1.000                 | 0.998                 |
| Flat non-idle AIB, $d = 0.20$                             | 95.08                      | 0.986                    | 0.997                 | 0.992                 | 0.970                 |

Table A.6: Full-test summary for the teacher-student distillation runs.

| Run type                                                  | Avg. episodic survival (%) | Avg. action-0 all agents | Avg. action-0 agent 0 | Avg. action-0 agent 1 | Avg. action-0 agent 2 |
|-----------------------------------------------------------|----------------------------|--------------------------|-----------------------|-----------------------|-----------------------|
| Do-nothing policy                                         | 20.25                      | 1.000                    | 1.000                 | 1.000                 | 1.000                 |
| Teacher: MAPPO with local rho<br>heuristic                | 97.07                      | 0.968                    | 0.992                 | 0.985                 | 0.928                 |
| Student BC, balanced non-idle 0.50,<br>weight 5, aux 0.50 | 92.14                      | 0.939                    | 0.974                 | 0.925                 | 0.919                 |
| Student BC, balanced non-idle 0.20,<br>weight 3, aux 0.25 | 90.76                      | 0.957                    | 0.982                 | 0.948                 | 0.940                 |

## A.2 Screen A: Encoder Depth and Width

Table A.7 gives the complete endpoint grid, while Table A.8 summarizes each factor across the other. Figures A.1 and A.2 provide the corresponding training-curve and parameter-count diagnostics.

Table A.7: Screen A. Full-test survival of the best checkpoint, one seed per cell, by message-passing depth and encoder width. Screen B reuses mp2\_h128 as its all-visible/mean reference.

| Layers $K$ | Encoder width $d_h$ |              |       |       |
|------------|---------------------|--------------|-------|-------|
|            | 16                  | 32           | 64    | 128   |
| 1          | 84.02               | 92.72        | 65.52 | 92.96 |
| 2          | 92.12               | 91.37        | 53.38 | 94.72 |
| 3          | 89.11               | <b>97.80</b> | 65.57 | 86.48 |

Table A.8: Screen A. Marginal effect of each factor, averaged over the other. Four runs per depth, three per width.

| Level                        | Mean (%) | Min (%) | Max (%) |
|------------------------------|----------|---------|---------|
| <i>Message-passing depth</i> |          |         |         |
| $K = 1$                      | 83.80    | 65.52   | 92.96   |
| $K = 2$                      | 82.90    | 53.38   | 94.72   |
| $K = 3$                      | 84.74    | 65.57   | 97.80   |
| <i>Encoder width</i>         |          |         |         |
| $d_h = 16$                   | 88.42    | 84.02   | 92.12   |
| $d_h = 32$                   | 93.97    | 91.37   | 97.80   |
| $d_h = 64$                   | 61.49    | 53.38   | 65.57   |
| $d_h = 128$                  | 91.39    | 86.48   | 94.72   |

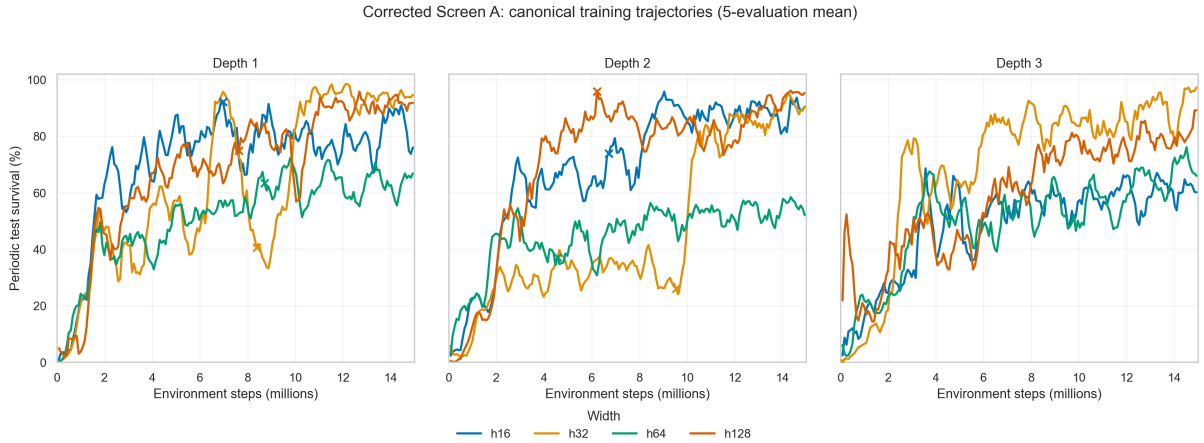


Figure A.1: Screen A. Test episodic survival during training, one panel per message-passing depth, with the four widths compared inside each panel. Restarted runs use the canonical history in which the continuation replaces the overlapping post-checkpoint branch.

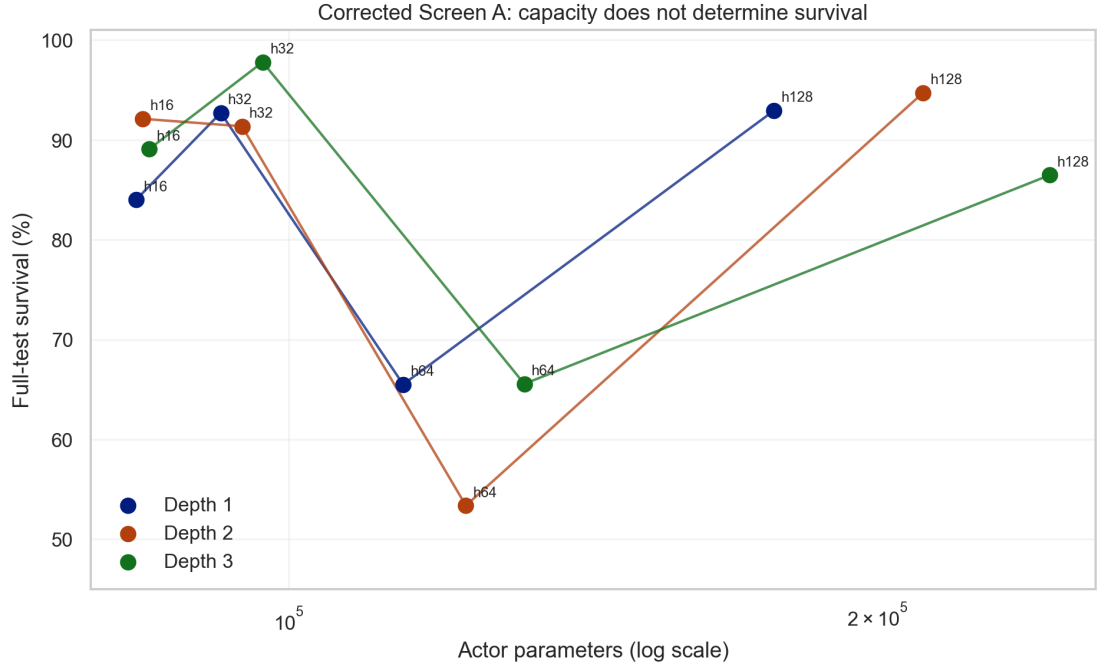


Figure A.2: Screen A. Full-test survival against actor parameter count on a logarithmic axis, joined within each depth. The grid spans 83.6k to 244.9k actor parameters, a factor of three.

### A.3 Screen B: Graph Readout Scope and Reducer

Figures A.3 and A.4 complement the endpoint comparison in Table 6.5 with the eight training trajectories and the full-test interaction between readout scope and reducer.

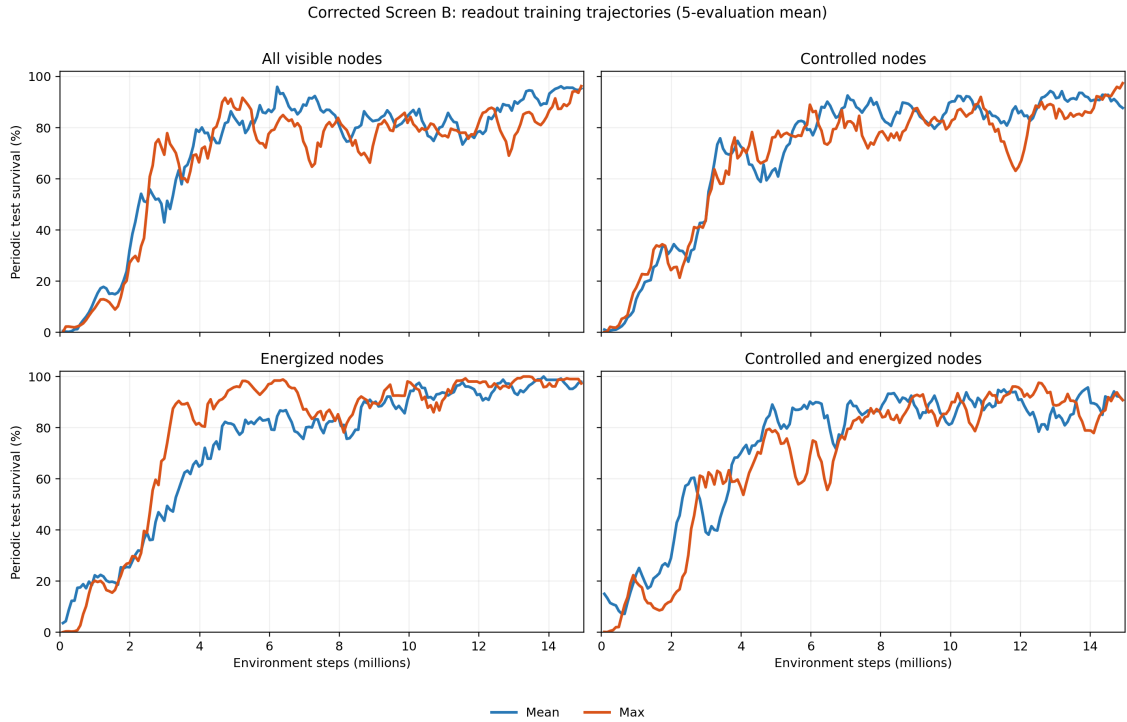


Figure A.3: Screen B. Smoothed training episodic survival curves by readout scope and reducer (seed 0).

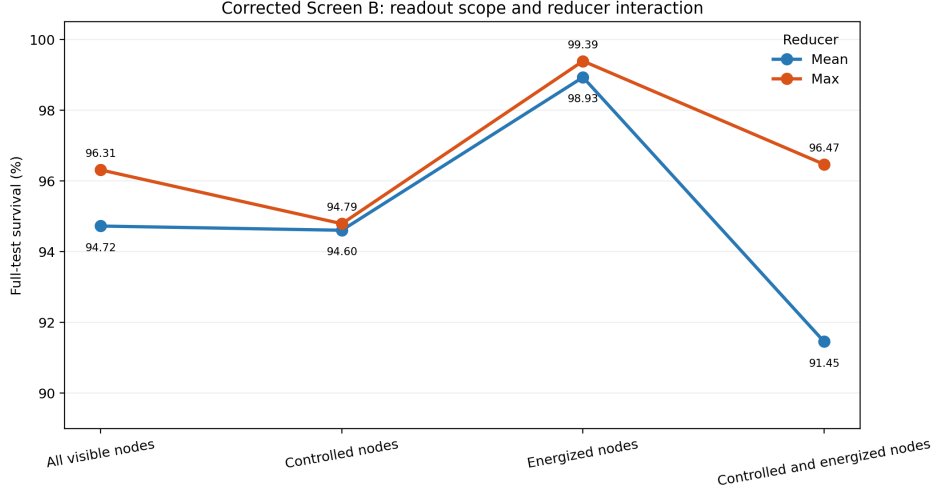


Figure A.4: Screen B. Full-test survival of each best checkpoint. Connecting the two reducers within a scope makes the scope-reducer interaction visible; energized-only readout leads under both reductions.

## A.4 Screen C: Structural Augmentations of the Busbar Graph

Tables A.9 and A.10 and Figures A.5 and A.6 complement the factorial cube retained in Section 6.5 with the complete results.

Table A.9: Screen C. Cells ranked by full-test survival, with the difference from the unaugmented `e0n0v0` baseline.

| Cell                | Mean (%) | Std (pp) | Min (%) | Max (%) | $\Delta$ baseline (pp) |
|---------------------|----------|----------|---------|---------|------------------------|
| <code>e0n0v0</code> | 97.31    | 1.22     | 96.55   | 98.72   | 0.00                   |
| <code>e1n0v0</code> | 96.20    | 3.94     | 91.66   | 98.68   | -1.11                  |
| <code>e0n0v1</code> | 95.63    | 6.19     | 88.52   | 99.82   | -1.68                  |
| <code>e0n1v0</code> | 93.51    | 6.39     | 86.72   | 99.40   | -3.80                  |
| <code>e1n1v0</code> | 90.01    | 4.92     | 85.42   | 95.19   | -7.30                  |
| <code>e1n0v1</code> | 89.92    | 9.42     | 80.48   | 99.33   | -7.39                  |
| <code>e0n1v1</code> | 68.91    | 5.77     | 62.59   | 73.91   | -28.40                 |
| <code>e1n1v1</code> | 66.11    | 33.95    | 27.86   | 92.68   | -31.20                 |

Table A.10: Screen C. Main effect of each factor (12 runs per level) and the four single-factor flips of the cube, each one enabling that factor with the other two held fixed. **Spread** is the range of those four flips.

| Factor                      | Off (%) | On (%) | Effect (pp) | Flips (pp)               | Mean flip (pp) | Spread (pp) |
|-----------------------------|---------|--------|-------------|--------------------------|----------------|-------------|
| <i>e</i> (substation edges) | 88.84   | 85.56  | -3.28       | -1.1, -5.7, -3.5, -2.8   | -3.28          | 4.60        |
| <i>v</i> (virtual readout)  | 94.26   | 80.14  | -14.11      | -1.7, -24.6, -6.3, -23.9 | -14.11         | 22.92       |
| <i>n</i> (summary nodes)    | 94.77   | 79.63  | -15.13      | -3.8, -26.7, -6.2, -23.8 | -15.13         | 22.92       |

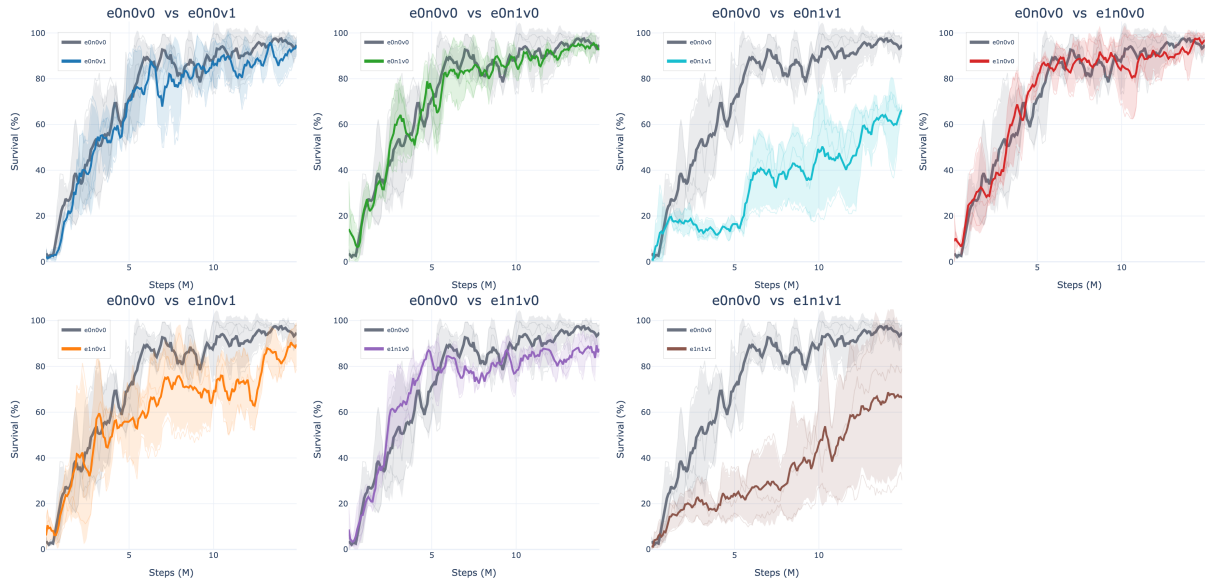


Figure A.5: Screen C. Test episodic survival during training, one panel per augmented cell against the unaugmented **e0n0v0** baseline in grey. Thin lines are the three seeds, the thick line is their mean, and the band is one standard deviation.



Figure A.6: Screen C. The same eight cells as a heatmap of the full-test survival of each run's best checkpoint, with the three individual seed values printed under each cell mean.

## A.5 Screen D: Generator and Load Message Direction

Figure A.7 shows the training trajectories behind the endpoint heatmap retained in Section 6.6, while Table A.11 gives the complete ranking.

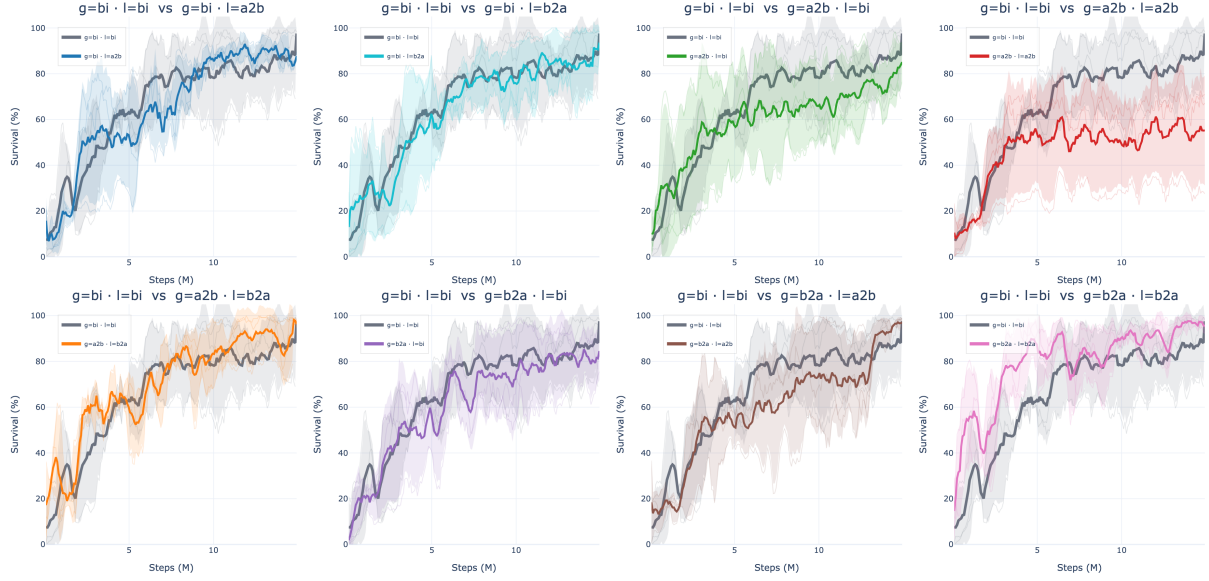


Figure A.7: Screen D. Test episodic survival during training, one panel per direction pair against the bidirectional baseline in grey. Thin lines are the three seeds, the thick line is their mean, and the band is one standard deviation.

Table A.11: Screen D. Direction pairs ranked by full-test survival, with the difference from the bi/bi baseline.

| Generator / load | Mean (%) | Std (pp) | Min (%) | Max (%) | $\Delta$ baseline (pp) |
|------------------|----------|----------|---------|---------|------------------------|
| b2a / b2a        | 97.94    | 2.20     | 95.40   | 99.25   | +5.30                  |
| b2a / a2b        | 97.43    | 1.42     | 96.32   | 99.03   | +4.79                  |
| a2b / b2a        | 94.47    | 7.53     | 85.78   | 99.03   | +1.83                  |
| bi / bi          | 92.65    | 8.24     | 83.21   | 98.40   | 0.00                   |
| bi / a2b         | 91.35    | 5.54     | 86.52   | 97.40   | -1.30                  |
| bi / b2a         | 87.76    | 11.04    | 77.64   | 99.53   | -4.89                  |
| b2a / bi         | 84.93    | 13.01    | 70.11   | 94.46   | -7.72                  |
| a2b / bi         | 83.16    | 12.75    | 72.27   | 97.19   | -9.49                  |
| a2b / a2b        | 56.41    | 22.12    | 30.88   | 69.60   | -36.24                 |

## A.6 Screen E: Candidate-Action Scoring

Figures A.8 and A.9 complement the full-test table retained in Section 6.7 with the training trajectories and endpoint ordering.

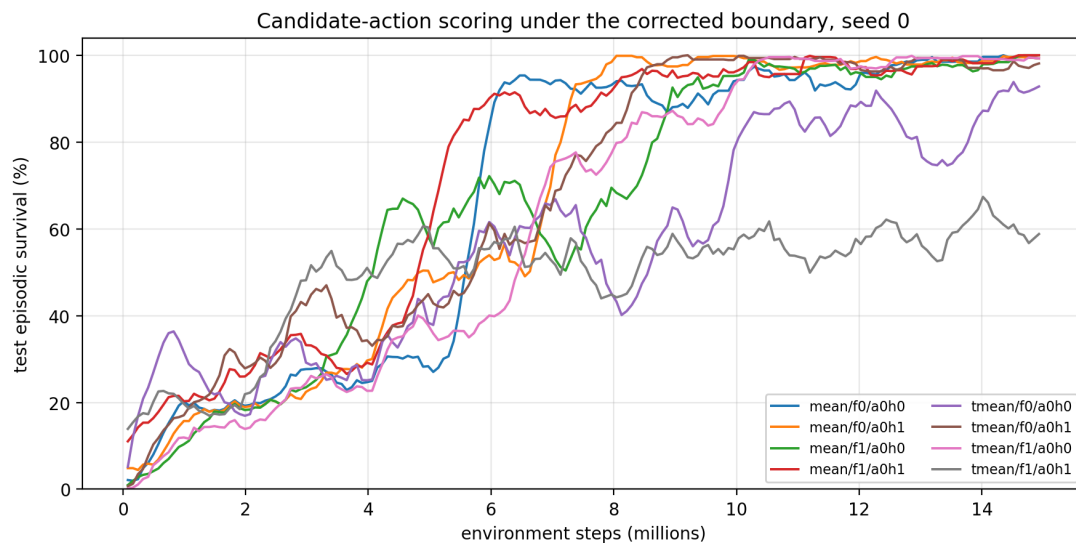


Figure A.8: Screen E. Test episodic survival during training, smoothed over nine evaluations. Six of the eight cells converge to the top of the range and stay there; `typed_mean/f1/a0h1` never does.

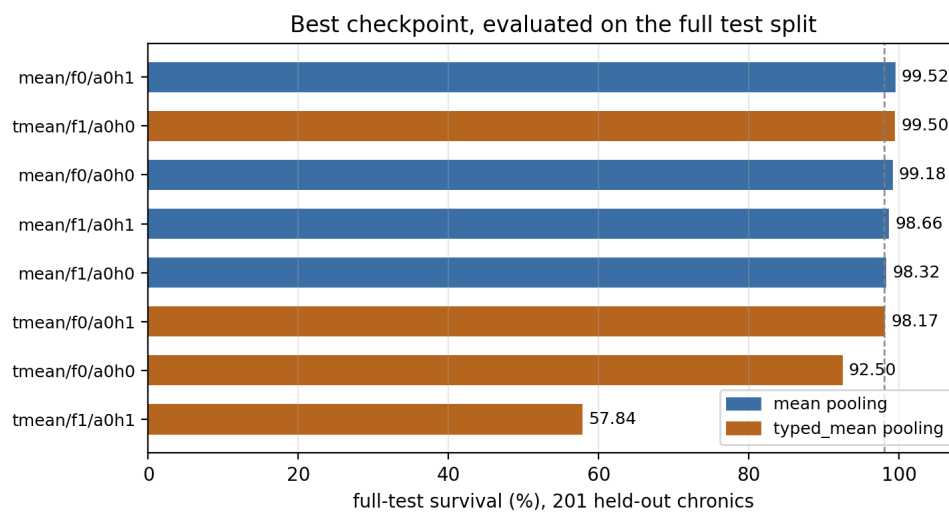


Figure A.9: Screen E. Full-test survival of each best checkpoint. The dashed line marks 98%.

## A.7 Encoder Input Distributions

Chapter 7 retains the compact shift table (Table 7.1) and the consolidated distribution panel (Figure 7.1). This section provides the raw moments behind those statistics and the tail behaviour that linear axes compress. All values are measured on the disaggregated line-node graph under the protocol of Section 7.1.

Table A.12: Raw moments behind Table 7.1, with the Wasserstein distance. “Owning” restricts each channel to the node type that carries it.  $W_1/\sigma_p$  tracks the offset almost exactly on the displaced channel, which is why the main table omits it: for distributions that differ mainly by a shift the two statistics carry the same information.

| Channel      | Rows   | bus14  |        | WCCI   |        | offset | ratio | $W_1/\sigma_p$ |
|--------------|--------|--------|--------|--------|--------|--------|-------|----------------|
|              |        | mean   | sd     | mean   | sd     |        |       |                |
| <b>p</b>     | all    | 6.538  | 17.615 | 5.461  | 37.718 | −0.037 | 2.141 | 0.204          |
|              | owning | 27.306 | 26.998 | 19.067 | 68.613 | −0.158 | 2.541 | 0.485          |
| <b>theta</b> | all    | −1.236 | 2.570  | 0.305  | 2.127  | +0.653 | 0.827 | 0.653          |
|              | owning | −5.162 | 2.707  | 1.064  | 3.871  | +1.864 | 1.430 | 1.864          |
| <b>rho</b>   | all    | 0.133  | 0.203  | 0.116  | 0.187  | −0.088 | 0.922 | 0.088          |
|              | owning | 0.364  | 0.168  | 0.319  | 0.176  | −0.260 | 1.050 | 0.261          |

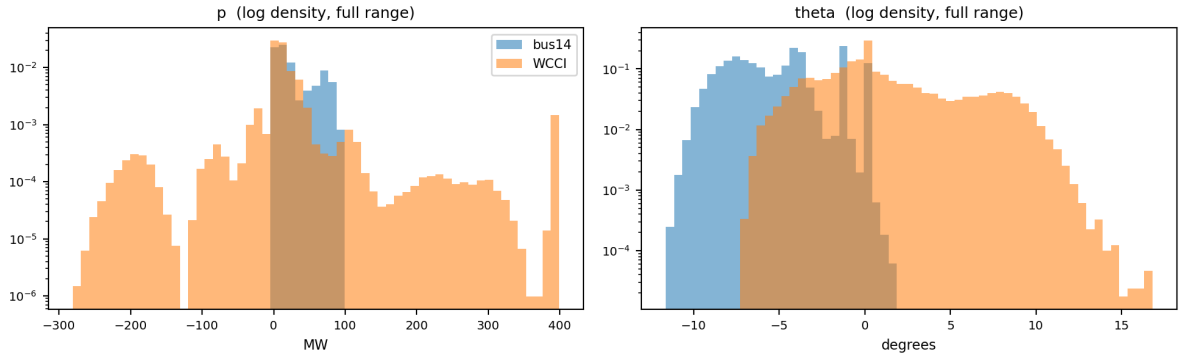


Figure A.10: Full range of the two asset channels on log density. WCCI carries a generator pinned near its 399 MW rating and a load region reaching −281 MW, neither of which occurs anywhere on **bus14**: a transferred encoder has no calibration for either. The angle panel shows the displacement rather than a difference in width.

## A.8 Deterministic Feature Scales

Section 5.1 gives the two references that carry physical content, the power base and the angular scale. The complete list of divisors, including the discrete counters, is below.

Table A.13: Deterministic physical scales used for continuous graph features. The power base is the largest installed generator rating on the grid; the remaining discrete channels use the corresponding Grid2Op parameter bound.

| Features                                                                                           | Scale $s_f$                    | Source                              |
|----------------------------------------------------------------------------------------------------|--------------------------------|-------------------------------------|
| <code>gen_p</code> , <code>load_p</code> , <code>p</code>                                          | Power base in MW               | Largest installed generator rating. |
| <code>gen_theta</code> , <code>load_theta</code> , <code>theta</code> ,<br><code>theta_diff</code> | 180 degrees                    | Fixed angular scale.                |
| <code>time_before_cooldown_sub</code>                                                              | Maximum substation<br>cooldown | Grid2Op environment parameter.      |
| <code>time_before_cooldown_line</code>                                                             | Maximum line cooldown          | Grid2Op environment parameter.      |
| <code>timestep_overflow</code>                                                                     | Allowed overflow duration      | Grid2Op environment parameter.      |
| Maintenance time and duration                                                                      | Episode horizon                | Grid2Op chronic horizon.            |

## A.9 WCCI Result Atlas

This section gives the detailed results behind Chapters 8 and 9. Every survival value uses the same 50 WCCI chronics and the easy/difficult split defined in Section 8.2.

### A.9.1 Evaluation coverage

Figure A.11 shows which result files are currently available. It is included so that incomplete blocks are visible rather than silently removed.

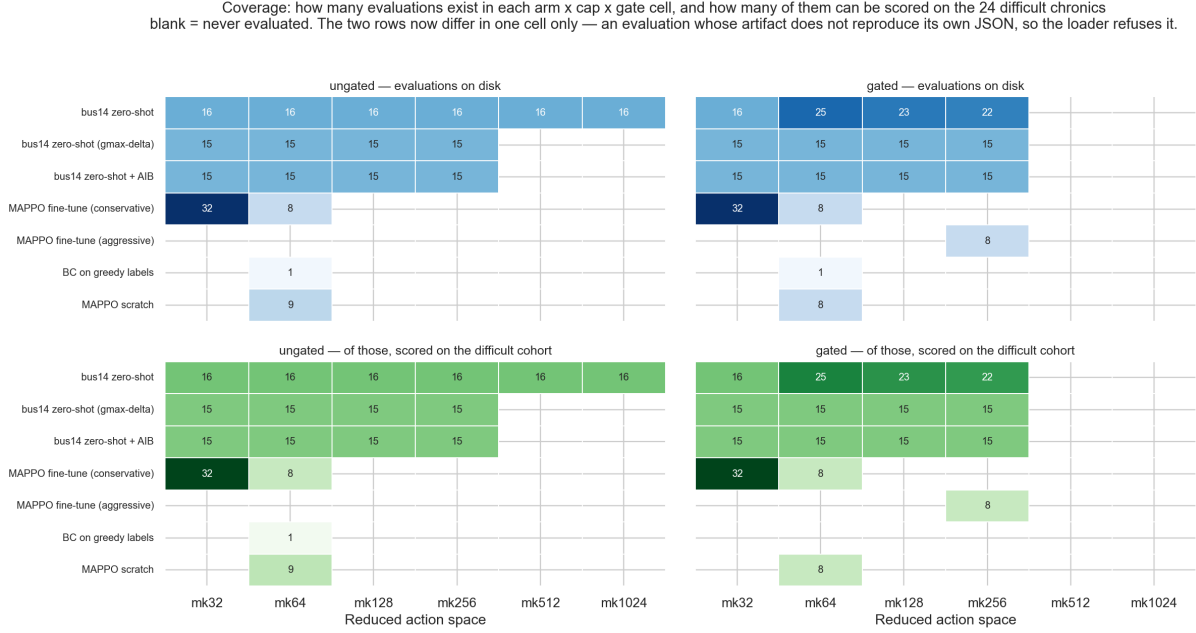


Figure A.11: Coverage of the WCCI evaluation matrix at the time of writing.

The core zero-shot block is complete: 16 architectures, four action-space sizes, and both gate conditions. The Gmax-delta and adaptive-budget blocks are also complete within their 15-architecture designs.

## A.9.2 Zero-shot matrices

Zero-shot transfer, shared-candidate 'cas\_hl' backbone — 16 architectures x 4 reduced action spaces  
survival on the 24 difficult chronics (%)

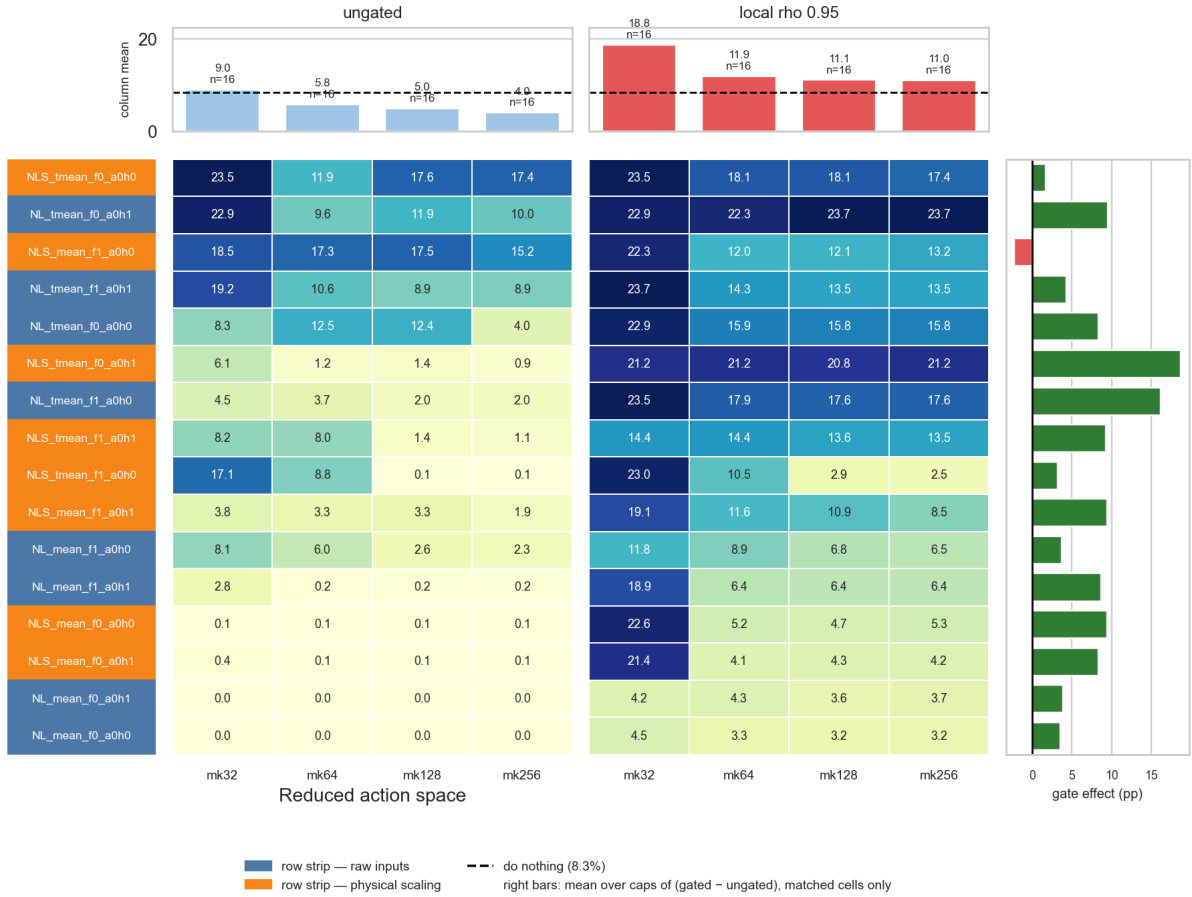


Figure A.12: Zero-shot difficult-cohort survival across input, pooling, descriptor, idle-head and action-space choices.

AIB zero-shot transfer — 15 architectures x 4 reduced action spaces, both gate conditions complete survival on the 24 difficult chronics (%)

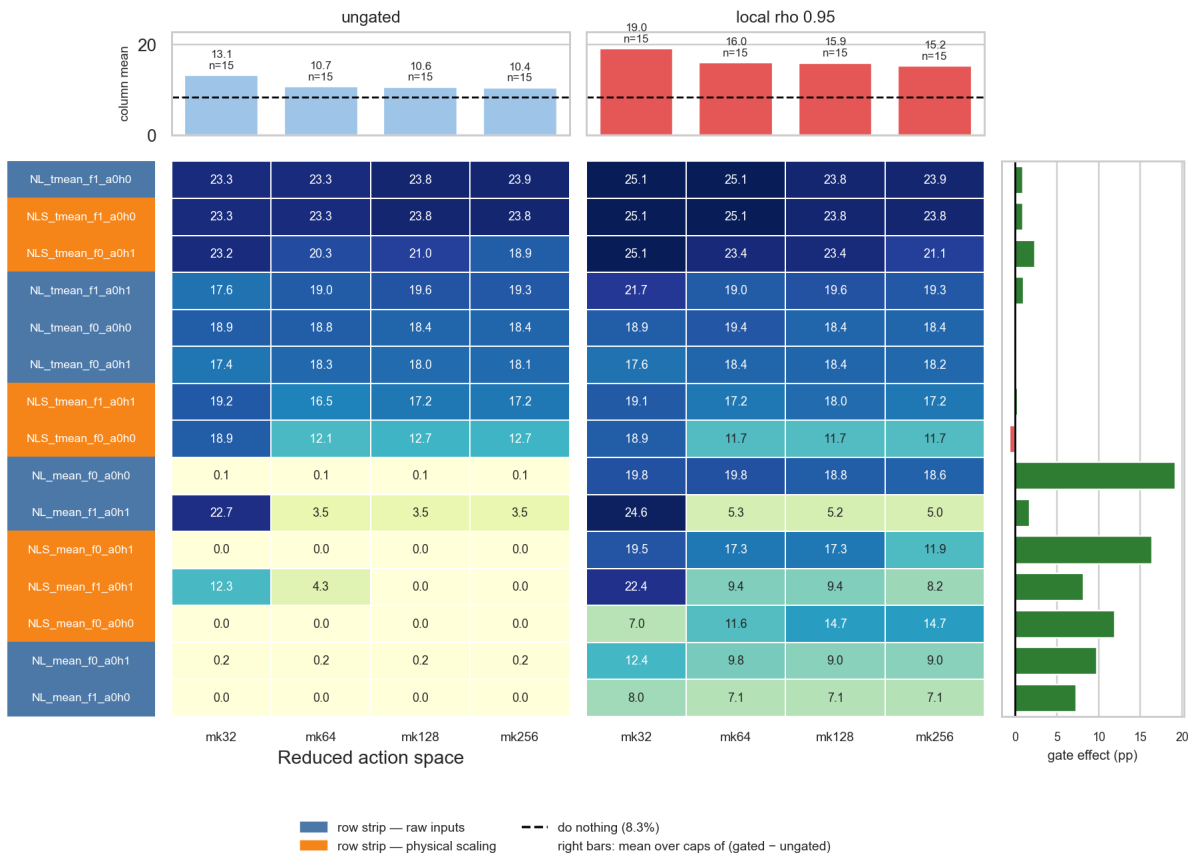


Figure A.13: Difficult-cohort survival with the adaptive intervention budget.

gmax-delta backbone, zero-shot transfer — 15 architectures x 4 reduced action spaces, both gate conditions complete survival on the 24 difficult chronics (%)

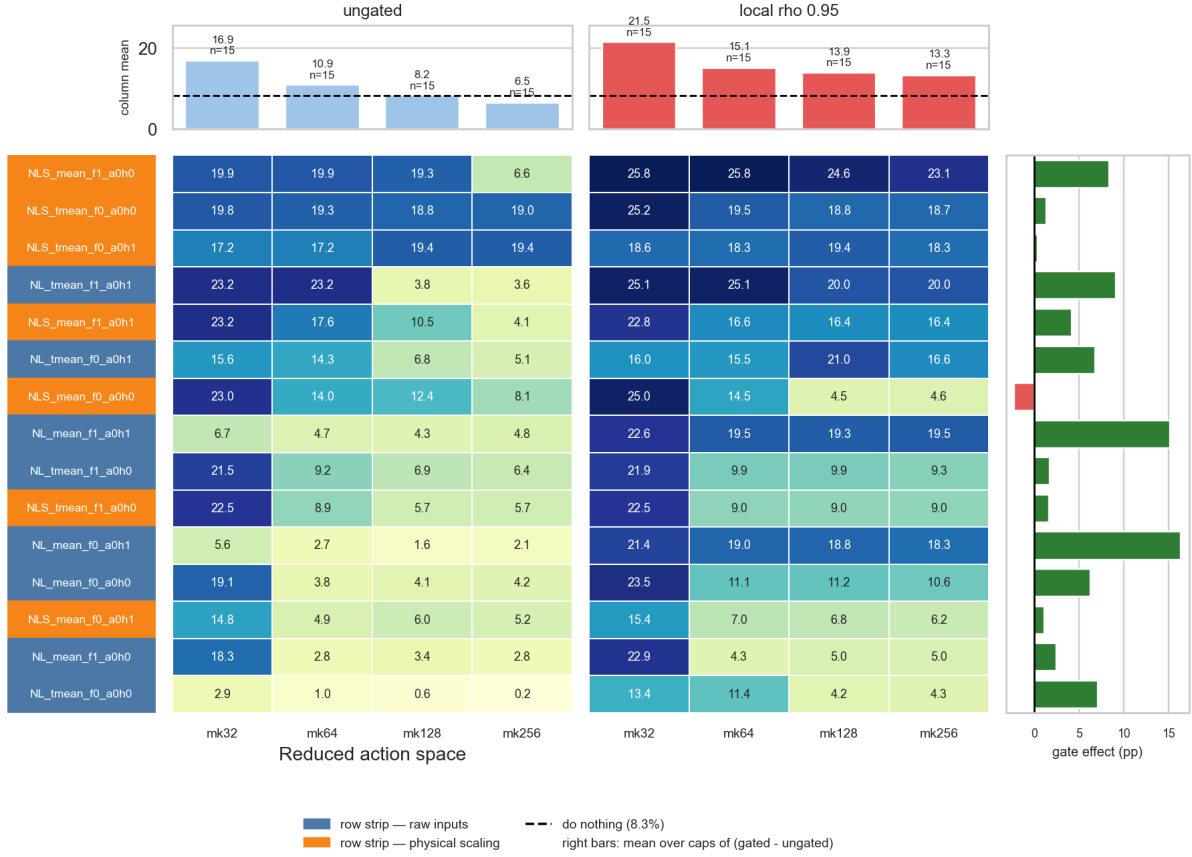


Figure A.14: Difficult-cohort survival with the Gmax-delta action feature.

### A.9.3 Target-trained controls

Table A.14: Flat-MLP control results. *Easy kept* counts complete episodes among the 26 chronics completed by do nothing.

| $k$ | Checkpoint | Heuristic           | Overall (%) | Difficult (%) | Rescues | Easy kept |
|-----|------------|---------------------|-------------|---------------|---------|-----------|
| 64  | best test  | local $\rho = 0.95$ | 43.20       | 9.79          | 1       | 15        |
| 64  | best test  | none                | 38.22       | 8.50          | 0       | 11        |
| 64  | final      | local $\rho = 0.95$ | 45.33       | 8.55          | 0       | 16        |
| 64  | final      | none                | 32.64       | 8.82          | 0       | 9         |
| 256 | best test  | local $\rho = 0.95$ | 29.80       | 5.39          | 0       | 10        |
| 256 | best test  | none                | 48.55       | 8.92          | 0       | 18        |
| 256 | final      | local $\rho = 0.95$ | 28.20       | 4.59          | 0       | 10        |
| 256 | final      | none                | 50.95       | 7.38          | 0       | 21        |

Table A.15: Fixed-list GNN controls trained directly on WCCI.

| Graph         | $k$ | Checkpoint | Overall (%) | Difficult (%) | Rescues | Easy kept |
|---------------|-----|------------|-------------|---------------|---------|-----------|
| bus_e0n0v0    | 64  | best test  | 20.05       | 3.55          | 0       | 2         |
| bus_e0n0v0    | 64  | final      | 29.47       | 4.50          | 0       | 4         |
| bus_e0n0v0    | 256 | best test  | 33.35       | 4.58          | 0       | 7         |
| bus_e0n0v0    | 256 | final      | 33.67       | 3.90          | 0       | 6         |
| bus_e1n0v0    | 64  | best test  | 37.94       | 5.10          | 0       | 12        |
| bus_e1n0v0    | 64  | final      | 39.63       | 5.15          | 0       | 9         |
| bus_e1n0v0    | 256 | best test  | 26.54       | 4.24          | 0       | 7         |
| bus_e1n0v0    | 256 | final      | 20.63       | 4.64          | 0       | 2         |
| het_ga2b_lb2a | 64  | best test  | 36.05       | 3.08          | 0       | 9         |
| het_ga2b_lb2a | 64  | final      | 39.54       | 4.59          | 0       | 10        |
| het_ga2b_lb2a | 256 | best test  | 31.31       | 4.41          | 0       | 8         |
| het_ga2b_lb2a | 256 | final      | 34.78       | 4.90          | 0       | 10        |
| het_gbi_lbi   | 64  | best test  | 51.35       | 7.21          | 0       | 21        |
| het_gbi_lbi   | 64  | final      | 44.95       | 6.80          | 0       | 15        |
| het_gbi_lbi   | 256 | best test  | 23.62       | 4.70          | 0       | 4         |
| het_gbi_lbi   | 256 | final      | 31.32       | 6.55          | 0       | 5         |

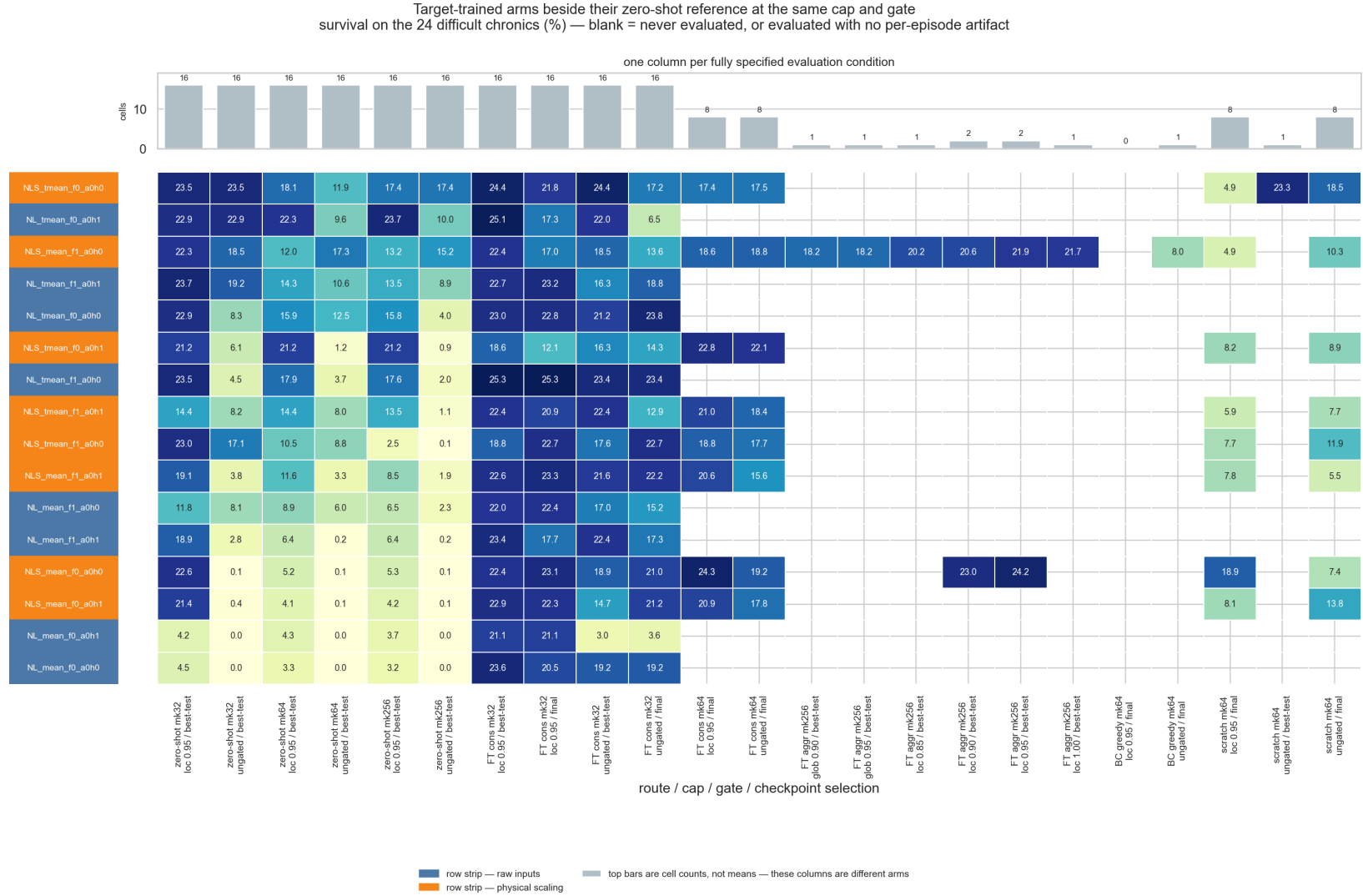


Figure A.15: Detailed comparison of target-trained WCCI controls.

## A.9.4 Per-chronic diagnostic

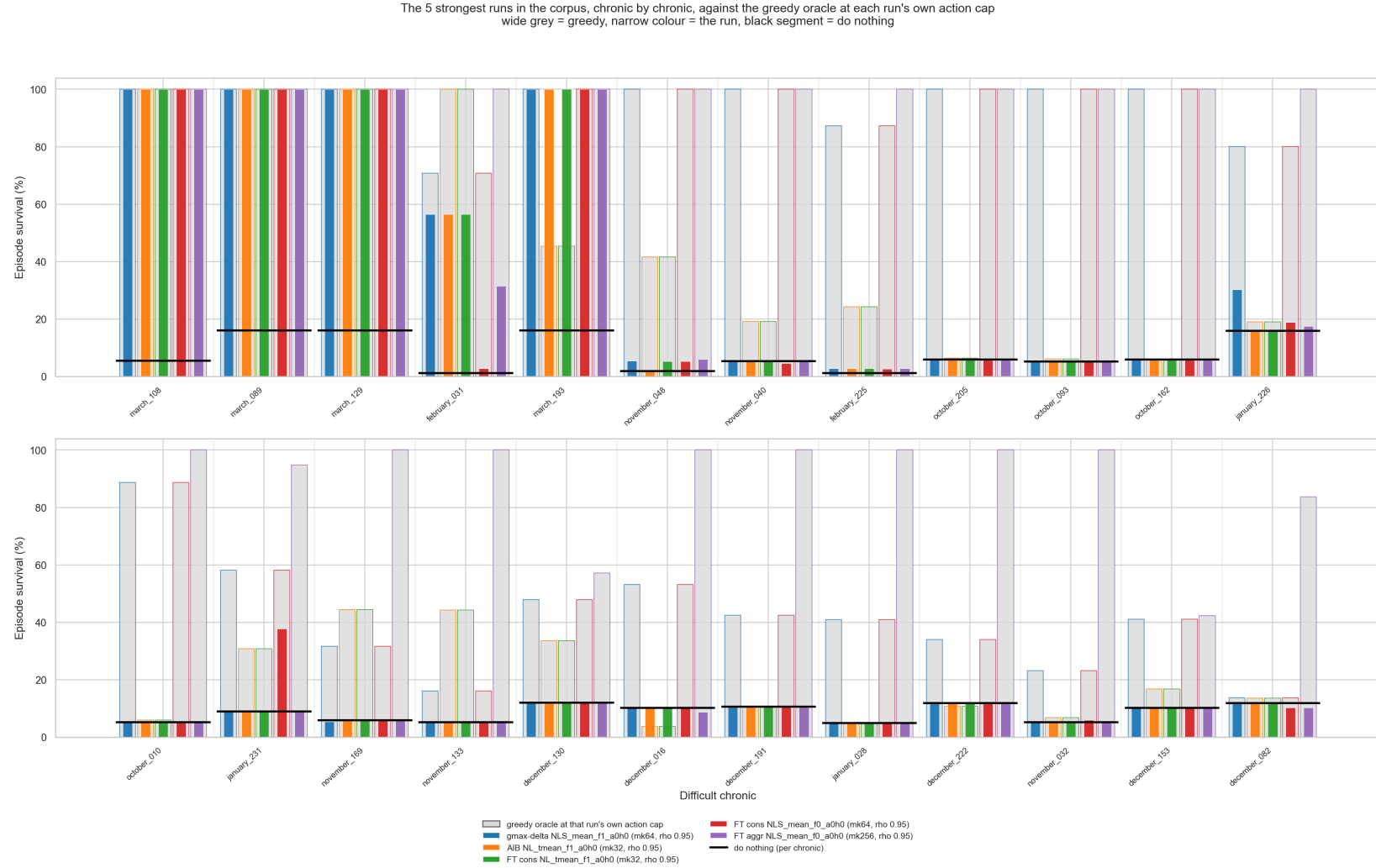


Figure A.16: Per-chronic survival for selected policies on the difficult cohort, with do nothing and the greedy reference.

## Appendix B

# GINE Graph-Actor Architecture

This appendix specifies the complete edge-aware actor used in the graph experiments: the shared input and encoder pipeline, the GINE message-passing update, the graph readout, and the agent-specific action head.

### B.1 Shared input and encoder pipeline

For agent  $a$ , the encoder receives the local graph  $\mathcal{G}_t^{(a)}$ . Candidate edges with `edge_mask`= 0 are removed before message passing, so the layer sees only the active directed set  $\mathcal{E}_t^{(a)}$ . The source of  $u \rightarrow v$  sends a message and  $v$  aggregates it. Node and edge masks do not become learned features.

Each raw node vector  $\mathbf{x}_v$  first passes through

$$\mathbf{h}_v^{(0)} = \text{LayerNorm}(\text{ReLU}(\mathbf{W}_x \mathbf{x}_v + \mathbf{b}_x)), \quad (\text{B.1})$$

which produces a 128-dimensional node state. Learned global node-identifier embeddings and the optional external edge pre-encoder are disabled. Every GINE layer nevertheless contains its own trainable projection of the edge vector to the dimension required by that layer.

Two message-passing layers are used. After each layer, the implementation applies ReLU followed by LayerNorm:

$$\mathbf{h}_v^{(k+1)} = \text{LayerNorm}\left(\text{ReLU}(\tilde{\mathbf{h}}_v^{(k+1)})\right). \quad (\text{B.2})$$

The same encoder parameters are shared by all agents. Each agent nevertheless passes its own local graph, with its own node set, active edges, and one-hop context, through that shared encoder.

### B.2 GINE message passing

For an active edge  $u \rightarrow v$ , layer  $k$  projects the complete edge feature vector and adds it to the sender state:

$$\tilde{\mathbf{e}}_{uv}^{(k)} = \mathbf{W}_e^{(k)} \mathbf{e}_{uv} + \mathbf{b}_e^{(k)}, \quad (\text{B.3})$$

$$\mathbf{m}_{u \rightarrow v}^{(k)} = \text{ReLU}\left(\mathbf{h}_u^{(k)} + \tilde{\mathbf{e}}_{uv}^{(k)}\right), \quad (\text{B.4})$$

$$\tilde{\mathbf{h}}_v^{(k+1)} = \text{MLP}^{(k)}\left((1 + \epsilon)\mathbf{h}_v^{(k)} + \sum_{u:(u,v) \in \mathcal{E}_t} \mathbf{m}_{u \rightarrow v}^{(k)}\right). \quad (\text{B.5})$$

The implementation uses the PyTorch Geometric default  $\epsilon = 0$  without a trainable epsilon parameter. Each internal GINE MLP is Linear–ReLU–Linear and outputs 128 features. Equation B.5 shows why edge attributes are part of the message content. Two identical node states can send different messages when one edge is an overloaded physical line and the other is an asset-attachment relation. Likewise,

reversing a typed disaggregated edge activates a different relation-one-hot column and can produce a different message.

GINE uses an unweighted sum over incoming active edges. It does not learn a softmax competition between neighbors. The edge embedding changes what is sent; the number of incoming messages and their vector sum determine the aggregate.

### B.3 Pooling and graph output

With mean pooling, the graph representation after the second message-passing layer is

$$\bar{\mathbf{h}}_G = \frac{\sum_v m_v \mathbf{h}_v^{(K)}}{\max(1, \sum_v m_v)}, \quad (\text{B.6})$$

where  $m_v$  is the dynamic visibility mask. This is one mean over all visible node types, not a separate mean per type. Therefore:

- the busbar graph pools controlled busbars and only the far-end busbars currently attached by live boundary lines;
- the disaggregated graph additionally pools controlled generators and loads, but never neighboring private assets;
- the disaggregated line-node graph additionally pools transmission-line nodes and contains no neighboring equipment.

An inactive contextual candidate contributes neither to the numerator nor to the denominator. The ordinary GINE mean readout includes visible context; `controlled_mean` replaces  $m_v$  by the product of the visibility and controlled-node masks. Visible context can still influence controlled nodes through message passing, but it is not aggregated directly. Sum and maximum pooling, including their `controlled_*` variants, change only the readout and do not change message passing or the information boundary.

The pooled 128-dimensional vector then passes through the shared graph output projection

$$\mathbf{z}_a = \text{ReLU}(\mathbf{W}_o \bar{\mathbf{h}}_G + \mathbf{b}_o), \quad \mathbf{z}_a \in \mathbb{R}^{128}. \quad (\text{B.7})$$

When virtual-node readout is selected, the final virtual-node state replaces  $\bar{\mathbf{h}}_G$ ; the output projection and its dimension stay the same.

### B.4 Agent-specific action head

The flat observation is not concatenated to  $\mathbf{z}_a$ . Each agent has its own two-layer MLP action head:

$$\mathbf{q}_a^{(1)} = \text{ReLU}(\mathbf{W}_a^{(1)} \mathbf{z}_a + \mathbf{b}_a^{(1)}), \quad (\text{B.8})$$

$$\mathbf{q}_a^{(2)} = \text{ReLU}(\mathbf{W}_a^{(2)} \mathbf{q}_a^{(1)} + \mathbf{b}_a^{(2)}), \quad (\text{B.9})$$

$$\boldsymbol{\ell}_a = \mathbf{W}_a^{(3)} \mathbf{q}_a^{(2)} + \mathbf{b}_a^{(3)}, \quad \boldsymbol{\ell}_a \in \mathbb{R}^{|\mathcal{A}_a|}, \quad (\text{B.10})$$

$$\pi_a(\cdot \mid \mathcal{G}_t^{(a)}) = \text{Categorical}(\text{logits} = \boldsymbol{\ell}_a). \quad (\text{B.11})$$

Both hidden layers contain 128 units. During training, an action is sampled from this categorical distribution. Deterministic evaluation selects  $\arg \max_j \ell_{a,j}$ . Action zero is the local do-nothing action.

Only the GNN encoder and graph output projection are shared across agents. The action heads are agent-specific because agents can have different local action sets and therefore different output widths  $|\mathcal{A}_a|$ . The critic is a separate MLP over the centralized flat state and does not reuse the actor graph embedding.

Equation B.11 is the default head. The optional candidate-action head of Section 5.5 keeps this pipeline up to  $\mathbf{z}_a$  and additionally consumes the post-message-passing node states  $\mathbf{h}_v^{(K)}$ , which are the same states that Equation B.6 averages.

## B.5 Exact node inputs by representation

All node types of one representation receive the same feature columns. Columns not listed for a node type are zero. Type identity reaches GINE through the `is_*` columns. The inventory below describes visible nodes. For an inactive contextual candidate, the complete row—including type and role columns—is zero, all incident edges are inactive, and the node is excluded from readout.

Table B.1: Node types and nonzero input features received by GINE, under the default `gnn_angle_representation = node`. Under `edge_diff` the `gen_theta`, `load_theta` and `theta` entries leave the node rows in every representation; the busbar and disaggregated graphs move the angle drop onto the physical line edge, while the disaggregated line-node graph writes it on the transmission-line row as `theta_diff` (Section 5.1.2).

| Representation          | Node type          | Populated input features                                                                                                                                                                                         |
|-------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bus                     | Busbar             | <code>gen_p</code> , <code>gen_theta</code> , <code>load_p</code> , <code>load_theta</code> , <code>time_before_cooldown_sub</code> , <code>domain_mask</code>                                                   |
| disaggregated           | Busbar             | <code>is_busbar</code> , <code>time_before_cooldown_sub</code> , <code>domain_mask</code>                                                                                                                        |
|                         | Generator          | <code>is_generator</code> , <code>p=gen_p</code> , <code>theta=gen_theta</code> , <code>domain_mask</code>                                                                                                       |
|                         | Load               | <code>is_load</code> , <code>p=load_p</code> , <code>theta=load_theta</code> , <code>domain_mask</code>                                                                                                          |
| disaggregated line-node | Busbar             | <code>is_busbar</code> , <code>time_before_cooldown_sub</code> , <code>domain_mask</code>                                                                                                                        |
|                         | Generator          | <code>is_generator</code> , <code>p=gen_p</code> , <code>theta=gen_theta</code> , <code>domain_mask</code>                                                                                                       |
|                         | Load               | <code>is_load</code> , <code>p=load_p</code> , <code>theta=load_theta</code> , <code>domain_mask</code>                                                                                                          |
|                         | Transmission line  | <code>is_transmission_line</code> , <code>line_status</code> , <code>rho</code> , <code>timestep_overflow</code> , <code>time_before_cooldown_line</code> , optional maintenance times, <code>domain_mask</code> |
| Any augmented schema    | Substation summary | Learned projection of the substation structural descriptor; no raw physical feature row                                                                                                                          |
|                         | Virtual node       | One learned vector shared across graphs; no raw physical feature row                                                                                                                                             |

## B.6 Exact edge inputs by representation

The edge vector contains all edge information consumed by GINE. Inactive candidates are filtered out before the vectors in Table B.2 reach a convolution.

Table B.2: Directed edge relations and edge-feature vectors received by GINE, under the default `gnn_angle_representation = node`. “Relation” denotes one active relation-one-hot column. Under `edge_diff` the physical-line block of the busbar and disaggregated graphs gains a `theta_diff` column; the line-node rows are unchanged, because there the drop is a line-node feature and the attachment edges stay free of physical channels.

| Representation          | Directed relation                                           | Edge features                                                                                                                                                           |
|-------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bus                     | Physical busbar $\leftrightarrow$ busbar                    | <code>line_status</code> , <code>rho</code> , <code>timestep_overflow</code> , <code>time_before_cooldown_line</code> , optional maintenance times; no relation one-hot |
|                         | Optional self or same-substation                            | Same width; all physical values zero; no relation one-hot                                                                                                               |
| disaggregated           | Physical busbar $\leftrightarrow$ busbar                    | Complete measured physical-line block plus <code>relation_physical_line</code>                                                                                          |
|                         | Generator $\leftrightarrow$ busbar                          | Neutral physical block plus direction-specific generator relation                                                                                                       |
|                         | Load $\leftrightarrow$ busbar                               | Neutral physical block plus direction-specific load relation                                                                                                            |
|                         | Optional self or same-substation                            | Neutral physical block plus the corresponding relation                                                                                                                  |
| disaggregated line-node | Busbar $\leftrightarrow$ line node                          | One origin/extremity-and-direction relation; no line-state attributes                                                                                                   |
|                         | Generator $\leftrightarrow$ busbar                          | One direction-specific generator relation                                                                                                                               |
|                         | Load $\leftrightarrow$ busbar                               | One direction-specific load relation                                                                                                                                    |
|                         | Optional self or same-substation                            | One corresponding relation                                                                                                                                              |
| Any augmented schema    | Busbar $\rightarrow$ summary/virtual, with optional reverse | Zero edge vector; the relation carries no flow                                                                                                                          |

The complete relation names and activation rules are given in Tables 5.5 and 5.7. Those tables distinguish, for example, generator-to-busbar from busbar-to-generator and origin from extremity line attachments. These distinctions matter directly to an edge-aware model such as GINE, which folds the edge vector into every message.

# Appendix C

## Graph Construction Details

This appendix collects the construction and sizing details of the three graph schemas of Section 5.2: how large each candidate graph is, how inactive candidates are neutralized, and how a local actor graph is cut out of the full grid. None of it changes the representations themselves.

### C.1 Candidate graph sizes

Let  $S$  be the number of included substations,  $B$  the busbars per substation,  $L$  the included physical lines, and  $N_g, N_d$  the generators and loads. Write  $d_g, d_d, d_\ell \in \{1, 2\}$  for the number of retained directions of the generator, load, and line-node attachments: 2 for **bidirectional** and 1 for either one-way mode. Let  $P_\ell$  be the number of represented line endpoints: an internal line contributes two and a boundary line in a local disaggregated line-node graph contributes only its controlled endpoint. Thus  $L \leq P_\ell \leq 2L$ . Table C.1 gives the size of the fixed candidate graph before any mask is applied.

Table C.1: Size of the candidate graph of each schema. Enabling self edges adds  $|\mathcal{V}|$  directed edges and enabling same-substation edges adds  $SB(B - 1)$ , in every schema.

| Schema                  | Candidate nodes      | Candidate directed edges              |
|-------------------------|----------------------|---------------------------------------|
| <b>bus</b>              | $SB$                 | $2B^2L$                               |
| disaggregated           | $SB + N_g + N_d$     | $2B^2L + B(d_gN_g + d_dN_d)$          |
| disaggregated line-node | $SB + N_g + N_d + L$ | $Bd_\ell P_\ell + B(d_gN_g + d_dN_d)$ |

Three consequences are worth naming. A busbar graph with both optional relations enabled therefore has  $2B^2L + SB + SB(B - 1)$  candidate directed edges. The disaggregated schema adds  $N_g + N_d$  nodes and between  $B(N_g + N_d)$  and  $2B(N_g + N_d)$  asset candidates to the busbar graph, and buys individual asset states with them. For a complete state graph,  $P_\ell = 2L$ , so the disaggregated line-node schema replaces the  $2B^2L$  busbar-to-busbar term by  $2Bd_\ell L$ : identical at  $B = 2$  and cheaper for  $B > 2$ . A local boundary line has candidates only at its controlled endpoint, which reduces  $P_\ell$  and prevents construction of a far-end attachment. The schema adds  $L$  line nodes and one message-passing hop for internal endpoint exchange.

The hierarchical augmentations of Section 5.3 add on top of any of these: one substation-summary node per substation, with  $SB$  directed edges for **toward\_summary** or  $2SB$  for **bidirectional**; and one virtual node, with  $N_{\text{bus}}$  or  $2N_{\text{bus}}$  edges, or  $S$  or  $2S$  when summary nodes are also enabled.

### C.2 Indexing and masking conventions

Grid2Op numbers busbars from one in its topology vector; those identifiers are converted to the zero-based index  $b$  used in the node identifier  $i(s, b)$ .

An inactive candidate edge keeps its row in the edge-feature tensor, so the tensor shape is fixed for an episode, but the row is zero-filled and its **edge\_mask** entry is 0. Masked edges are removed before the

encoder is called, so those zero rows never generate a message. This is what allows a topology change to alter connectivity without rebuilding the graph. An inactive contextual node likewise retains its candidate row, but its complete feature vector is zero and its `node_mask` entry is 0. Every incident physical or structural edge is deactivated. The row therefore participates in neither message passing nor graph readout; a mean readout excludes it from both numerator and denominator.

### C.3 Local actor graphs

A local actor graph always contains the controlled substations and every line touching them. Additional membership is schema-specific. The `bus` graph contains only the far-end busbar currently attached by each live boundary line, and that busbar exposes aggregated neighboring power and angle. The direct disaggregated graph contains the same far-end busbar, but never its generator or load nodes. The disaggregated line-node graph represents the shared line itself and therefore constructs no neighboring busbar or private asset and no far-end attachment edge. The centralized state graph is intentionally unrestricted for centralized training. Ordinary pooling aggregates all visible nodes, whereas `controlled_*` pooling restricts readout to action-domain nodes; neither choice changes the construction boundary.

### C.4 Boundary verification

The regression suite checks that contextual visibility follows a live busbar attachment, moves when the attachment is switched, and disappears after a line outage; controlled nodes remain present; structural relations cannot make a hidden candidate visible; inactive rows are zero; neighboring private assets are absent from disaggregated graphs; and all neighboring equipment is absent from disaggregated line-node local graphs. Candidate-action tests additionally verify that a boundary line remains addressable through its line node without adding far-end equipment. Controlled-readout tests verify both that the pooled set is fixed and that gradients can still reach visible context through message passing.

## Appendix D

# Experiment Configurations

This appendix provides the complete run configurations for the GINE encoder and heavy GINE ablations.

Table D.1: GINE rerun configuration summary. The reference run is Heavy GINE Baseline.

| Run                             | Actor/critic encoder   | GNN hidden/layers | Actor/critic heads            | Concat flat | Edge pre-enc. | Entropy | Init action-0 | Opt. critic | Difference from Heavy GINE Baseline                                                                      |
|---------------------------------|------------------------|-------------------|-------------------------------|-------------|---------------|---------|---------------|-------------|----------------------------------------------------------------------------------------------------------|
| Heavy GINE Baseline             | <code>gnn / gnn</code> | 128 / 2           | $3 \times 256 / 3 \times 256$ | true        | true          | 0.02    | 0.7           | false       | Baseline: heavy concat-flat GINE actor, GNN critic, and legacy critic updates                            |
| Light GINE (Concat, MLP Critic) | <code>gnn / mlp</code> | 64 / 1            | $2 \times 128 / 2 \times 128$ | true        | false         | 0.02    | 0.7           | true        | Light GINE actor, MLP critic, smaller heads, no edge pre-encoder, optimized critic updates               |
| Heavy GINE (No Bias)            | <code>gnn / gnn</code> | 128 / 2           | $3 \times 256 / 3 \times 256$ | true        | true          | 0.02    | 0.0           | false       | Bias ablation: same as baseline, but the initial do-nothing prior is 0.0                                 |
| No Bias, Opt Critic             | <code>gnn / gnn</code> | 128 / 2           | $3 \times 256 / 3 \times 256$ | true        | true          | 0.02    | 0.0           | true        | Bias 0.0 plus optimized critic updates; otherwise heavy concat-flat GINE with GNN critic                 |
| Optimized Heavy GINE            | <code>gnn / gnn</code> | 128 / 2           | $3 \times 256 / 3 \times 256$ | true        | true          | 0.01    | 0.0           | true        | Optimized heavy GINE: bias 0.0, lower entropy 0.01, optimized critic updates                             |
| Optimized Light GINE            | <code>gnn / mlp</code> | 64 / 1            | $2 \times 128 / 2 \times 128$ | false       | false         | 0.01    | 0.0           | true        | Light optimized GINE: MLP critic, no concat-flat, lower entropy 0.01, bias 0.0, optimized critic updates |

Table D.2: Heavy-GINE parameter comparison from configs/gine\_s0\_s1\_s2.

| Run                                | Main change from baseline                                           | GNN  | GNN size       | Critic | Actor/critic heads            | Concat | Shared | Edge pre. | Node ID | Entropy   | Init action-0 | Opt. critic |
|------------------------------------|---------------------------------------------------------------------|------|----------------|--------|-------------------------------|--------|--------|-----------|---------|-----------|---------------|-------------|
| No Flat Concat                     | Removes flat-observation concatenation                              | gine | $2 \times 128$ | gnn    | $3 \times 256 / 3 \times 256$ | no     | yes    | yes       | yes     | 0.02–0.02 | 0.7           | no          |
| Heavy GINE Baseline                | Reference: heavy concat-flat GINE with GNN critic                   | gine | $2 \times 128$ | gnn    | $3 \times 256 / 3 \times 256$ | yes    | yes    | yes       | yes     | 0.02–0.02 | 0.7           | no          |
| Opt. Critic Updates                | Enables optimized critic updates                                    | gine | $2 \times 128$ | gnn    | $3 \times 256 / 3 \times 256$ | yes    | yes    | yes       | yes     | 0.02–0.02 | 0.7           | yes         |
| MLP Critic                         | Replaces the GNN critic with an MLP critic                          | gine | $2 \times 128$ | mlp    | $3 \times 256 / 3 \times 256$ | yes    | yes    | yes       | yes     | 0.02–0.02 | 0.7           | no          |
| Unshared Actor                     | Uses non-shared actor GNN weights                                   | gine | $2 \times 128$ | gnn    | $3 \times 256 / 3 \times 256$ | yes    | no     | yes       | yes     | 0.02–0.02 | 0.7           | no          |
| Light GINE Encoder                 | Uses a lighter GINE encoder                                         | gine | $1 \times 64$  | gnn    | $3 \times 256 / 3 \times 256$ | yes    | yes    | yes       | yes     | 0.02–0.02 | 0.7           | no          |
| Zero Entropy Decay                 | Decays entropy from 0.02 to 0.0                                     | gine | $2 \times 128$ | gnn    | $3 \times 256 / 3 \times 256$ | yes    | yes    | yes       | yes     | 0.02–0.00 | 0.7           | no          |
| No Node-ID Embs                    | Removes learned node-ID embeddings                                  | gine | $2 \times 128$ | gnn    | $3 \times 256 / 3 \times 256$ | yes    | yes    | yes       | no      | 0.02–0.02 | 0.7           | no          |
| GAT Message Passing                | Replaces GINE with GAT message passing                              | gat  | $2 \times 128$ | gnn    | $3 \times 256 / 3 \times 256$ | yes    | yes    | yes       | yes     | 0.02–0.02 | 0.7           | no          |
| Weighted GCN                       | Replaces GINE with weighted GCN message passing                     | gcn  | $2 \times 128$ | gnn    | $3 \times 256 / 3 \times 256$ | yes    | yes    | yes       | yes     | 0.02–0.02 | 0.7           | no          |
| Light GINE (No Concat, MLP Critic) | Light no-concat GINE with MLP critic and optimized critic updates   | gine | $1 \times 64$  | mlp    | $2 \times 128 / 2 \times 128$ | no     | yes    | no        | yes     | 0.02–0.02 | 0.7           | yes         |
| Light GINE (Concat, MLP Critic)    | Light concat-flat GINE with MLP critic and optimized critic updates | gine | $1 \times 64$  | mlp    | $2 \times 128 / 2 \times 128$ | yes    | yes    | no        | yes     | 0.02–0.02 | 0.7           | yes         |
| No Initial Bias                    | Removes the initial do-nothing bias                                 | gine | $2 \times 128$ | gnn    | $3 \times 256 / 3 \times 256$ | yes    | yes    | yes       | yes     | 0.02–0.02 | 0.0           | no          |

## Appendix E

# Full-Test Checkpoint Registry

Table [E.1](#) reports the checkpoint step used for each full-test result table. The values are the `checkpoint_global_step` fields stored in the corresponding JSON files under `Topology_Task/outputs/full_test_eval`. The do-nothing policy has no learned checkpoint. For the behavioral cloning students, the stored step is the student checkpoint’s optimizer-step count rather than a MAPPO environment step.

Table E.1: Checkpoint steps used for the full-test evaluations.

| Run type                                               | Seed 0 checkpoint step | Seed 1 checkpoint step | Seed 2 checkpoint step |
|--------------------------------------------------------|------------------------|------------------------|------------------------|
| Do-nothing policy                                      | —                      | —                      | —                      |
| Baseline flat policy                                   | 13,600,000             | 13,040,000             | 13,760,000             |
| Global rho heuristic, $\rho_{safe} = 0.90$             | 13,920,000             | 14,320,000             | 14,800,000             |
| Local rho heuristic, $\rho_{safe} = 0.90$              | 14,160,000             | 14,400,000             | 14,160,000             |
| Intervention gate, final-action MAP                    | 12,000,000             | 12,720,000             | 13,760,000             |
| Intervention gate, hierarchical greedy                 | 7,120,000              | 14,720,000             | 13,760,000             |
| Flat Sparse16, $\lambda_{fixed} = 0.000$               | 9,600,000              | 12,160,000             | 7,920,000              |
| Flat Sparse16, $\lambda_{fixed} = 0.001$               | 10,640,000             | 11,520,000             | 11,600,000             |
| Flat Sparse16, $\lambda_{fixed} = 0.003$               | 12,000,000             | 12,960,000             | 12,480,000             |
| Flat Sparse16, $\lambda_{fixed} = 0.010$               | 8,960,000              | 5,760,000              | 12,560,000             |
| Gated Sparse16, $\lambda_{fixed} = 0.000$              | 8,480,000              | 4,320,000              | 6,000,000              |
| Gated Sparse16, $\lambda_{fixed} = 0.001$              | 5,760,000              | 7,600,000              | 3,280,000              |
| Gated Sparse16, $\lambda_{fixed} = 0.003$              | 8,960,000              | 7,120,000              | 7,600,000              |
| Gated Sparse16, $\lambda_{fixed} = 0.010$              | 12,000,000             | 7,840,000              | 7,920,000              |
| Flat local-safe AIB, $d = 0.20$                        | 13,120,000             | 9,360,000              | 12,160,000             |
| Flat local-safe AIB, $d = 0.10$                        | 12,880,000             | 13,200,000             | 12,720,000             |
| Flat local-safe AIB, $d = 0.35$                        | 12,800,000             | 5,360,000              | 11,520,000             |
| Gated local-safe AIB, $d = 0.20$ , hierarchical greedy | 14,960,000             | 5,280,000              | 16,800,000             |
| Flat non-idle AIB, $d = 0.20$                          | 12,480,000             | 14,880,000             | 11,920,000             |
| Teacher: MAPPO with local rho heuristic                | 14,160,000             | 14,400,000             | 14,160,000             |
| Student BC, balanced non-idle 0.50, weight 5, aux 0.50 | 12,375                 | 14,994                 | 13,617                 |
| Student BC, balanced non-idle 0.20, weight 3, aux 0.25 | 12,375                 | 14,994                 | 13,617                 |

# Bibliography

- [1] E. B. Fisher, R. P. O'Neill, and M. C. Ferris, "Optimal transmission switching," *IEEE Transactions on Power Systems*, vol. 23, no. 3, pp. 1346–1355, 2008.
- [2] J. Viebahn, M. Naglic, A. Marot, B. Donnot, and S. H. Tindemans, "Potential and challenges of AI-powered decision support for short-term system operations," in *CIGRE Paris Session*, 2022.
- [3] B. Donnot, "Grid2op: A testbed platform to model sequential decision making in power systems," *arXiv preprint arXiv:2011.07929*, 2020.
- [4] A. Marot et al., "Learning to run a power network challenge for training topology controllers," *Electric Power Systems Research*, 2021.
- [5] M. Tan, "Multi-agent reinforcement learning: Independent vs. cooperative agents," in *International Conference on Machine Learning*, 1993.
- [6] R. Lowe, Y. Wu, A. Tamar, J. Harb, P. Abbeel, and I. Mordatch, "Multi-agent actor-critic for mixed cooperative-competitive environments," in *Advances in Neural Information Processing Systems*, 2017.
- [7] C. Yu et al., "The surprising effectiveness of ppo in cooperative multi-agent games," *Advances in Neural Information Processing Systems*, 2022.
- [8] M. Hassouna, C. Holzhüter, P. Lytaev, J. Thomas, B. Sick, and C. Scholz, *Graph reinforcement learning for power grids: A comprehensive survey*, 2024. arXiv: [2407.04522](https://arxiv.org/abs/2407.04522). [Online]. Available: <https://arxiv.org/abs/2407.04522>
- [9] M. Subramanian, J. Viebahn, S. H. Tindemans, B. Donnot, and A. Marot, "Exploring grid topology reconfiguration using a simple deep reinforcement learning approach," in *IEEE Madrid PowerTech*, 2021.
- [10] D. Yoon, S. Hong, B.-J. Lee, and K.-E. Kim, "Winning the L2RPN challenge: Power grid management via semi-Markov afterstate actor-critic," in *International Conference on Learning Representations*, 2021.
- [11] A. Chauhan, M. Baranwal, and A. Basumatary, "PowRL: A reinforcement learning framework for robust management of power networks," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, 2023, pp. 14 757–14 764.
- [12] M. Lehna, J. Viebahn, A. Marot, S. Tomforde, and C. Scholz, "Managing power grids through topology actions: A comparative study between advanced rule-based and reinforcement learning agents," *Energy and AI*, vol. 14, p. 100 276, 2023.
- [13] B. Manczak, J. Viebahn, and H. van Hoof, *Hierarchical reinforcement learning for power network topology control*, 2023. arXiv: [2311.02129](https://arxiv.org/abs/2311.02129). [Online]. Available: <https://arxiv.org/abs/2311.02129>
- [14] E. van der Sar, A. Zocca, and S. Bhulai, *Multi-agent reinforcement learning for power grid topology optimization*, 2023. arXiv: [2310.02605](https://arxiv.org/abs/2310.02605). [Online]. Available: <https://arxiv.org/abs/2310.02605>
- [15] B. de Mol, D. Barbieri, J. Viebahn, and D. Grossi, "Centrally coordinated multi-agent reinforcement learning for power grid topology control," in *Proceedings of the 16th ACM International Conference on Future and Sustainable Energy Systems (e-Energy)*, 2025.
- [16] E. Marchesini et al., "MARL2Grid-TR: A multi-agent RL benchmark in power grid operations," in *International Conference on Learning Representations*, 2026.
- [17] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," *International Conference on Learning Representations*, 2017.

- [18] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, “Graph attention networks,” in *International Conference on Learning Representations*, 2018.
- [19] K. Xu, W. Hu, J. Leskovec, and S. Jegelka, “How powerful are graph neural networks?” *International Conference on Learning Representations*, 2019.
- [20] M. de Jong, J. Viebahn, and Y. Shapovalova, *Generalizable graph neural networks for robust power grid topology control*, 2025. arXiv: [2501.07186](https://arxiv.org/abs/2501.07186). [Online]. Available: <https://arxiv.org/abs/2501.07186>
- [21] E. Marchesini et al., *RL2Grid: Benchmarking reinforcement learning in power grid operations*, 2025. arXiv: [2503.23101](https://arxiv.org/abs/2503.23101). [Online]. Available: <https://arxiv.org/abs/2503.23101>
- [22] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal policy optimization algorithms,” *arXiv preprint arXiv:1707.06347*, 2017.
- [23] J. Achiam, D. Held, A. Tamar, and P. Abbeel, “Constrained policy optimization,” *International Conference on Machine Learning*, 2017. [Online]. Available: <https://arxiv.org/abs/1705.10528>
- [24] A. Stooke, J. Achiam, and P. Abbeel, “Responsive safety in reinforcement learning by PID lagrangian methods,” *International Conference on Machine Learning*, 2020. [Online]. Available: <https://arxiv.org/abs/2007.03964>
- [25] N. Carrara, E. Leurent, R. Laroche, T. Urvoy, and O.-A. Maillard, “Budgeted reinforcement learning in continuous state space,” *Advances in Neural Information Processing Systems*, 2019. [Online]. Available: <https://arxiv.org/abs/1903.01004>
- [26] L. Lautenbacher et al., *Multi-objective reinforcement learning for power grid topology control*, 2025. arXiv: [2502.00040](https://arxiv.org/abs/2502.00040). [Online]. Available: <https://arxiv.org/abs/2502.00040>
- [27] W. Hu et al., “Strategies for pre-training graph neural networks,” in *International Conference on Learning Representations*, 2020.